

Dynamics and Guidance of Reentry Spacecraft

Submitted in partial fulfilment of the requirements

for the degree of

Master of Technology

By

Pragati Pradip Joshi

Roll No.-15210059

Under guidance of

Prof. Hari Hablani

Space Flight Vehicles Guidance, Navigation and Control

Center of Astronomy

Indian Institute of Technology

Indore

Prof. Harish Palanthandalam-Madapusi

Associate Professor

Mechanical Engineering Department

Indian Institute of Technology

Gandhinagar



Department of Mechanical Engineering

Indian Institute of Technology Gandhinagar

Dynamics and Guidance of Reentry Spacecraft

Submitted in partial fulfilment of the requirements

for the degree of

Master of Technology

By

Pragati Pradip Joshi

Roll No.-15210059

Under guidance of

Prof. Hari Hablani

Space Flight Vehicles Guidance, Navigation and Control

Center of Astronomy

Indian Institute of Technology

Indore

Prof. Harish Palanthandalam-Madapusi

Associate Professor

Mechanical Engineering Department

Indian Institute of Technology

Gandhinagar



Department of Mechanical Engineering

Indian Institute of Technology Gandhinagar

Advisory Committee:

1. Prof. Harish PM
2. Prof. Atul Bhargav
3. Prof. Nihar Mohapatra

Declaration

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included; I have adequately cited the referenced. I also declare that I adhere to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will cause disciplinary action from institute and evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Pragati Pradip Joshi

15210059

Certificate

It is certified that the work contained in the thesis titled “**Dynamics and Guidance of Reentry Spacecraft**” by PRAGATI PRADIP JOSHI (15210059), has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

Prof. Harish Palanthandalam-Madapusi

Associate Professor

Indian Institute of Technology Gandhinagar

Acknowledgement

I would like to express my deep and sincere gratitude to my guide Prof. H. B. Hablani who has been constantly guiding and encouraging throughout the project work. I am thankful to my supervisor Prof. Harish PM who gave me opportunity to continue with research.

I am grateful to Mr. Madhav Dhekane, Director , IISU, Dr. U.P Rajeev, Head ,Guidance and Design Division, VSSC, Dr. Sriram Swaminathan , Scientist, Guidance and Design Division , VSSC who gave me golden opportunity to be a part of Indian Space Research Organisation .Finally I extend my thanks to my family, friends who have been supported me throughout.

Pragati Pradip Joshi

Contents

Declaration	i
Certificate	ii
Acknowledgement	iii
List of figures	vi
List of tables	viii
List of symbols	ix
Abstract	xi
Chapter	
1. Introduction	1
1.1. Motivation	1
1.2. Literature Survey	1
1.3. Report Layout	2
2. Reentry Dynamics	3
2.1. Atmospheric reentry	3
2.2. Reentry dynamics	3
2.3. Reentry of Apollo command module	9
2.3.1. Calculation of angle of attack for trim condition	11
2.4. Entry Corridor	12
2.4.1. Equations involved in entry corridor design	12
2.4.2. Simulation results	13
3. Unmanned Apollo Command Module: Ballistic Reentry	15
3.1. Ballistic reentry of unmanned Apollo command module	15
3.2. Initial values	16
3.3. Frame transformation	17
3.3.1. Frame transformation from RTN frame to ECI frame	18
3.3.2. Frame transformation from ECI frame to wind axes frame	20
3.4. Simulation results for ballistic reentry	23
4. Unmanned Apollo Command Module: Low-lifting Reentry	31
4.1. Low-lifting reentry	31
4.2. Unguided low-lifting reentry	33
4.2.1. Governing equations and initial state vector	33
4.2.2. Simulation results	34
4.3. Guided low-lifting reentry	41
4.3.1. Guidance laws	41
4.3.2. Generation of nominal trajectory	45
4.3.2.1. Governing equations and initial state vector	45
4.3.2.2. Implementation of constant drag deceleration law	45
4.3.2.3. Implementation of constant descent rate law	51
4.4. Closed-loop guidance	53
4.4.1. Closed-loop guidance laws	54
4.4.2. Determination of C_1 and C_2	55

4.4.3. Roll angle modulation	56
4.5. Rigid body Dynamics	57
5. X-33 Vehicle Reentry Problems	58
5.1. Equations of motion for reentry	58
5.2. Parameters for simulation	60
5.3. Simulation results	60
5.3.1. Variation of Angle of attack	61
5.4. Optimization problem	64
5.4.1. Optimization of single angle of attack for whole flight to land the vehicle at a particular location	64
5.4.1.1. Problem definition	64
5.4.1.2. Objective function	64
5.4.1.3. Optimization technique	64
5.4.1.4. Results	66
5.4. 2. Optimization of angle of attack using <i>fmincon</i> to maximize range	67
5.4.2.1. Problem definition	67
5.4.2.2. Objective function	67
5.4.2.3. Nonlinear constraints	67
5.4.2.4. Initialization and boundary conditions	68
5.4.2.5. Optimization technique	68
5.4.2.6. Results	68
6. Conclusion and Future Work	72
References	73
Appendix A	74
Standard Atmosphere model	

List of Figures:

1. Body centered frame
2. Velocity components in ENU frame
3. Aerodynamic force components in wind axes frame
4. Apollo command module
5. Trim condition
6. Entry corridor for unmanned capsule
7. Entry corridor for manned reentry module
8. Approach entry into atmosphere
9. Conic orbit
10. Nodal and ECI frame
11. Density model
12. Ballistic reentry: Altitude vs time
13. Ballistic reentry: Velocity vs time
14. Ballistic reentry: Flight path angle vs time
15. Ballistic reentry: Latitude vs longitude
16. Ballistic reentry: Downrange vs time
17. Ballistic reentry: Azimuth angle variation
18. Ballistic reentry: Aerodynamic coefficients vs time
19. Ballistic reentry: L/D ratio vs time
20. Ballistic reentry: Gravitational acceleration vs time
21. Ballistic reentry: Aerodynamic acceleration vs time
22. Ballistic entry heating and acceleration
23. Maximum heating and maximum acceleration variation with initial flight path angle
24. Low-lifting reentry: Aerodynamic coefficients and L/D vs angle of attack (α)
25. Unguided lifting reentry: Altitude variation
26. Unguided lifting reentry: Velocity vs time
27. Unguided lifting reentry: Flight path angle vs time
28. Unguided lifting reentry: Latitude vs longitude
29. Unguided lifting reentry: Azimuth angle variation
30. Unguided lifting reentry: Downrange variation
31. Unguided lifting reentry: Aerodynamic coefficients vs time
32. Unguided lifting reentry: L/D vs time
33. Unguided lifting reentry: Aerodynamic acceleration vs time
34. Unguided lifting reentry: Gravitational acceleration vs time
35. Unguided lifting reentry: Heat generated vs time
36. Drag deceleration comparison to switch to constant drag deceleration law
37. Constant drag deceleration law: Altitude vs time
38. Constant drag deceleration law: Velocity variation
39. Constant drag deceleration law : Flight path angle vs time
40. Constant drag deceleration law: Latitude vs longitude
41. Constant drag deceleration law: Azimuth angle vs time
42. Constant drag deceleration law: Downrange variation
43. Constant drag deceleration law: Comparison of required angle of attack and trim angle of attack for desired L/D
44. Constant drag deceleration law: Drag deceleration vs time
45. Open-loop guided nominal trajectory: Aerodynamic acceleration vs time
46. Constant drag deceleration law: Descent rate vs time
47. Guided L/D for constant drag deceleration law vs altitude
48. Guided L/D for constant drag deceleration law and constant descent rate law
49. Constant descent rate law: descent rate vs time
50. Closed-loop guidance law: constant drag deceleration law
51. Closed-loop guidance law: constant descent rate law
52. Roll modulation
53. Rotational dynamics of Apollo
54. Three-view of X-33 vehicle
55. Variation of angle of attack: Altitude vs time
56. Variation of angle of attack: Velocity vs time

57. Variation of angle of attack: Flight path angle vs time
58. Variation of angle of attack: Heading angle vs time
59. Variation of angle of attack: Latitude vs longitude
60. Quadratic steepest descent method
61. Steepest descent method with simple linear interpolation
62. Optimization of angle of attack using *fmincon* to maximize range: Altitude vs time
63. Optimization of angle of attack using *fmincon* to maximize range: Velocity vs time
64. Optimization of angle of attack using *fmincon* to maximize range: Flight path angle vs time
65. Optimization of angle of attack using *fmincon* to maximize range: Heading angle vs time
66. Optimization of angle of attack using *fmincon* to maximize range: Latitude vs longitude

List of Tables:

1. Abbreviations
2. Apollo parameters
3. Orbital elements
4. Harris-Priester density model
5. X-33 Parameters
6. Initial parameters for X-33
7. Gliding condition for various α
8. Details of optimization
9. Optimised α details

List of Symbols:

τ - Longitude

δ - Latitude

r - Radial distance from center of the earth

v - Velocity of vehicle

γ - Flight path angle

χ - Azimuth angle

do - Downrange

W - Gravitational force vector

A - Aerodynamic force vector

α - Angle of attack

g - Gravitational acceleration

L/D - Lift to drag ratio

$(L/D)_v$ - Vertical component of L/D

D - Drag force

v_c -Velocity of vehicle in circular orbit around earth

μ - Gravitational parameter of earth

w_e - Rotational velocity of earth

h_s - Atmospheric scale height of earth

B_c -Ballistic coefficient

r_0 - Earth's radius

ρ –Density of atmosphere

S - Reference cross-sectional area

m - Mass of vehicle

dia - Diameter of vehicle

p - Semi-latus rectum of orbit

e - Eccentricity of orbit

θ - True anomaly of orbit

Ω - Right ascension of ascending node

i - Inclination of orbit

ω - Argument of perigee of orbit

C_L - Coefficient of lift

C_D - Coefficient of drag

C_A - Axial force component

C_N - Normal force component

L- Lift force

β_c - Command roll angle

θ_c – Spherical cone angle

C_1, C_2 - Control gains

d_0 - Nominal drag deceleration

$\dot{h}_0$ - Nominal descent rate

Γ_1, Γ_2 - Rotation matrices

Abstract

The diversity of reentry space missions of recent times is astonishing. Soyuz (Russian) spacecraft bring astronauts from International Space Station (ISS) to earth and land in the deserts of Kazakhstan in Central Asia. Reentry Module of Chinese Shenzhou (Divine) spacecraft, designed for human space flight, lands on the ground or in ocean. SpaceX Dragon ferries cargo between the USA and the ISS and it splashes in the Pacific Ocean, analogous to Apollo landings decades earlier but it is, remarkably, reusable. ISRO's Space Recovery Experiment splashed down in the Bay of Bengal also. ISRO is conducting flight tests of its own reusable launch vehicle. Its reentry module resembles now retired Space Shuttle which entered the atmosphere like an airplane. Cygnus (American) reentry craft is designed to transport cargo to ISS but it burns during its fiery atmospheric reentry, similar to the Automated Transfer Vehicles of European Space Agency. The present research on reentry dynamics and guidance is inspired by this multifaceted global scenario.

The objectives of thesis are to develop understanding and simulation of reentry dynamics and the classic guidance laws constant drag deceleration and constant descent rate for landing. Identify the limitations of the classic guidance laws. Overcome by treating reentry as a two-point boundary value problem with initial conditions at the entry and return trajectory interface, and the final conditions as the desired landing conditions under the constraint of constant angle of attack and limits on the footprint size and touchdown velocity. This work also focuses on verifying the superiority of treating reentry problem as two-point boundary value optimisation problem with the help of two examples.

CHAPTER 1

Introduction

1.1. Motivation

Space industry is evolving day by day to implement new and innovative technologies. Indian space research organisation (ISRO) is working on various space problems. It has been setting new records every day. It includes launching of Carosat-2 along with 103 satellites in one go using launch vehicle PSLV-C37, test of hypersonic flight experiment (HEX) of reusable launch vehicle technology demonstrator (RLV-TD) and many more. Russia's Soyuz reentry missions and return of Dragon cargo craft launched by Falcon 9 of SpaceX are remarkable achievements in space technology. Recently Soyuz MS-202 returned with three crew member to the earth from International Space Station successfully. The reentry vehicle with crew members is rescued from Kazakhstan.

Considering rate of launching of satellites into space, large amount of cost is associated with this. In order to reduce launch cost, reusable launch vehicle technology is evolved. Reentry is one of the intricate problems as vehicle re-enters with high speeds of the order 10km/s. The type of reentry, the way to dissipate high kinetic energy and the planning of reentry trajectory is to be done carefully.

The thesis focuses on study of reentry dynamics and development of technique for safe reentry and landing of unmanned Apollo command module returning from lunar mission. It is based on classical approach. Reference trajectory is designed based on constant drag deceleration law followed by constant descent rate law for low-lifting reentry. These guidance laws are designed to satisfy heating constraint and maximum 'g' load constraint on vehicle. In order to follow reference trajectory, closed-loop guidance is also implemented. It focuses on complexities involved in classical approach and hence need of treating reentry problem as two-point boundary value optimisation problem.

1.2. Literature survey

[2] describes conventional shuttle entry guidance technique. It is based on analytic solution of governing equations in terms of drag profile which meets flight constraints. Control law ensures damping of oscillatory type trajectory motion and steers shuttle orbiter according to drag profile.

[3] explains targeting philosophy implemented in Apollo 11 mission which is divided in two categories as corridor requirement and range requirement. It also focuses on post flight analysis of Apollo 10.

[4] gives information regarding geometry and simulation results of aerodynamics of the Apollo command module at high altitudes.

Reentry dynamics, associated aerodynamics, governing equations of motion and solution to these equations are given in [6]. It also gives gravity model, density model utilised while solving reentry problem.

[7] deals with two-body problem, types of orbits and determination of the orbital elements.

[11] describes perspective for on-board, real-time, optimal trajectory generation for reusable launch vehicle using pseudo-spectral algorithm. It is shown that PS guidance is capable of accommodating greater uncertainties and disturbances. This is a single guidance algorithm which is implemented for entry to landing site with provision of safety.

[12] gives the study of reentry dynamics, types of reentry, development of governing equations and concept of entry corridor. It addresses frame transformation from perifocal frame to inertial frame to frame in which governing equations are derived in. Generation of reference trajectory for low-lifting entry is based on guidance laws: constant drag deceleration law followed by constant descent rate law and implementation of closed-loop guidance to closely follow reference trajectory is explained in this.

Rotational dynamics involved in reentry phenomenon is explained in [14]. The rotational governing equations of motion and simplified form to control vehicle are explained. Equations to determine rotational aerodynamic coefficients involved in rotational dynamics of unmanned Apollo command module are given.

1.3. Report layout

Chapter 2 describes basics of reentry dynamics, development of the governing equations of motion for unmanned Apollo command module. It presents design of the entry corridor for unmanned Apollo command module and manned Apollo module based on heating constraints and maximum 'g' load constraints. Unmanned module and manned module have the same geometry but different ballistic coefficients, different allowable 'g' load and also different maximum heat limits which the vehicle can withstand because of presence of crew in the manned module.

Chapter 3 presents study of ballistic reentry of unmanned module, frame transformation of state variables to the frame in which equations are defined, study of orbital mechanics. It includes simulation results of dynamics of state variables for ballistic reentry.

Chapter 4 explains lifting reentry of unmanned command module. It gives classical approach of generation of reference trajectory based on guidance laws: constant drag deceleration law followed by constant descent rate law and implementation of closed-loop guidance to follow reference trajectory. It deals with complexities involved in it and need of treating reentry problem as two-point boundary value optimisation problem.

In Chapter 5 reentry dynamics, governing equations of motion, effect of varying angle of attack and effect of varying bank angle for spaceplane X-33 is presented. Reentry vehicle X-33 is a winged unmanned rocket plane developed in 1990's by NASA which was planned to be the next generation, commercially operated reusable launch vehicle. It includes optimization of angle of attack to land X-33 at a particular location using steepest descent method and optimization of angle of attack to maximise the range using "*fmincon*" function on MATLAB.

Chapter 6 summarizes the work and discusses the possible avenues for future work.

CHAPTER 2

Reentry Dynamics

2.1. Atmospheric reentry

Reentry is return of space vehicle manned or unmanned, after accomplishing its space missions. Reentry is intricate because velocity of returning spacecraft's is greater than 10 km/s and it must pass through Earth's atmosphere. Some of the reentry missions are:

- a. NASA's Space Shuttle Program, Apollo series
- b. ISRO's Space capsule Recovery Experiment (SRE), Reusable Launch Vehicle- Technology Demonstrator (RLV-TD)
- c. Russia's Soyuz

2.2. Reentry Dynamics

Reentry dynamics is governed by non-linear differential equations of motion in suitable frame of reference. These equations of motion are same as that of for launch phase. The only difference is in the direction of velocity vector, corresponding angle of attack and flight path angle. Angle of attack is angle made by the velocity vector with x-axis of body (reentry vehicle). Flight path angle is angle made by velocity vector with local horizontal.

According to Newton's second law of motion basic vector equation in inertial frame of reference is given as,

$$m \frac{dv}{dt} = F + W + A \dots \dots \dots (1)$$

F : Effective thrust force vector

W : Gravitational force vector

A : Aerodynamic force vector

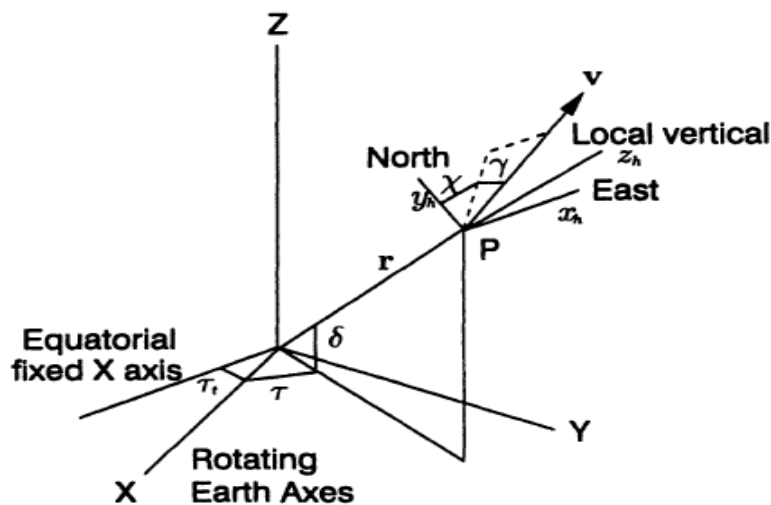


Fig.1: Body centered frame

Frame XYZ is rotating with angular velocity ω_e w.r.t. the center of the earth and vehicle centered ENU frame is rotating with angular velocity ω_h w.r.t. ECEF frame (XYZ) as shown in above figure. Velocity vector of vehicle (V) in inertial frame is given as:

$$V = v + \omega_e \times r \dots \dots \dots (2)$$

v: Velocity relative rotating earth (frame XYZ) in the absence of winds

Acceleration of spacecraft in ECEF frame is given as,

$$\frac{dV}{dt} = \frac{dv}{dt} + \omega_e \times v + \omega_e \times (v + \omega_e \times r) \dots \dots \dots \dot{\omega}_e = 0$$

$$\frac{dV}{dt} = \frac{dv}{dt} + 2\omega_e \times v + \omega_e \times (\omega_e \times r) \dots \dots (3)$$

Governing equations are derived in ENU frame which is rotating frame w.r.t. ECEF (XYZ) frame.

Rate of change of velocity relative to ENU frame is given as:

$$\frac{dv}{dt} = \dot{v} + \omega_h \times v$$

From eqn. (3)

$$\frac{dV}{dt} = \dot{v} + \omega_h \times v + 2\omega_e \times v + \omega_e \times (\omega_e \times r) \dots (4)$$

$$= (F + W + A)/m \dots \dots \dots (5)$$

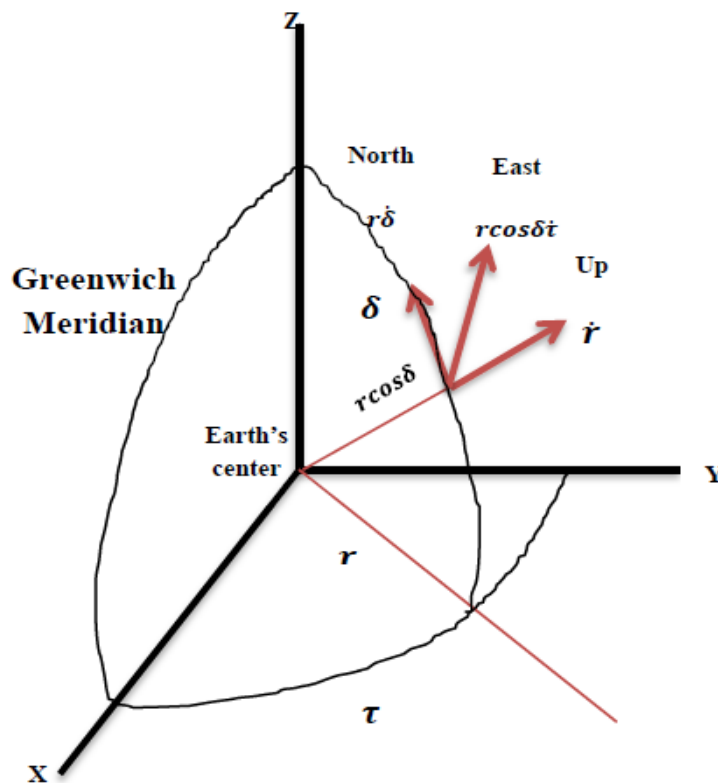


Fig.2: Velocity components in ENU frame

From fig.1 and fig.2, velocity vector components in ENU frame are given as:

$$\mathbf{v} = \dot{\mathbf{r}} = \begin{bmatrix} r\dot{\delta}\cos\gamma \\ r\dot{\delta}\sin\gamma \\ \dot{r} \end{bmatrix} = v \begin{bmatrix} \cos\gamma\sin\chi \\ \cos\gamma\cos\chi \\ \sin\gamma \end{bmatrix} \dots\dots\dots (6)$$

On comparison,

$$\dot{\delta} = \frac{v\cos\gamma\sin\chi}{r\cos\delta} \dots\dots\dots (7)$$

$$\dot{\gamma} = \frac{v\cos\gamma\cos\chi}{r} \dots\dots\dots (8)$$

$$\dot{r} = v\sin\gamma \dots\dots\dots (9)$$

R.H.S. of eqn. (4) is calculated with,

$$\mathbf{r} = (0, 0, r) \dots\dots\dots (10)$$

$$\text{Angular velocity of the earth in ENU frame: } \omega_e = \omega_e(0, \cos\delta, \sin\delta) \dots\dots\dots (11)$$

$$\text{Angular speed of ENU frame: } \omega_h = (-\dot{\delta}, \dot{\delta}\cos\delta, \dot{\delta}\sin\delta)$$

$$\omega_h = v/r(-\cos\gamma\cos\chi, \cos\gamma\sin\chi, \cos\gamma\sin\chi\tan\delta) \dots\dots\dots (12) \text{ From eqns. (7-8)}$$

$$\begin{aligned} \omega_e \times \mathbf{v} &= \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & \omega_e\cos\delta & \omega_e\sin\delta \\ v\cos\gamma\sin\chi & v\cos\gamma\cos\chi & v\sin\gamma \end{bmatrix} \dots\dots\dots \text{From eqn. (6) \& eqn. (11)} \\ &= \omega_e v \begin{bmatrix} \cos\delta\sin\gamma - \sin\delta\cos\gamma\cos\chi \\ \sin\delta\cos\gamma\sin\chi \\ -\cos\delta\cos\gamma\sin\chi \end{bmatrix} = \omega_e v \mathbf{q} \dots\dots\dots (13) \end{aligned}$$

$$\mathbf{q} = \begin{bmatrix} \cos\delta\sin\gamma - \sin\delta\cos\gamma\cos\chi \\ \sin\delta\cos\gamma\sin\chi \\ -\cos\delta\cos\gamma\sin\chi \end{bmatrix}$$

$$\omega_e \times \mathbf{r} = \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & \omega_e\cos\delta & \omega_e\sin\delta \\ 0 & 0 & r \end{bmatrix} = \omega_e r \begin{bmatrix} \cos\delta \\ 0 \\ 0 \end{bmatrix} \dots\dots\dots (14) \text{ From eqns. (10-11)}$$

$$\begin{aligned} \omega_e \times (\omega_e \times \mathbf{r}) &= \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & \omega_e\cos\delta & \omega_e\sin\delta \\ \omega_e r\cos\delta & 0 & 0 \end{bmatrix} \dots\dots\dots \text{From eqn. (14)} \\ &= \omega_e^2 r \begin{bmatrix} 0 \\ \sin\delta\cos\delta \\ -\cos^2\delta \end{bmatrix} \end{aligned}$$

$$\omega_e \times (\omega_e \times \mathbf{r}) = \omega_e^2 \mathbf{p} \dots\dots\dots (15)$$

$$\mathbf{p} = \begin{bmatrix} 0 \\ \sin\delta\cos\delta \\ -\cos^2\delta \end{bmatrix}$$

In order to determine $\dot{v}$ we differentiate eqn. (6),

$$\dot{v} = \begin{bmatrix} \dot{v}\cos\gamma\sin\chi + v\cos\gamma\cos\chi\dot{\chi} - v\sin\gamma\dot{\gamma}\sin\chi \\ \dot{v}\cos\gamma\cos\chi - v\cos\gamma\sin\chi\dot{\chi} - v\sin\gamma\dot{\gamma}\cos\chi \\ \dot{v}\sin\gamma + v\cos\gamma\dot{\gamma} \end{bmatrix} = \begin{bmatrix} \cos\gamma\sin\chi & \cos\chi & -\sin\gamma\sin\chi \\ \cos\gamma\cos\chi & -\sin\chi & -\sin\gamma\cos\chi \\ \sin\gamma & 0 & \cos\gamma \end{bmatrix} \begin{bmatrix} \dot{v} \\ v\dot{\chi}\cos\gamma \\ v\dot{\gamma} \end{bmatrix}$$

$$\dot{v} = \varphi \begin{bmatrix} \dot{v} \\ v\dot{\chi}\cos\gamma \\ v\dot{\gamma} \end{bmatrix} \dots\dots\dots (16)$$

$$\varphi = \begin{bmatrix} \cos\gamma\sin\chi & \cos\chi & -\sin\gamma\sin\chi \\ \cos\gamma\cos\chi & -\sin\chi & -\sin\gamma\cos\chi \\ \sin\gamma & 0 & \cos\gamma \end{bmatrix}$$

φ is an orthogonal matrix.

$$\begin{aligned} \omega_h \times v &= \begin{bmatrix} \ddot{r} & \ddot{\gamma} & \ddot{\chi} \\ -v/r\cos\gamma\cos\chi & v/r\cos\gamma\sin\chi & v/r\cos\gamma\sin\chi\tan\delta \\ v\cos\gamma\sin\chi & v\cos\gamma\cos\chi & v\sin\gamma \end{bmatrix} \\ &= (v^2/r)\cos\gamma \begin{bmatrix} \sin\chi\sin\gamma - \cos\chi\cos\gamma\sin\chi\tan\delta \\ \cos\chi\sin\gamma + \sin^2\chi\tan\delta\cos\gamma \\ \cos\gamma \end{bmatrix} \\ &= \left(\frac{v^2}{r}\right)\cos\gamma s \end{aligned}$$

$$s = \begin{bmatrix} \sin\chi\sin\gamma - \cos\chi\cos\gamma\sin\chi\tan\delta \\ \cos\chi\sin\gamma + \sin^2\chi\tan\delta\cos\gamma \\ \cos\gamma \end{bmatrix}$$

Eqn. (4) becomes,

$$\frac{dv}{dt} = \varphi \begin{bmatrix} \dot{v} \\ v\dot{\chi}\cos\gamma \\ v\dot{\gamma} \end{bmatrix} + \left(\frac{v^2}{r}\right)\cos\gamma s + 2\omega_e vq + \omega_e^2 rp \dots\dots\dots (17)$$

From eqn. (5) and eqn. (17),

$$\begin{bmatrix} \dot{v} \\ v\dot{\chi}\cos\gamma \\ v\dot{\gamma} \end{bmatrix} = -\varphi^T \left[\left(\frac{v^2}{r}\right)\cos\gamma s + 2\omega_e vq + \omega_e^2 rp \right] + \varphi^T (F + W + A)/m \dots\dots\dots (18)$$

Thrust force vector for reentry: $F = 0$

φ^T : Transforms from ENU frame to wind axes (w frame).

It gives,

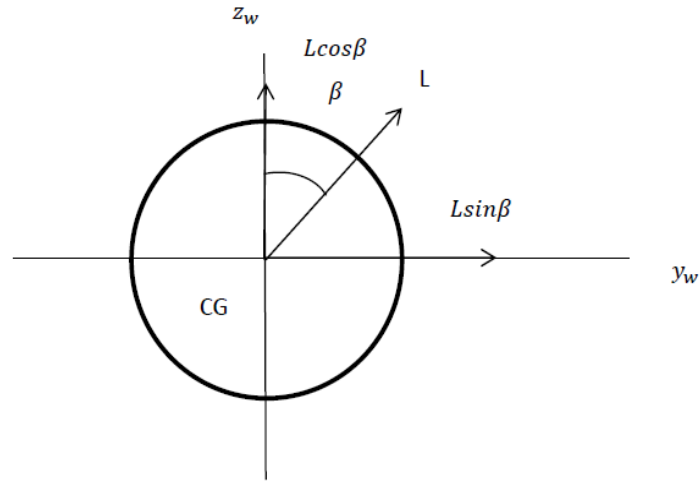
$$\dot{v} = w_e^2 r \cos\delta (-\cos\gamma \cos\chi \sin\delta + \sin\gamma \cos\delta) + (W_1 + A_1)/m \dots\dots\dots (19)$$

$$\begin{aligned} \dot{\chi} &= 2w_e (-\tan\gamma \cos\chi \cos\delta + \sin\delta) + \frac{w_e^2 r}{v\cos(\gamma)} \sin\chi \cos\delta \sin\delta + \frac{v}{r} \cos\gamma \sin\chi \tan\delta + (W_2 + A_2)/(m\cos\gamma) \dots\dots\dots (20) \end{aligned}$$

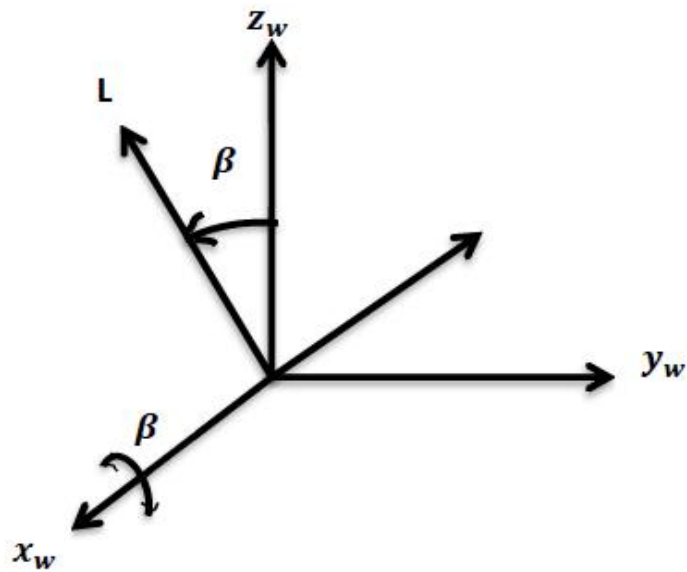
$$\dot{\gamma} = 2w_e \sin \chi \cos \delta + \frac{w_e^2 r}{v} \cos \delta (\sin \gamma \cos \chi \sin \delta + \cos \gamma \cos \delta) + \frac{v}{r} \cos \gamma + (W_3 + A_3)/(mv) \dots (21)$$

$$\dot{\delta} = v \cos \gamma \dots (22)$$

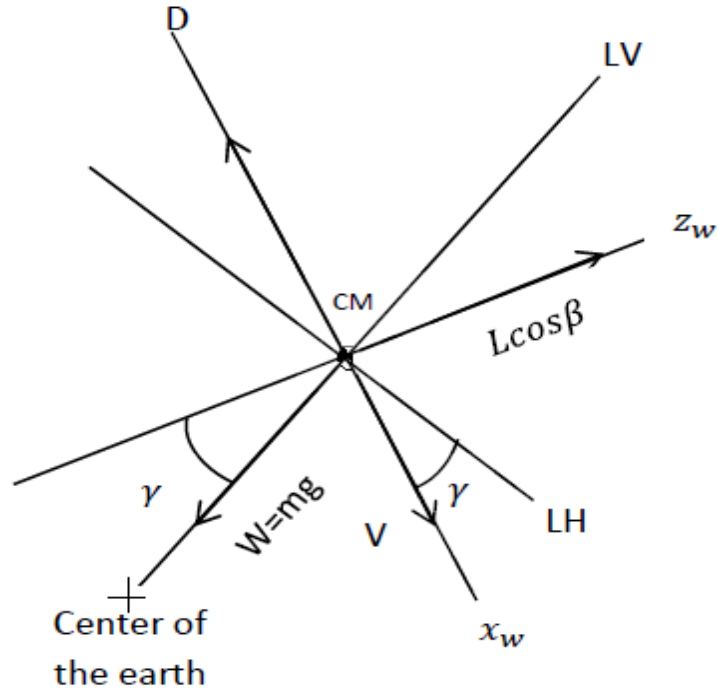
The force components in wind axes are given as:



(a)[3]



(b)



(c)

Fig.3: Aerodynamic force component in the trajectory plane and across the trajectory plane

x_w, y_w, z_w	Wind frame axes
LV	Local vertical
LH	Local horizontal
CM	Center of mass of vehicle

Table 1: Meaning of abbreviations

Gravitational force: $W = \begin{bmatrix} -mg \sin \gamma \\ 0 \\ -mg \cos \gamma \end{bmatrix}$

From figure 3 (a) for positive clockwise roll angle [12],

Aerodynamic force: $A = \begin{bmatrix} -D \\ L \sin \beta \\ L \cos \beta \end{bmatrix}$

If we consider anticlockwise roll angle as positive then from figure 3 (b),

Aerodynamic force: $A = \begin{bmatrix} -D \\ -L \sin \beta \\ L \cos \beta \end{bmatrix}$

We consider aerodynamic force calculated from figure 3 (a).

Eqns. (7-9) and (19-22) are governing equations for reentry dynamics in wind axes.

2.3. Reentry of Apollo command module

NASA's Apollo mission, specifically Apollo 11-17 was designed to land humans on moon and bring them back. Apollo-11 was first manned mission for landing moon. We consider Apollo command module as reentry vehicle which is returning from lunar missions. The shape of Apollo command module is conical shape with blunt nose as shown in figure below. Unmanned capsule and manned Apollo capsule has the same shape as shown in fig. 4.

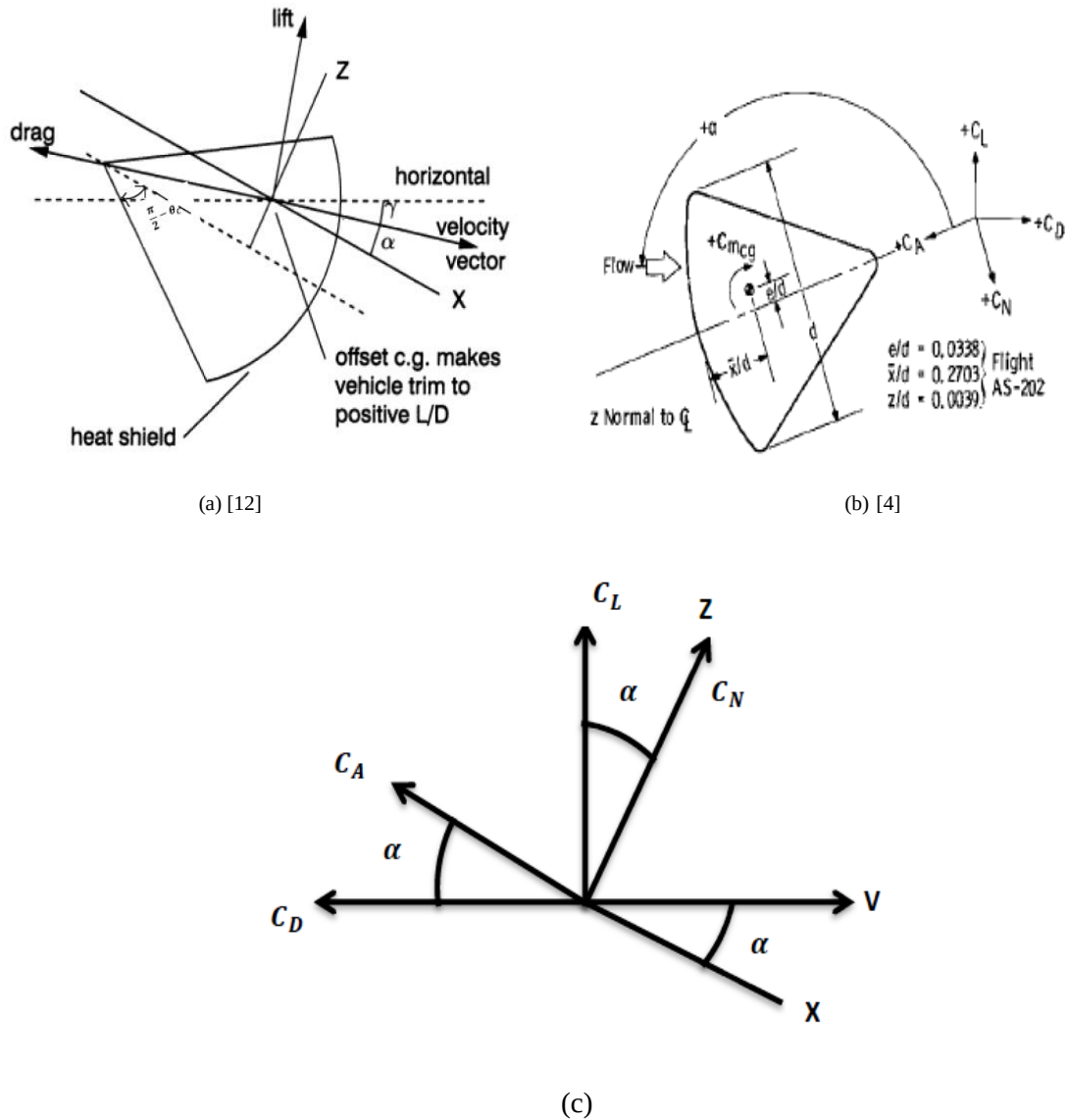


Fig.4: Apollo command module

Centre of gravity is offset so that vehicle is able to generate lift force as seen in fig. 3(b). Lift vector (L) is perpendicular to velocity vector as seen in above figure and drag vector (D) is acting in the direction opposite to velocity vector (v). The vehicle is entering with angle of attack (α) made by velocity vector with longitudinal x axis of body. It is entering with flight path angle (γ) made by velocity vector with local horizontal axis. In general lift vector can be rolled about velocity vector with roll angle (β).

Regan's formulae [6] for coefficient of normal force and coefficient of axial force are derived for conical shape. Apollo command module has conical shape with blunt nose,

After reworking on Regan's formulae,

$$\text{Coefficient of normal force: } C_N = 0.5\cos^2\theta_c\sin 2\alpha \dots\dots\dots (23)$$

$$\text{Coefficient of axial force: } C_A = \cos^2\alpha(1 + \sin^2\theta_c) + 0.5\cos^2\theta_c\sin^2\alpha \dots\dots (24)$$

Blunt nose angle for the heat shield of Apollo command module $\theta_c = 60^\circ$

Resolving total force vector into components of figure 4 (c) for $\alpha < 0$,

$$\text{Coefficient of lift force: } C_L = \sin\alpha C_A + \cos\alpha C_N$$

$$\text{Coefficient of drag force: } C_D = \cos\alpha C_A - \sin\alpha C_N$$

We consider $\alpha > 0$ and coefficient of lift and coefficient of drag are given as [12],

$$\text{Coefficient of lift force: } C_L = -\sin\alpha C_A + \cos\alpha C_N\dots\dots\dots (25)$$

$$\text{Coefficient of drag force: } C_D = \cos\alpha C_A + \sin\alpha C_N\dots\dots\dots (26)$$

Lift force is computed as,

$$L = \frac{1}{2}\rho(h)v^2C_LS\dots\dots\dots (27)$$

$$S = \frac{\pi}{4}\text{dia}^2$$

Drag force is computed as,

$$D = \frac{1}{2}\rho(h)v^2C_DS\dots\dots\dots (28)$$

2.3.1. Calculation of angle of attack for trim condition

The static stability of Apollo command module is ensured by keeping pitch moment as zero about center of mass.

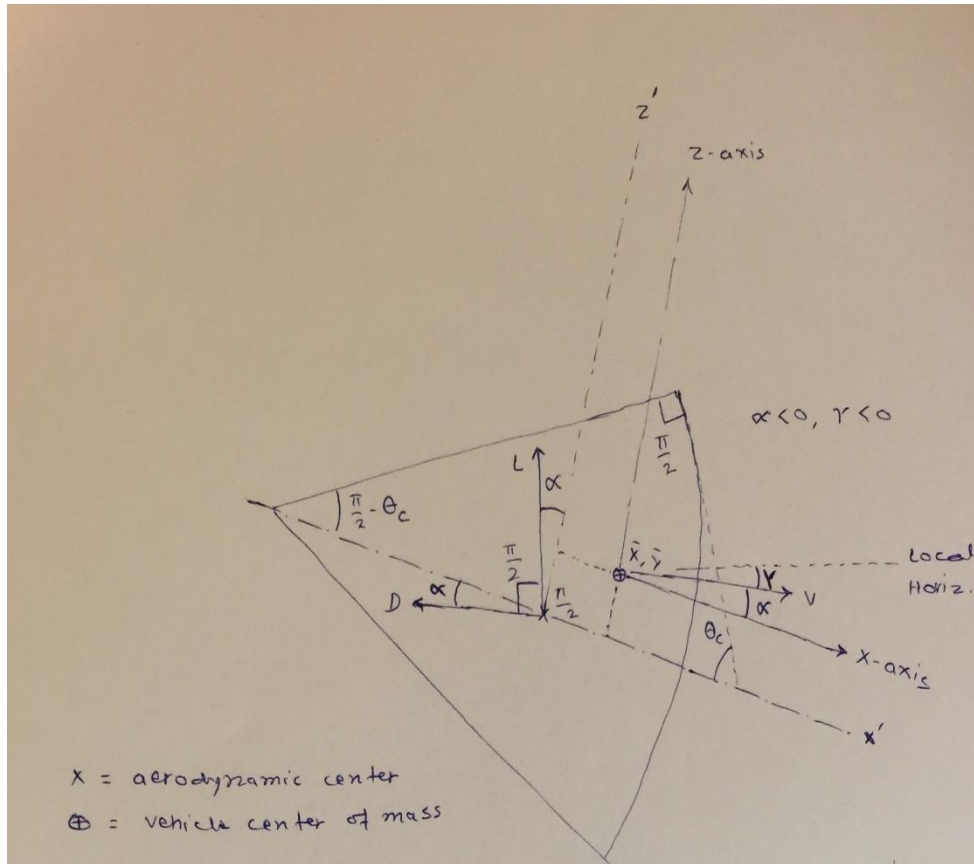


Fig.5: Trim condition

Above figure shows aerodynamic center where the aerodynamic forces act. X', Z' are cone's longitudinal and transverse axes and XZ are cone's axes passing through cone's center of mass. The aerodynamic forces along X' and Z' axes are given as:

Along X' axes: $-D \cos \alpha - L \sin \alpha$

Along Z' axes: $-D \sin \alpha + L \cos \alpha$

The pitch moment (M_y) is positive in anticlockwise direction and is given as:

$$(D \cos \alpha + L \sin \alpha) \bar{y} + (-D \sin \alpha + L \cos \alpha) \bar{x} = M_y$$

For trim condition $M_y = 0$

For negative angle of attack in clockwise direction,

$$(D \cos \alpha - L \sin \alpha) \bar{y} + (D \sin \alpha + L \cos \alpha) \bar{x} = 0$$

$$\bar{y} = \bar{x} \frac{D \sin \alpha + L \cos \alpha}{L \sin \alpha - D \cos \alpha}$$

$$\bar{y} = \bar{x} \frac{\tan \alpha + (L/D)}{(L/D) \tan \alpha - 1} \dots \dots \dots (29)$$

From these equations, trim angle of attack required for desired L/D can be determined with $(\bar{x}, \bar{y})$ is relative position of center of mass with respect to aerodynamic center.

From figure 4 (b),

$$\bar{x} = 3.6213 \times 0.5 - 0.2703 \times 3.9116 = 0.7533 \text{ m}$$

$$\bar{y} = 0.0338 \times 3.91160 = 0.1322 \text{ m}$$

2.4. Entry Corridor

Entry corridor is an envelope for which reentry constraints are satisfied. During reentry there are two constraints on vehicles which are to be satisfied. One is the limit of rate of heating and another is the rate of total acceleration. Heating is considered to design heat shield for the reentry vehicle so that it can enter safely. Total acceleration refers to maximum ‘g’ load acting on vehicle. In order to design entry corridor for small unmanned capsule and manned Apollo reentry module following data are utilised [12].

No.	Parameter	Small unmanned capsule	Manned Apollo reentry module
1.	Ballistic coefficient (kg/m ²) ($B_c = \frac{m}{C_{D \text{ at } \alpha=0} S_{ref}}$)	72.8	280
2.	Maximum total acceleration (g) (a_{max})	20	4
3.	Maximum heating (W/cm ²) (Q_{max})	500	60
4.	Nose Radius (m) (R_n , dia)	2	3.9
5.	L/D	0.3	0.3

Table 2: Apollo parameters

2.4.1. Equations involved in entry corridor design

- A. Maximum total acceleration: The acceleration of vehicle is due to aerodynamic forces. It is given as:

$$\begin{aligned}
 a_{max} &= \sqrt{L^2 + D^2} / m \\
 &= \sqrt{\left(\frac{1}{2} \rho(h) v^2 S_{ref}\right)^2 (C_L^2 + C_D^2)} / m \dots\dots\dots \text{From eqns. (27-28)} \\
 &= \frac{1}{2} \rho(h) v^2 S_{ref} C_D / m \sqrt{1 + \left(\frac{C_L}{C_D}\right)^2} \\
 a_{max} &= \frac{1}{2} \rho(h) v^2 S_{ref} C_D / m \sqrt{1 + \left(\frac{C_L}{C_D}\right)^2} \\
 v_{max} &= \left[\frac{2 B_c a_{max}}{\rho(h) \sqrt{1 + \left(\frac{C_L}{C_D}\right)^2}} \right]^{1/2} \dots\dots\dots (30)
 \end{aligned}$$

- B. Maximum heating: The total energy dissipated is generated due to multiplication of drag force and velocity of vehicle. Most of this energy is dissipated into atmosphere through which it is entering. Radiant heating is dominant at higher vehicle speed over convective heating. Approximate formula for heat flux (W/m²) due to convective heating at hottest part is given as [12] :

$$Q_{max} = 1.45 \times 10^{-4} \sqrt{\rho(h) / R_n} v^3$$

$$v_{\max} = \left[\frac{Q_{\max}}{0.000145} \sqrt{\frac{R_n}{\rho(h)}} \right]^{1/3} \dots\dots\dots (31)$$

C. Level flight: Vertical acceleration of vehicle by neglecting the earth's rotational effect and no roll condition ($\beta = 0 \text{ deg}$) is given by eqn. (21) as:

$$\begin{aligned} \dot{\gamma} &= \frac{v}{r} \cos \gamma + (-mg \cos \gamma + L)/(mv) \\ v\dot{\gamma} &= \frac{v^2}{r} \cos \gamma - g \cos \gamma + \frac{L}{m} \\ &= \frac{v^2}{r} \cos \gamma - \frac{\mu}{r^2} \cos \gamma + \frac{\frac{1}{2}\rho(h)v^2 C_L S_{\text{ref}}}{m} \dots\dots\dots \text{From Eqn. (27)} \\ &= \frac{v^2}{r} \cos \gamma - \frac{\mu}{r^2} \cos \gamma + \frac{1}{2}\rho(h)v^2 \frac{C_D S_{\text{ref}}}{m} \frac{C_L}{C_D} \\ v\dot{\gamma} &= \frac{v^2}{r} \cos \gamma - \frac{\mu}{r^2} \cos \gamma + \frac{1}{2}\rho(h)v^2 \frac{1}{B_C} \frac{C_L}{C_D} \dots\dots\dots (32) \end{aligned}$$

Ballistic coefficient: $B_C = \frac{m}{C_D S_{\text{ref}}}$ at $\alpha = 0^\circ$

For level flight $\dot{\gamma} = 0$ and flight path angle is zero. Eqn. (32) becomes,

$$\begin{aligned} 0 &= \frac{v^2}{r} - \frac{\mu}{r^2} + \frac{1}{2}\rho(h)v^2 B_C \frac{C_L}{C_D} \\ v^2 \left(\frac{1}{r} + \frac{1}{2}\rho(h) B_C \frac{C_L}{C_D} \right) - \frac{\mu}{r^2} &= 0 \\ v_{\text{lim}} &= v_c \left[1 + \frac{\frac{C_L}{C_D} \rho(h)}{2B_C} \right]^{-\frac{1}{2}} \dots\dots\dots (33) \end{aligned}$$

Speed of vehicle in circular orbit around the earth $v_c = \sqrt{\frac{\mu}{r}}$

2.4.2. Simulation results

The following plots are limiting curves for maximum acceleration and maximum heat generated on vehicle for small unmanned capsule and manned reentry module. These curves are generated based on equations 30, 31, 33 using parameters from table 2.

Region between three curves of maximum acceleration and level flight is the permissible region to satisfy constraints. The results are qualitatively same but as per plots below, the permissible region is small in case of manned reentry module than unmanned capsule, so trajectory design and entry condition are more critical.

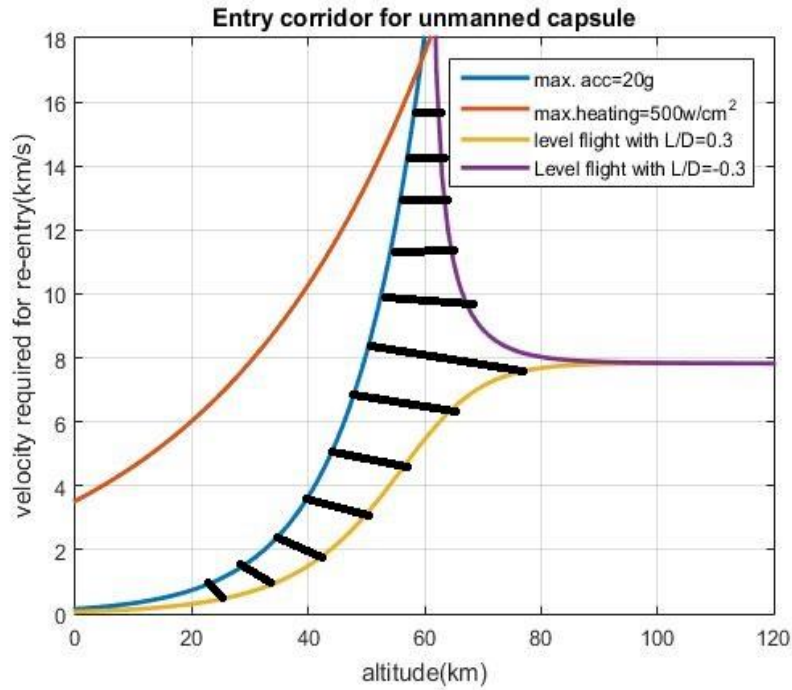


Fig.6: Entry Corridor for unmanned capsule

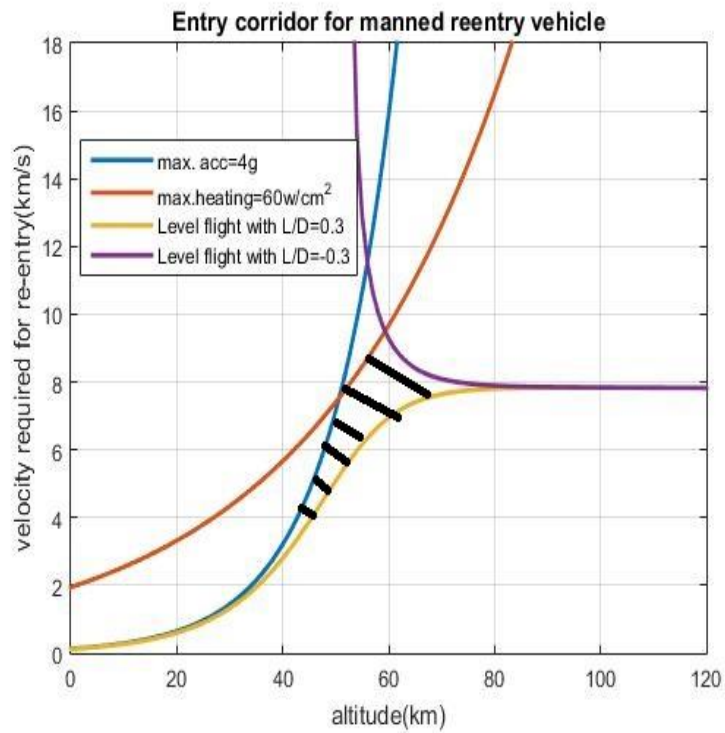


Fig.7: Entry Corridor for manned reentry module

For $L/D=-0.3$, the lift vector is downward and if the velocity is greater than maximum velocity then vehicle will descend. For $L/D=0.3$, the lift vector is upward and if the velocity is greater than maximum velocity then vehicle will ascend as shown in figure 5 and 6.

CHAPTER 3

Unmanned Apollo Command Module: Ballistic Reentry

Apollo command module re-enters into earth's atmosphere from interplanetary mission which carries high kinetic energy. As it passes through earth's atmosphere, it experiences large amount of drag force. Due to this drag force large amount of heat is generated and also velocity of vehicle reduces. With such a high orbital energy, trajectory followed by vehicle is hyperbolic path so that the energy decreases enough and vehicle can be captured in the planet's atmosphere. So reentry trajectory analysis is one of the complicated problems in terms of dynamics and thermal aspect.

Systematic classification of reentry problem is required on the basis of purpose of reentry as ballistic, semi-ballistic, skip, lifting. We are considering ballistic and lifting reentry for Apollo command module.

3.1. Ballistic reentry of unmanned Apollo command module

Most of the unmanned vehicles return to the earth following ballistic reentry. Apollo unmanned module also follows ballistic reentry. During this reentry, vehicle generates zero lift force. It is dominated by drag force and gravity force. Drag force reduces its speed and with the help of parachute deployment it touches down ground. Landing point is determined depending upon conditions during reentry. There is no control over vehicle as it starts ballistic reentry. Following figure shows how vehicle approaches and enters earth's atmosphere.

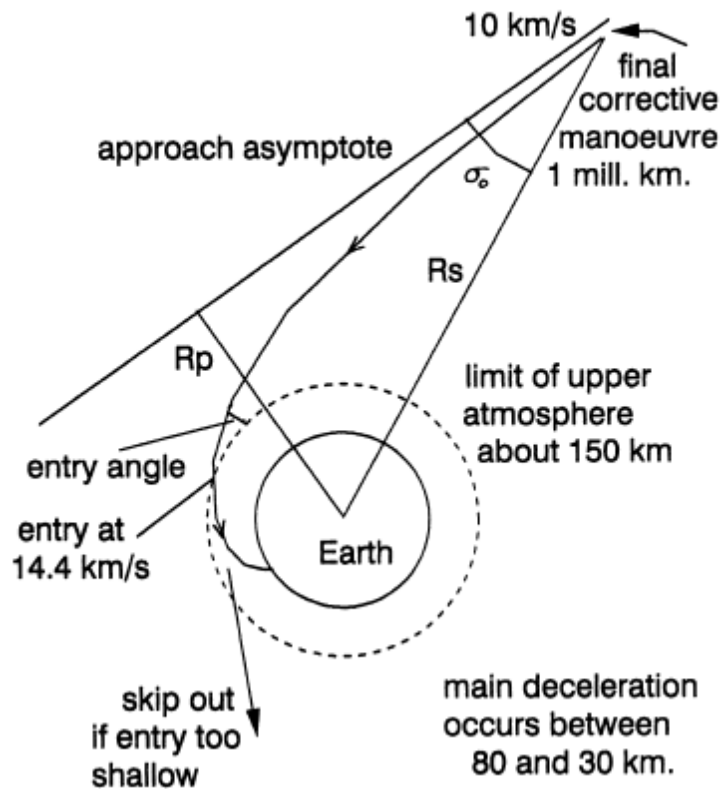


Fig.1: Approach and entry into atmosphere [12]

During ballistic reentry angle of attack (α) is equal to 0° , so for ballistic reentry aerodynamic coefficients are computed as :

Eqns. (23-26) from section 2.3 with $\alpha = 0^\circ$,

Blunt nose angle for the heat shield of Apollo command module $\theta_c = 60^\circ$

Coefficient of normal force: $C_N = 0$ (1)

Coefficient of axial force: $C_A = 1.75$ (2)

Coefficient of lift force: $C_L = 0$ (3)

Coefficient of drag force: $C_D = 1.75$ (4)

Lift to drag ratio $= \frac{L}{D} = \frac{C_L}{C_D} = 0$ (5)

3.2. Initial values

Initial Values before entering upper atmosphere:

$R_s = 1 \text{ million km}$

$R_p = 9621 \text{ km}$

$V_s = 10 \text{ km/s}$

R_p : Perpendicular distance by which vehicle would miss if it does nothing and no forces act on it

R_s : Radial distance

3.3. Frame Transformation

Governing equations (eqns. (7-9) and eqns. (19-22)) of motion are explained in section 2.2 which are computed in wind axes frame. Initial radius vector and velocity vector are expressed in perifocal frame. In order to simulate governing equations these velocity and radius vector are transformed into the wind axes frame as explained below.

The frames which are involved in frame transformations are:

- A. Radius- transverse –normal (RTN) spacecraft centered frame (i_r, i_θ, i_z)

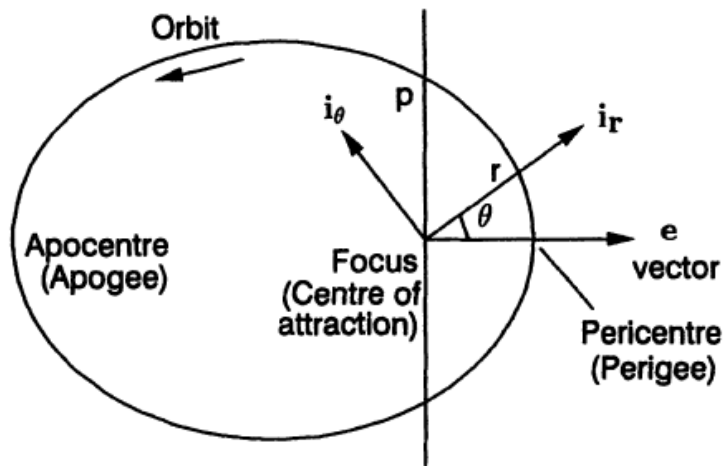


Fig.2: Conic Orbit [12]

i_r : along radius vector (r) and directed away from center of attraction

i_θ : in the plane of motion and perpendicular to radius vector in the direction of increasing θ

i_z : normal to plane of motion along angular momentum direction

- B. Nodal frame spacecraft orbit centered (i_x, i_y, i_z)

i_x : defines the ascending node

i_z : in the direction of the orbit angular momentum

i_y : in the equatorial plane and completes the triad

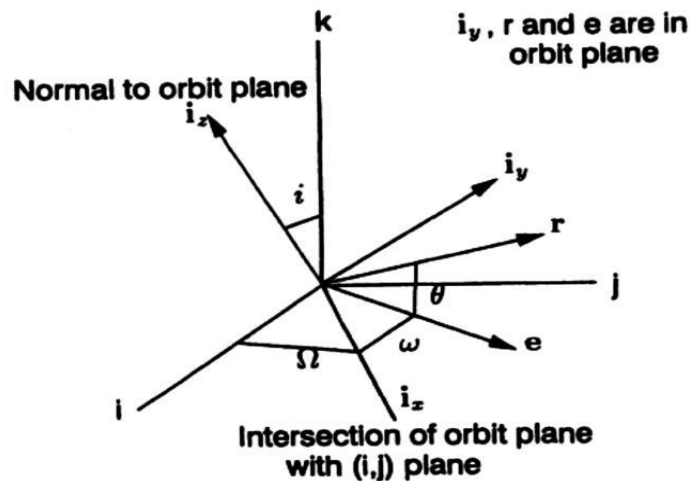


Fig.3: Nodal and ECI frame [12]

- C. Earth centered earth fixed (ECEF) frame (X,Y,Z) (fig.1)
 - X: passes through intersection of equatorial plane and Greenwich meridian
 - Y: in the direction to complete the triad
 - Z: along axis of rotation of earth i.e. perpendicular to equatorial plane
- D. Earth centered inertial (ECI) frame (i,j,k)
 - i: points toward vernal equinox lies in equatorial plane
 - j: in the direction to complete triad lies in equatorial plane
 - k: along axis of rotation of the earth i.e. perpendicular to equatorial plane
- E. East-North-Up (ENU) frame (x_h, y_h, z_h)
 - East: points to east (x_h)
 - North: points to true north along local horizontal (y_h)
 - Up: completes the right handed co-ordinate system by pointing up along local vertical (z_h)
- F. Wind axes frame (x_w, y_w, z_w)
 - x_w : along velocity vector v relative to rotating axes fixed in the earth's center
 - y_w : in the direction to complete triad
 - z_w : perpendicular to velocity vector in the plane of symmetry of vehicle

Frame transformation takes place from RTN frame to ECI frame to ECEF frame to ENU frame to wind axes frame.

3.3.1. Frame transformation from RTN frame to ECI frame (sec. 1.3.2 [12])

From figure 1, velocity and radius vector is given as:

$$\text{Velocity } V = \begin{bmatrix} -10\cos\sigma_0 \\ 10\sin\sigma_0 \\ 0 \end{bmatrix} \text{ km/s}, \text{ Radius } R = \begin{bmatrix} 1\text{million} \\ 0 \\ 0 \end{bmatrix} \text{ km}$$

where

$$\sigma_0 = \sin^{-1}\left(\frac{R_p}{R_s}\right)$$

In order to calculate radius vector and velocity vector in RTN frame, orbital elements are computed using Velocity V and radius R .

Using these radius vector and velocity vector, angular momentum vector is calculated as,

$$\mathbf{h} = (\mathbf{R} \times \mathbf{V})$$

With the help of magnitude of angular momentum orbital elements are calculated as follows [12]:

A. Eccentricity (e): It is deciding element of the shape of orbit which describes how much orbit is elongated as compared to circle.

$$e = \sqrt{1 - \frac{p}{a}}$$

$$\text{Semi-latus rectum } (p): p = \frac{h^2}{\mu}$$

B. Semi major axis (a): It is half of sum of distance between apogee and perigee. For hyperbolic orbit $a < 0$.

$$a = \frac{-\mu}{(v^2 - \frac{\mu}{2r})} \dots v = |V|, r = |R|$$

C. True anomaly (θ): It is angle measured between direction of perigee (axis e) and current position of vehicle as shown in fig.3.

$$\theta = \tan^{-1} \left(\frac{s \sqrt{\frac{\mu}{p}}}{p-r} \right)$$

$$s = \text{resin}(\theta) \sqrt{\frac{\mu}{p}}$$

D. Right ascension of ascending node (Ω): It is angle measured from i axis to intersection of orbit plane when its direction is upwards as shown in fig. 3.

$$\Omega = \tan^{-1} \left(\frac{i_z(1)}{-i_z(2)} \right)$$

$$i_z = \frac{(R \times V)}{|(R \times V)|}$$

E. Inclination (i): It measures how much orbital plane is inclined with ij plane i.e. equatorial plane as shown in fig. 3.

$$i = \cos^{-1}(i_z(3))$$

F. Argument of perigee (ω): It is angle measured in counter clockwise direction from reference axis i_x as shown in fig.3.

$$\tan(\theta + \omega) = \frac{i_z \cdot (i_x \times R)}{(i_x \cdot R)}$$

Components of i_x along i, j, k = $[\cos(\Omega) \quad \sin(\Omega) \quad 0]$

Orbital elements obtained from above calculations are:

p	$2.3262 \times 10^7 \text{ m}$
a	$-4.0120 \times 10^6 \text{ m}$
e	2.6073(hyperbolic)
θ	-157.99°
Ω	0°
i	90°
ω	67.99°

Table 1: Orbital elements

Radius and velocity vector in RTN frame (i_r, i_θ, i_z) are given as:

$$R = \left[\left(\frac{p}{1 + e \cos(\theta)} \right) \quad 0 \quad 0 \right]$$

$$V = \left[\sqrt{\frac{\mu}{p}} e \sin(\theta) \quad (1 + e \cos(\theta)) \sqrt{\frac{\mu}{p}} \quad 0 \right]$$

Γ_1 is transformation matrix which transforms RTN frame (i_r, i_θ, i_z) to nodal frame (i_x, i_y, i_z) and Γ_2 is transformation matrix which transforms nodal frame (i_x, i_y, i_z) frame to ECI frame (i, j, k) as follows.

$$R_{ECI} = \Gamma_2 \Gamma_1 R \dots \dots \dots (6)$$

$$V_{ECI} = \Gamma_2 \Gamma_1 V$$

$$\Gamma_1 = \begin{bmatrix} \cos(\theta + \omega) & -\sin(\theta + \omega) & 0 \\ \sin(\theta + \omega) & \cos(\theta + \omega) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Gamma_2 = \begin{bmatrix} \cos\Omega & -\sin i \cos\Omega & \sin i \sin\Omega \\ \sin\Omega & \cos i \cos\Omega & -\sin i \sin\Omega \\ 0 & \sin i & \cos i \end{bmatrix}$$

3.3.2. Frame transformation from ECI frame to wind axes frame

The transformation matrix from ECI frame to ECEF frame is,

$$C_{ECEF,ECI} = \begin{bmatrix} \cos(\tau_0 + \omega_e t) & -\sin(\tau_0 + \omega_e t) & 0 \\ \sin(\tau_0 + \omega_e t) & \cos(\tau_0 + \omega_e t) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

τ_0 : Right ascension at zero time

Radius vector transformed to ECEF frame is given as,

$$R_{ECI}^{ECEF} = C_{ECI,ECEF} \times R_{ECI}$$

Inertial velocity vector transformed to ECEF frame is given as,

$$V_{ECI}^{ECEF} = C_{ECI,ECEF} \times V_{ECI}$$

ECEF frame is rotating with angular velocity ω_e w.r.t inertial frame. Velocity vector in ECI frame with rotating ECEF frame is given as,

$$V_{ECI}^{ECEF} = V_{ECEF}^{ECEF} + \omega_e \times R_{ECI}^{ECEF} \dots \dots \text{(Section 2.4.1. [12])}$$

$$V_{ECEF}^{ECEF} = V_{ECI}^{ECEF} - \omega_e \times R_{ECI}^{ECEF}$$

Other state variables are calculated as,

$$R_{ECI}^{ECEF} = [R_x \quad R_y \quad R_z]$$

$$\text{Latitude: } \delta = \sin^{-1} \left(\frac{R_z}{|R|} \right) \dots \dots \dots (7)$$

$$\text{Longitude: } \tau = \tan^{-1} \left(\frac{R_y}{R_x} \right) \dots \dots \dots (8)$$

Frame transformation from ECEF frame to ENU frame is given as (eqn.2.77 [12]):

$$V_h = \begin{bmatrix} -\sin(\tau') & -\sin(\delta)\cos(\tau') & \cos(\delta)\cos(\tau') \\ \cos(\tau') & -\sin(\delta)\sin(\tau') & \cos(\delta)\sin(\tau') \\ 0 & \cos(\delta) & \sin(\delta) \end{bmatrix}^T V_{ECI}^{ECEF} \dots \dots \dots (9)$$

$$\tau' = \tau + \tau_0 + \omega_e t$$

$$\text{Flight path angle: } \gamma = \sin^{-1} \left(\frac{V_h(3)}{|V_h|} \right) \dots \dots \dots (10)$$

Azimuth angle: $\chi = \tan^{-1} \left(\frac{V_h(1)}{V_h(2)} \right) \dots \dots \dots (11)$

Eqn. (6-11) give initial values for state variables as,

$$X_0 = [\tau \quad \delta \quad r \quad v \quad \chi \quad \gamma]_{t=0}^T$$

$$X_0 = [0^\circ \quad 0^\circ \quad (150000 + 6378140)\text{m} \quad 1.4873 \times 10^4 \text{ m/s} \quad 88.9116^\circ \quad -8.99^\circ]^T$$

Numerical integration of governing equations of motion explained in section 2.2 (eqns. (7-9) and (19-22) using RK4 method is done on MATLAB. Following models are used for calculation:

Density model [12]:

$$\rho = \rho_0 e^{-(h/h_s)} \dots \dots \dots \text{valid below 100 km altitude}$$

ρ_0 : Density of atmosphere on the earth's surface

$$h_s = 6240 \text{ m [20]}$$

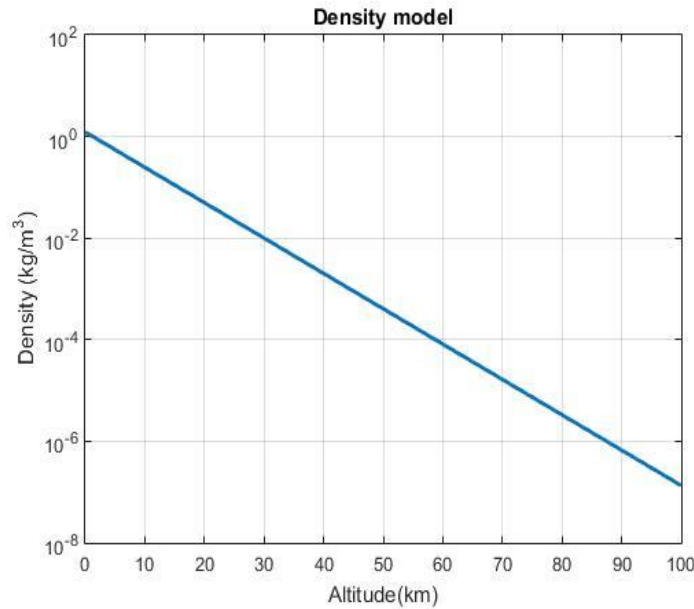


Fig. 4: Density Model

For altitude above 100 km altitude is given by Harris-Priester atmospheric density model as given in table [14]:

Altitude (km)	Minimum density $\left(\frac{\text{kg}}{\text{m}^3}\right)$	Maximum density $\left(\frac{\text{kg}}{\text{m}^3}\right)$
100	0.497×10^{-6}	0.497×10^{-6}
120	0.249×10^{-7}	0.249×10^{-7}
130	0.8377×10^{-8}	0.8710×10^{-8}
140	0.3899×10^{-8}	0.4059×10^{-8}
150	0.2122×10^{-8}	0.2215×10^{-8}

Table 2: Harris-Priester Density model

Gravity Model with latitude dependence and most predominant no spherical gravity term J_2 [12]:

$$g = \frac{\mu}{r^2} \left(1 + \frac{3J_2}{2} \left(\frac{R_0}{r} \right)^2 (1 - 3 \sin^2(\delta)) \right)$$

Zonal harmonic coefficient $J_2 = 0.0010826$ ([8]section 5.2)

Oblate Earth's radius model [12]:

$$R(\delta) = R(1 - f \sin^2(\delta))$$

$$f = 0.00335281 \text{ ([8]section 4.3)}$$

Lift and drag force are calculated as explained in section 2.3 on the basis of eqns. (1-4).

3.4. Simulation results for ballistic reentry

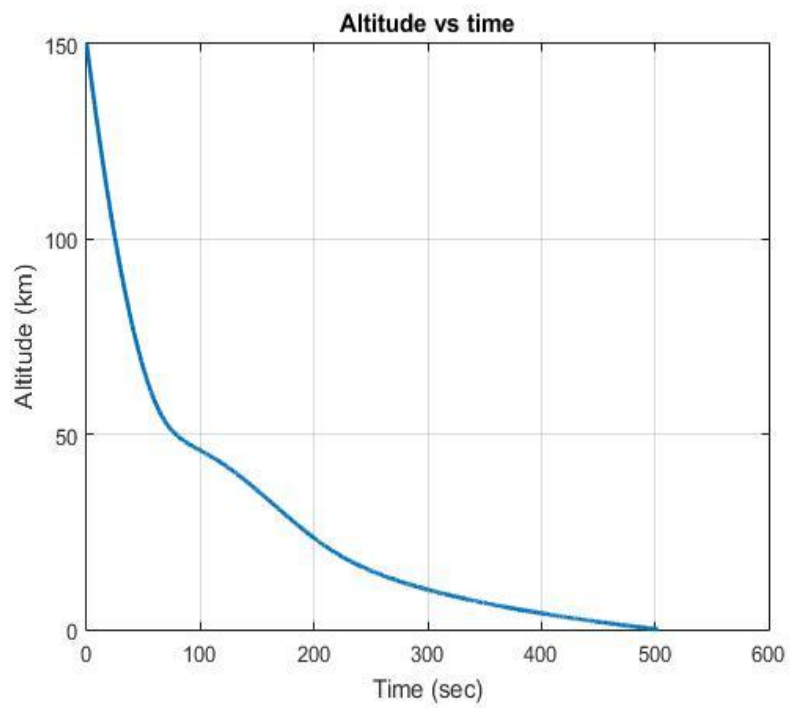
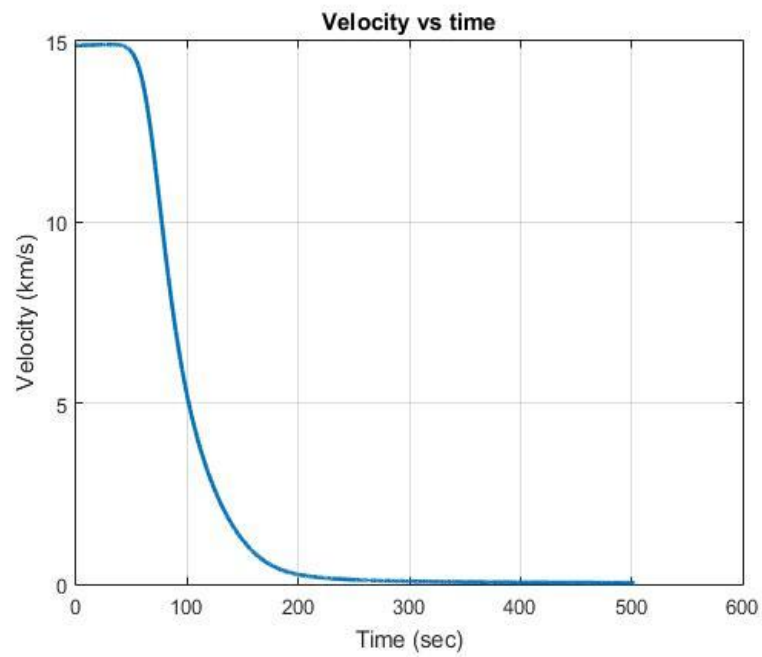
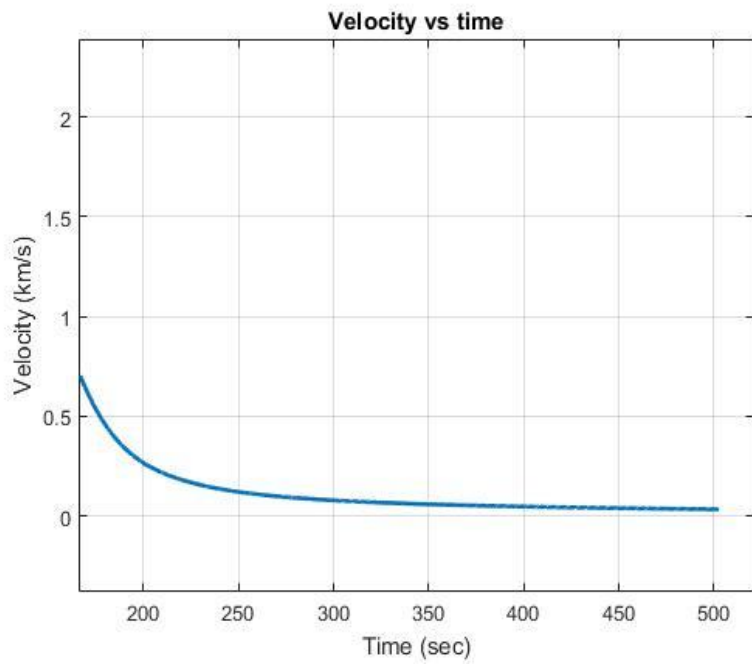


Fig.5: Altitude vs time



(a)



(b)

Fig.6: Velocity vs time

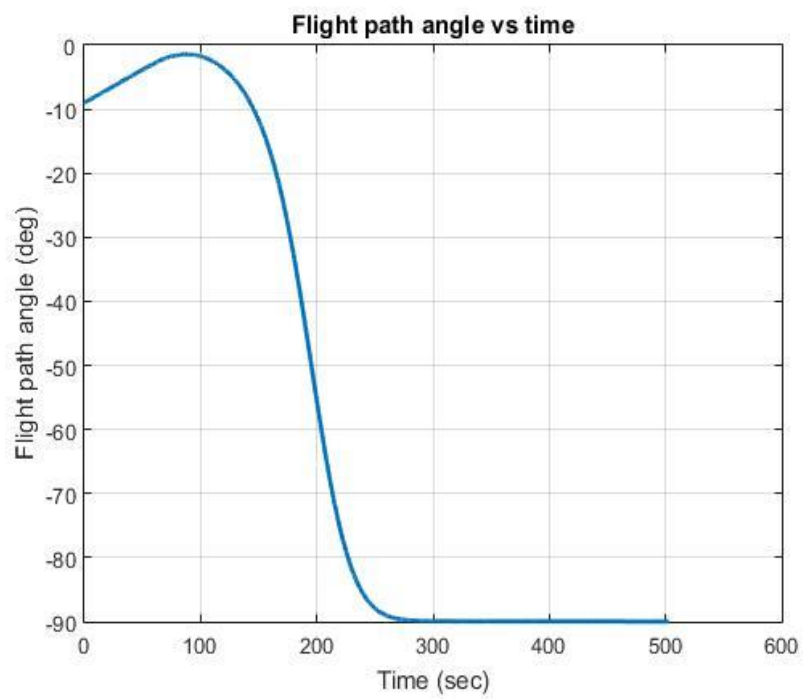


Fig.7: Flight path angle vs time

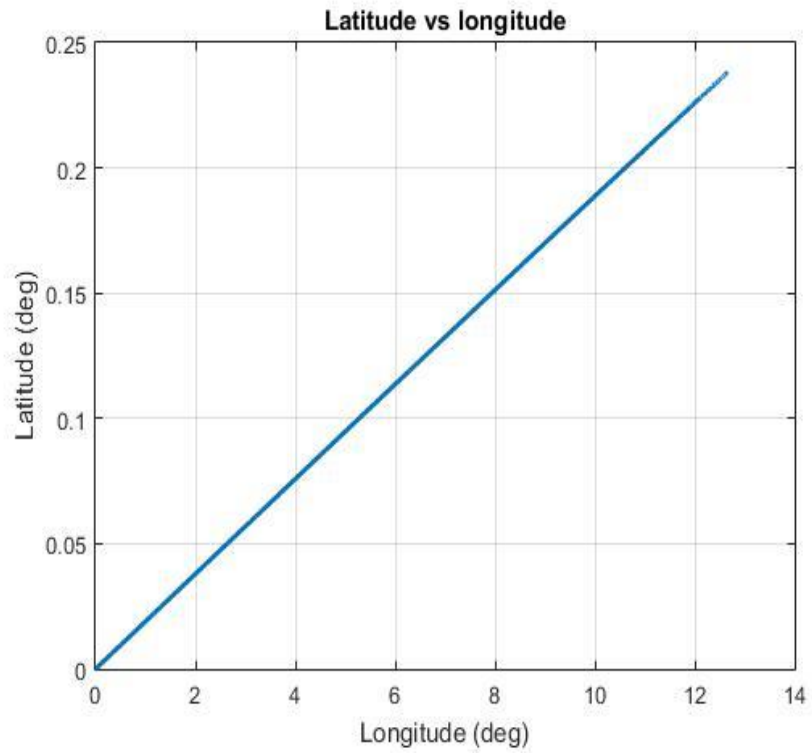


Fig.8: Latitude vs longitude

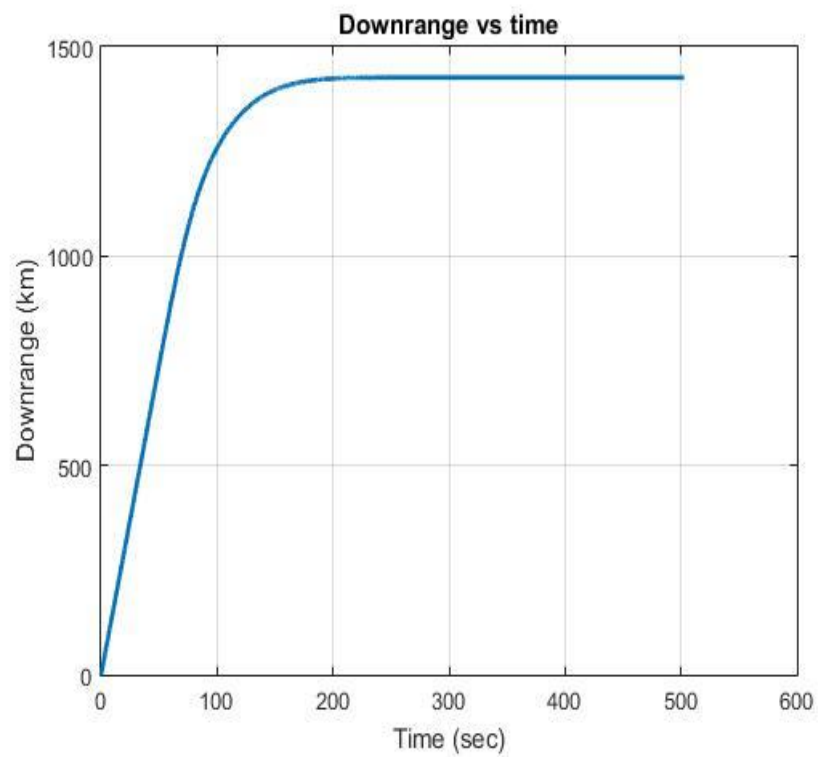
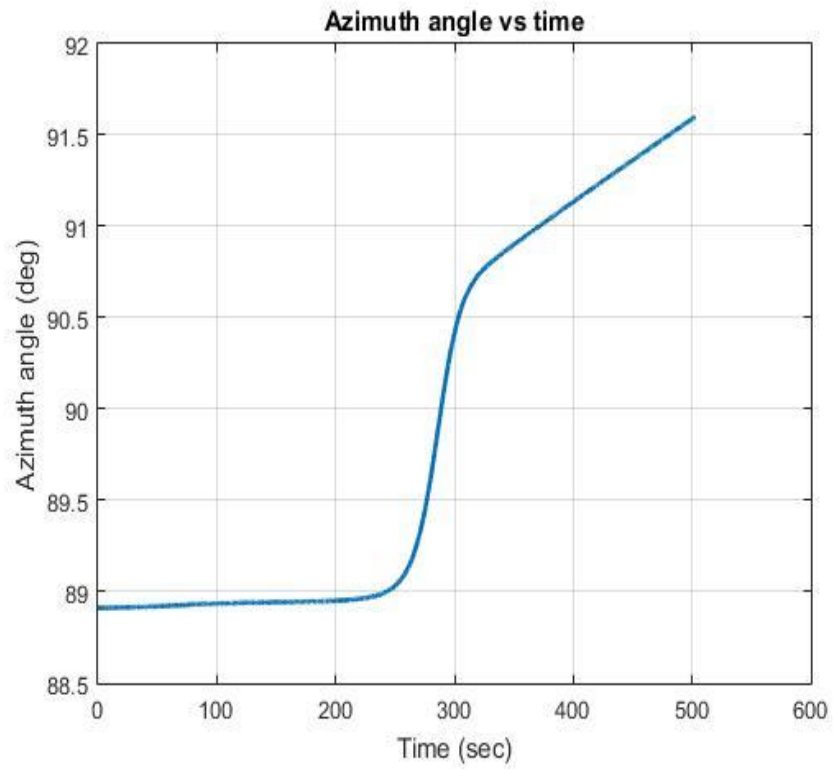
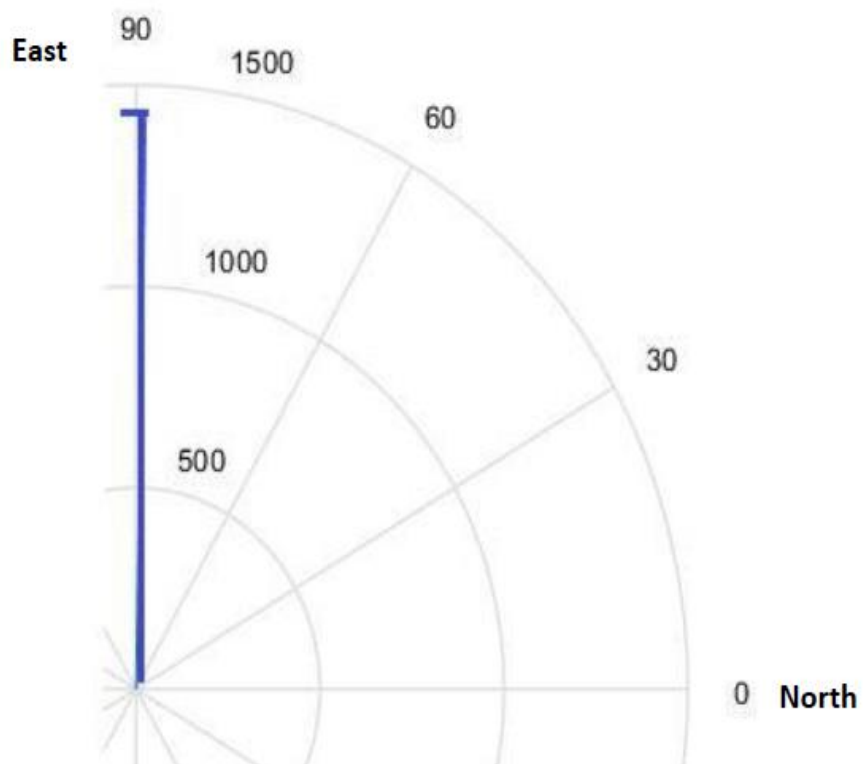


Fig.9: Downrange vs time



(a)



(b)

Fig.10: Azimuth angle variation

From above fig. 11, it is seen that azimuth angle varies from 89° to 91.5° because vehicle isn't rolled. There is no side force acting on the vehicle.

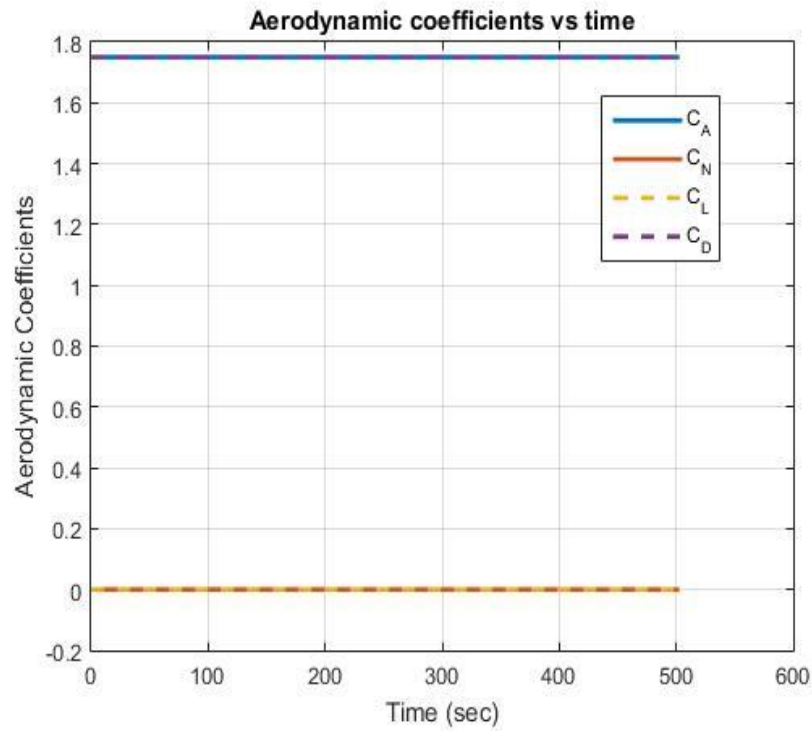


Fig.11: Aerodynamic coefficients vs time

Above plot verifies aerodynamic coefficients given by eqns. (1-4) for angle of attack as zero deg.

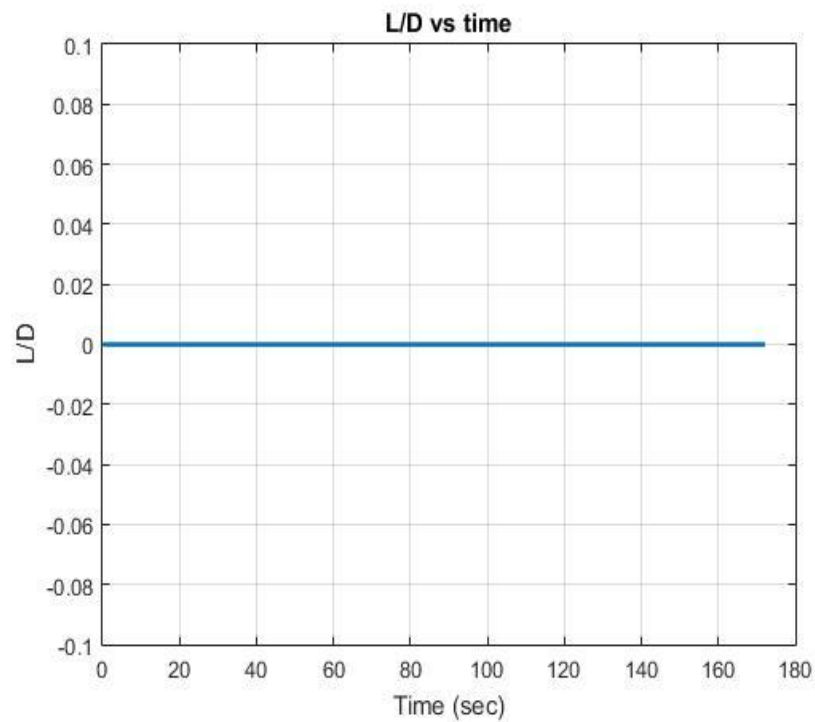


Fig.12: L/D ratio vs time

From above fig.13, L/D ratio is equal to zero as coefficient of lift C_L is 0 as shown in fig. 12.

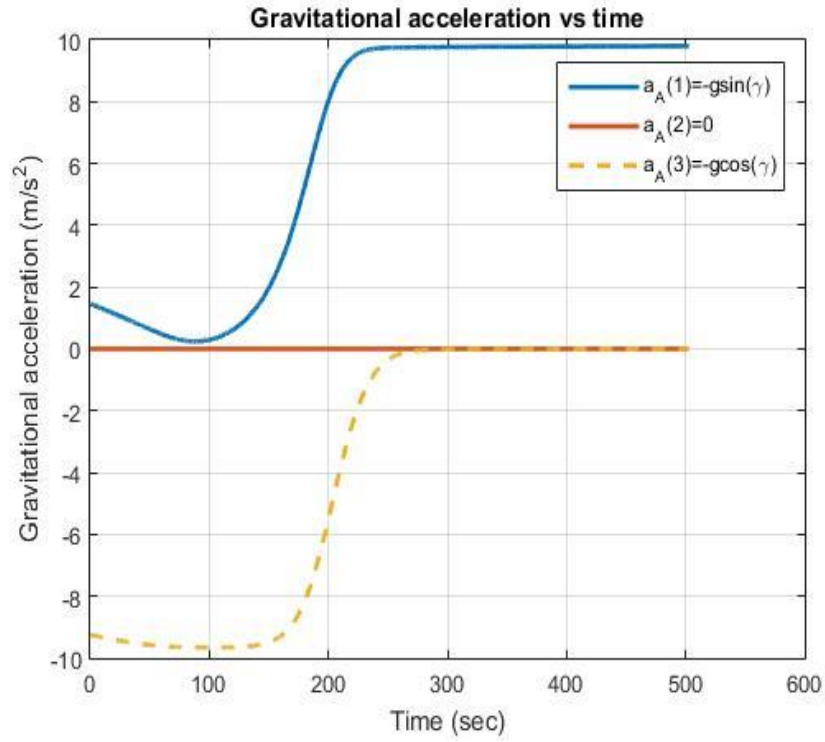


Fig. 13: Gravitational acceleration (wind axes frame) vs time

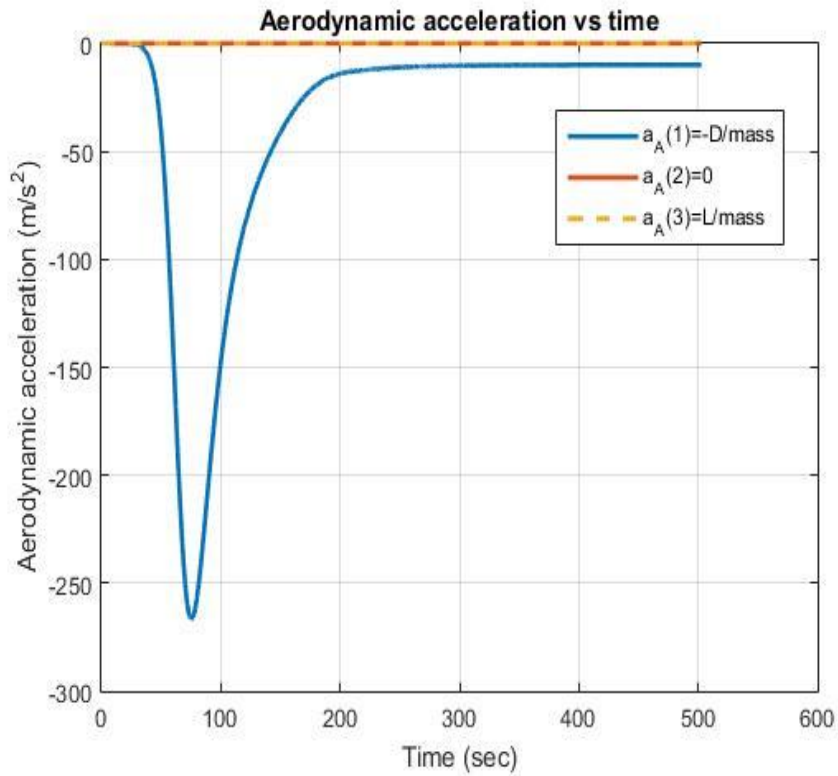


Fig.14: Aerodynamic acceleration (wind axes frame) vs time

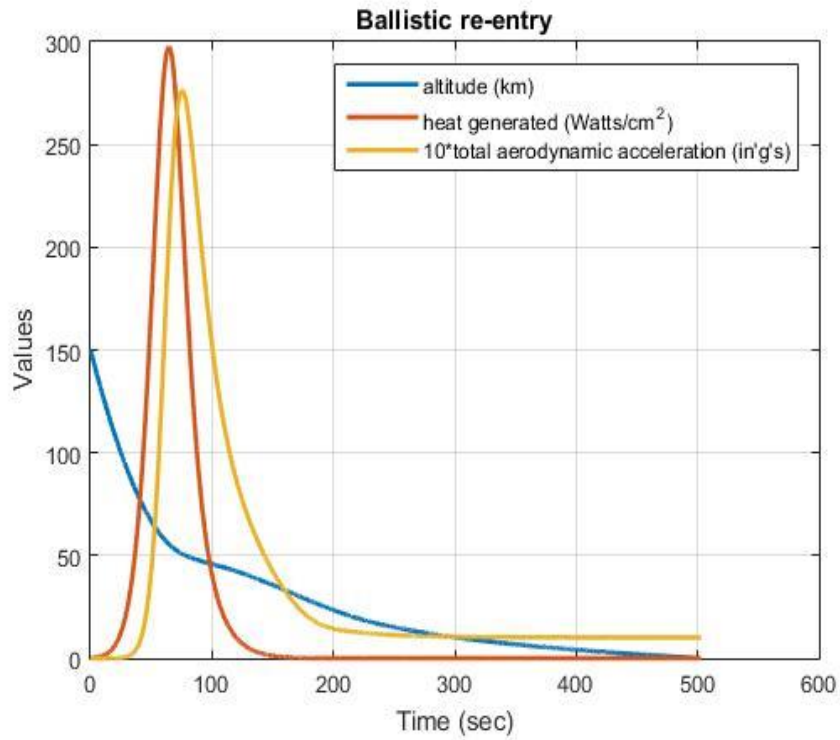


Fig.15: Ballistic entry heating and acceleration

From above fig. 16, it is seen that peak of heating and acceleration is attained in the altitude range of 80 km to 30 km.

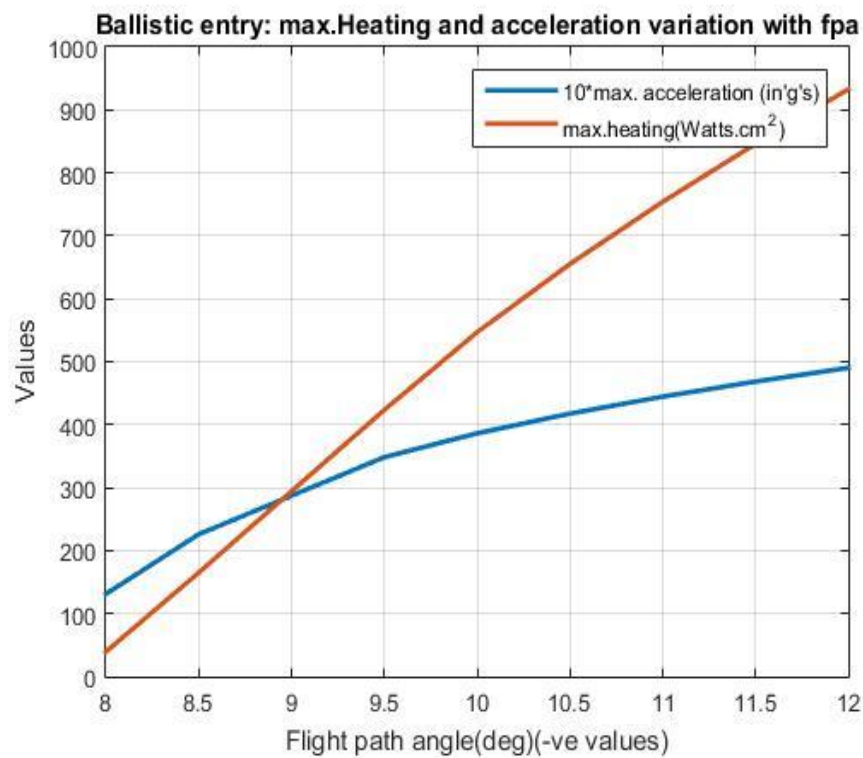


Fig. 16: Maximum heating and maximum acceleration variation with initial flight path angle

Figure 17 shows variation of maximum acceleration and maximum heating with various initial flight path angles. In above plot the ballistic reentry is simulated for various initial flight path angles and corresponding maximum heat generated and maximum acceleration is plotted. Maximum acceleration is multiplied by 10 and plotted in figure to match up with maximum heating values. If initial flight path angle is more positive than -8.1° then vehicle skips out of earth's atmosphere. From fig. 17, it seen than in order to limit maximum acceleration to 30g entry flight path angle should in the range of -8° to -9° .

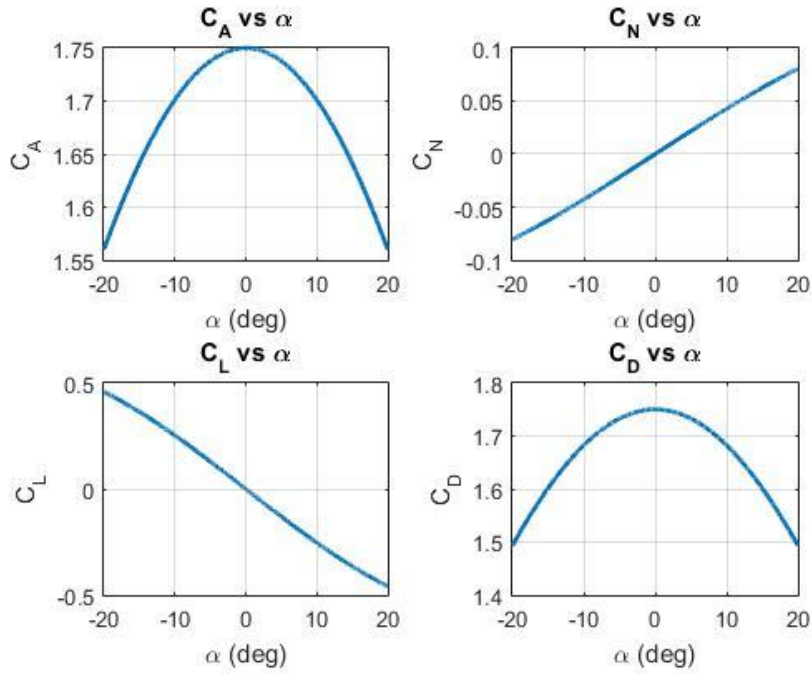
CHAPTER 4

Unmanned Apollo Command Module: Low-lifting Reentry

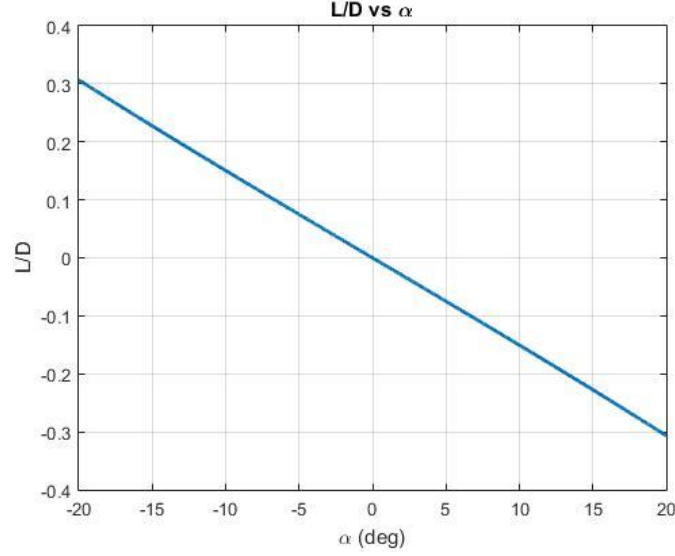
4.1. Low- lifting reentry

In case of lifting reentry, vehicle enters into the atmosphere at some angle of attack so that lift force is generated. In order to satisfy maximum heating constraint and maximum 'g' load constraint, vehicle is to be controlled. Generated lift force is used to control vehicle by rolling it through roll angle (β).

We consider the same unmanned Apollo command module as considered for ballistic reentry with $\theta_c = 60^\circ$. It is re-entering into atmosphere at some angle of attack. The variation of aerodynamic coefficients and L/D ratio with angle of attack ($-20^\circ \leq \alpha \leq 20^\circ$)[4] is given in following figure which is computed using eqns. (23-26) as explained in section 2.3.



(a)



(b)

Fig.1: Aerodynamic coefficients and L/D vs angle of attack (α)

Here we consider L/D ratio generated due to angle of attack is as low as 0.0749. +ve L/D ratio shows that the lift vector is upward. Unmanned Apollo command module is controlled to dissipate high energy associated with it and to land it at preselected location for retrieval. So the type of reentry it follows is low-lifting reentry.

Calculation of the angle of attack corresponding to low-lifting reentry is given as:

From section 2.2,

$$C_N = 0.5\cos^2\theta_c\sin 2\alpha \dots\dots\dots (1)$$

$$C_A = \cos^2\alpha(1 + \sin^2\theta_c) + 0.5\cos^2\theta_c\sin^2\alpha \dots\dots\dots (2)$$

$$C_L = -\sin\alpha C_A + \cos\alpha C_N \dots\dots\dots (3)$$

$$C_D = \cos\alpha C_A + \sin\alpha C_N \dots\dots\dots (4)$$

Substituting above equations in $\frac{L}{D} = \frac{C_L}{C_D}$ in we get,

$$\begin{aligned} \frac{L}{D} &= \frac{-\sin\alpha C_A + \cos\alpha C_N}{\cos\alpha C_A + \sin\alpha C_N} \\ &= \frac{-\sin\alpha(\cos^2\alpha(1+\sin^2\theta_c)+0.5\cos^2\theta_c\sin^2\alpha)+\cos\alpha(0.5\cos^2\theta_c\sin 2\alpha)}{\cos\alpha(\cos^2\alpha(1+\sin^2\theta_c)+0.5\cos^2\theta_c\sin^2\alpha)+\sin\alpha(0.5\cos^2\theta_c\sin 2\alpha)} \\ &= \frac{-\sin\alpha\cos^2\alpha(1+\sin^2\theta_c)-0.5\cos^2\theta_c\sin^3\alpha+\cos\alpha(0.5\cos^2\theta_c\sin 2\alpha)}{\cos^3\alpha(1+\sin^2\theta_c)+0.5\cos\alpha\cos^2\theta_c\sin^2\alpha+\sin\alpha(0.5\cos^2\theta_c\sin 2\alpha)} \end{aligned}$$

In the above equation, $\theta_c = 60^\circ$.

$$\frac{L}{D} = \frac{-1.75\sin\alpha\cos^2\alpha-0.125\sin^3\alpha+0.125\cos\alpha\sin 2\alpha}{1.75\cos^3\alpha+0.125\cos\alpha\sin^2\alpha+0.125\sin\alpha\sin 2\alpha}$$

$$\frac{L}{D} = \frac{-1.75\sin\alpha\cos^2\alpha - 0.125\sin^3\alpha + 0.25\cos^2\alpha\sin\alpha}{1.75\cos^3\alpha + 0.125\cos\alpha\sin^2\alpha + 0.25\sin^2\alpha\cos\alpha}$$

Dividing both the sides by $\cos^3\alpha$,

$$\frac{L}{D} = \frac{-1.75\tan\alpha - 0.125\tan^3\alpha + 0.25\tan\alpha}{1.75 + 0.125\tan^2\alpha + 0.25\tan^2\alpha}$$

$$0.125\tan^3\alpha + 0.375\frac{L}{D}\tan^2\alpha + 1.5\tan\alpha + 1.75\frac{L}{D} = 0$$

Here low refers to low L/D ratio (-0.3) given above.

Solving cubic equation in term of $\tan\alpha$ with $\frac{L}{D} = 0.0749$

$$\tan\alpha = -0.08748, -0.07 + 3.46i, -0.07 - 3.46i$$

$$\alpha = -5^\circ \dots\dots\dots\text{considering real root}$$

4.2. Unguided low-lifting reentry

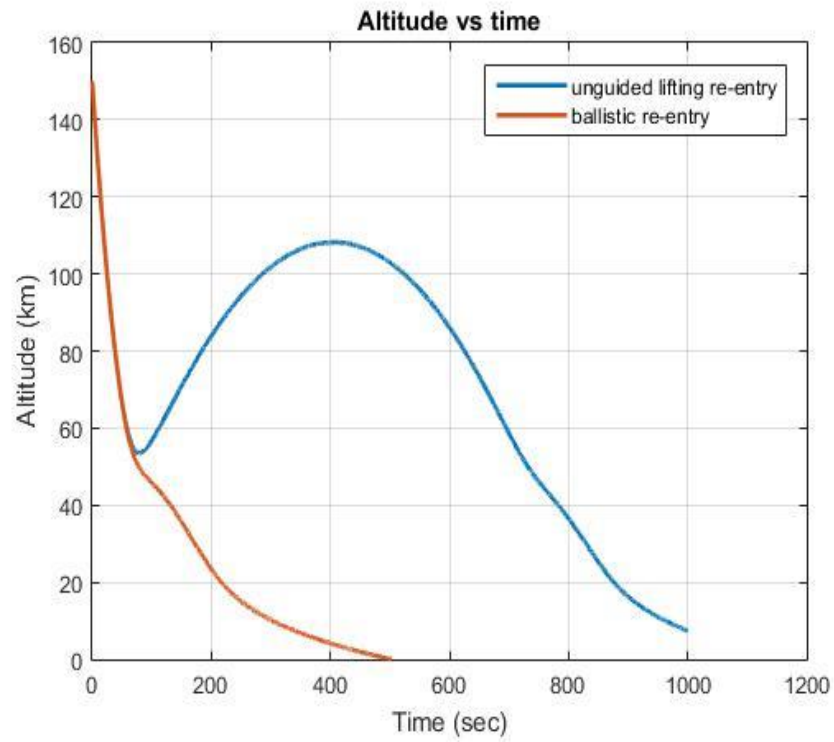
4.2.1. Governing equations and initial state vector

Governing eqns. (eqns. (7-9) and (19-22)) as explained in section 2.2 are numerically integrated using RK4 method in Matlab with time step of 0.01 sec. In these equations aerodynamic coefficients are computed using eqn. (23-26) as explained in section 2.3 with angle of attack $\alpha = -5^\circ$. In all computation exponential density model is utilised which is given in section 3.3. For calculation of the gravitational constant (g) gravitational model is used and the earth's radius is calculated using earth's radius model as explained in section 3.3. As guidance is not applied on unmanned module, roll angle β is taken as zero deg. Initial state vector is obtained by frame transformation as explained in section 3.3. and given by eqn. (12) in section 3.3.

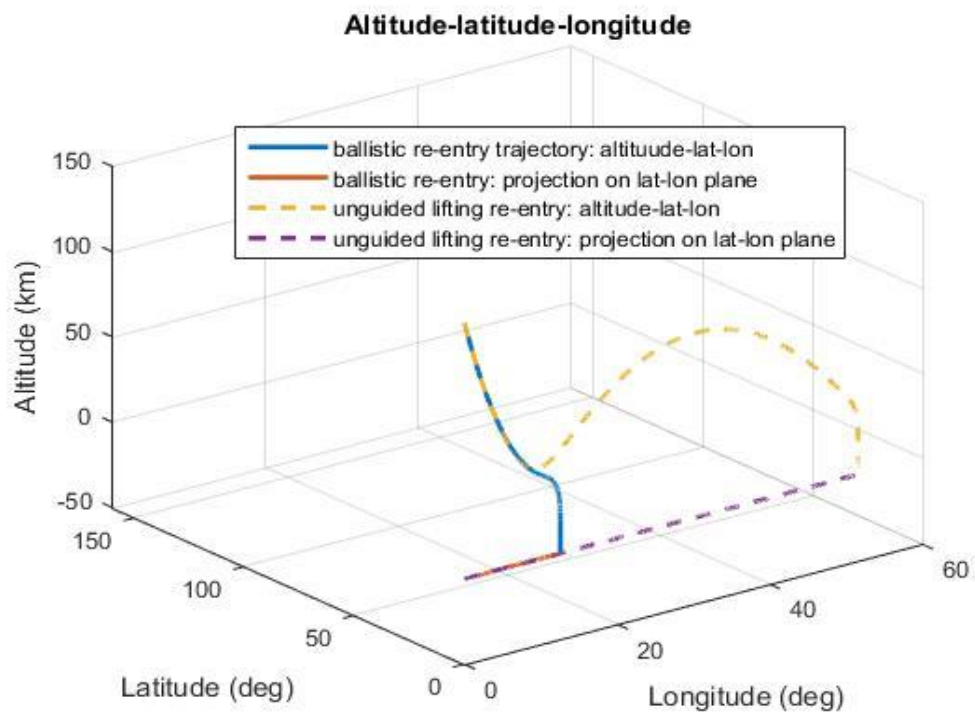
Simulation is stopped when unguided altitude touch down the ground.

4.2.2. Simulation results

A. Altitude vs time



(a)



(b)

Fig.2: Altitude variation

From above fig. 2 (a), it is seen than altitude drops in case of ballistic reentry because there is no lift force. In case of unguided lifting reentry, the altitude plot does have a peculiar feature of diminishing to nearly 50 km, and then rising to 110 km, and then start decreasing again.

B. Velocity vs time

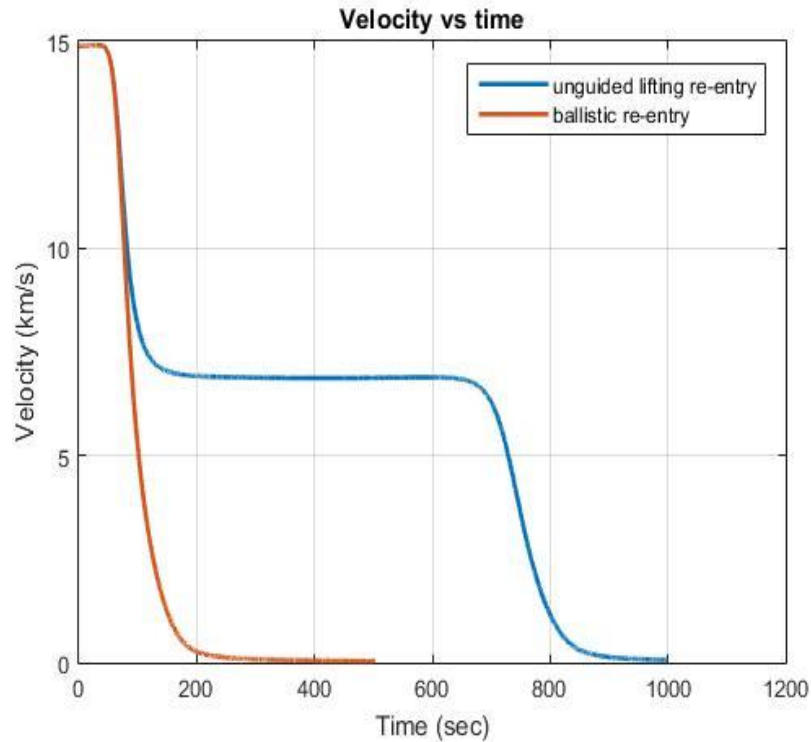


Fig.3: Velocity vs time

From above fig. 3 it is seen than velocity suddenly drops in case of ballistic reentry because lift force is zero.

C. Flight path angle vs time

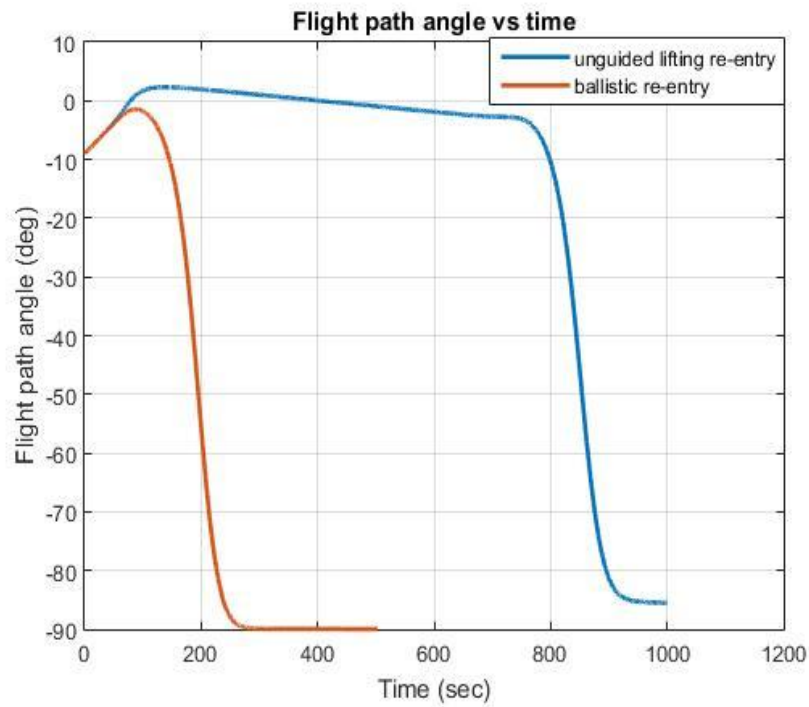


Fig.4: Flight path angle vs time

From above fig.4, it is seen that flight path angle stabilises to -85° in case of unguided low-lifting reentry and to -90° in case of ballistic reentry. This is because governing equation for flight path angle (eqn. (21), section 2.2) is affected by lifting force. In ballistic reentry lift force is zero whereas low-lifting reentry generates significant lifting force.

D. Latitude vs longitude

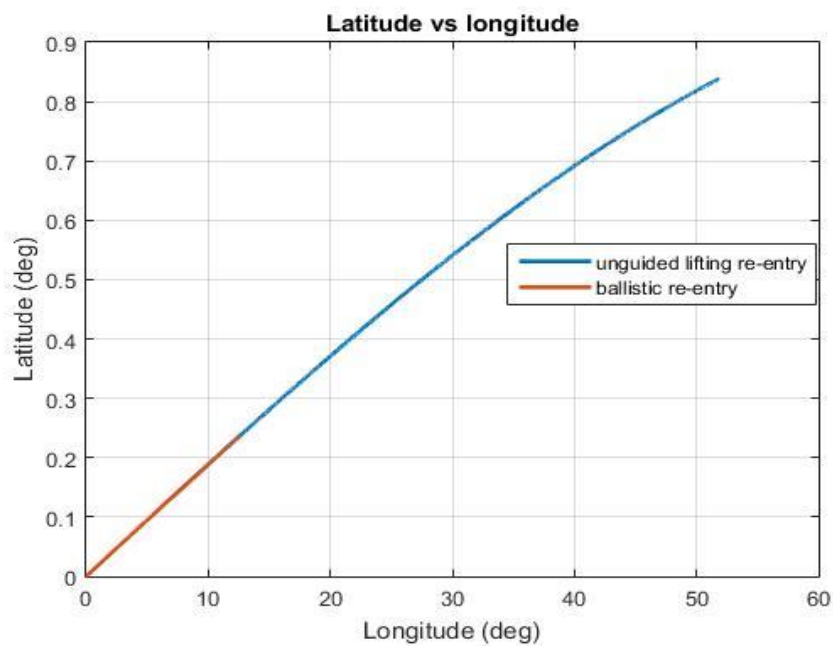
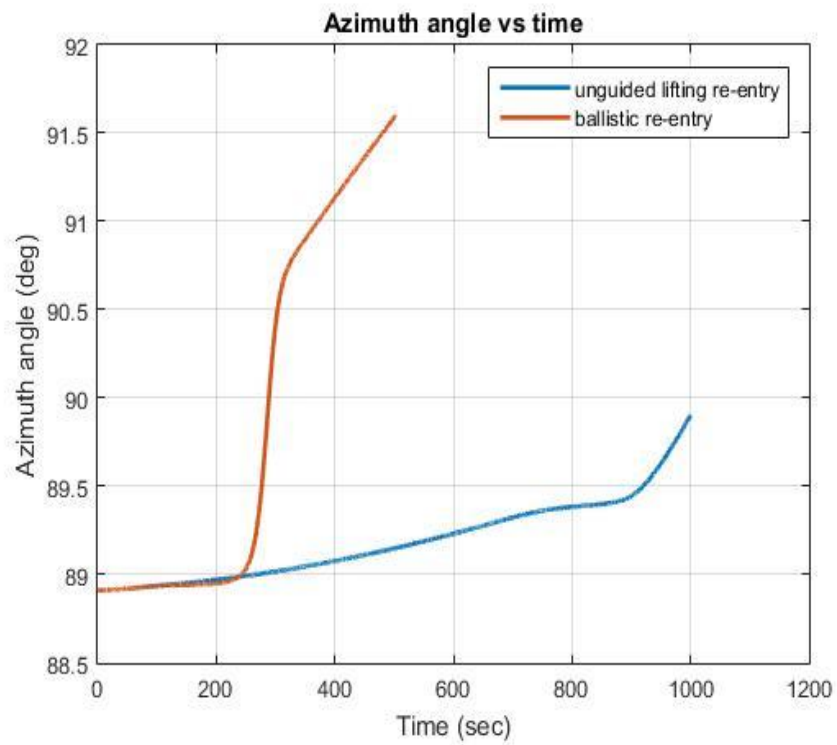
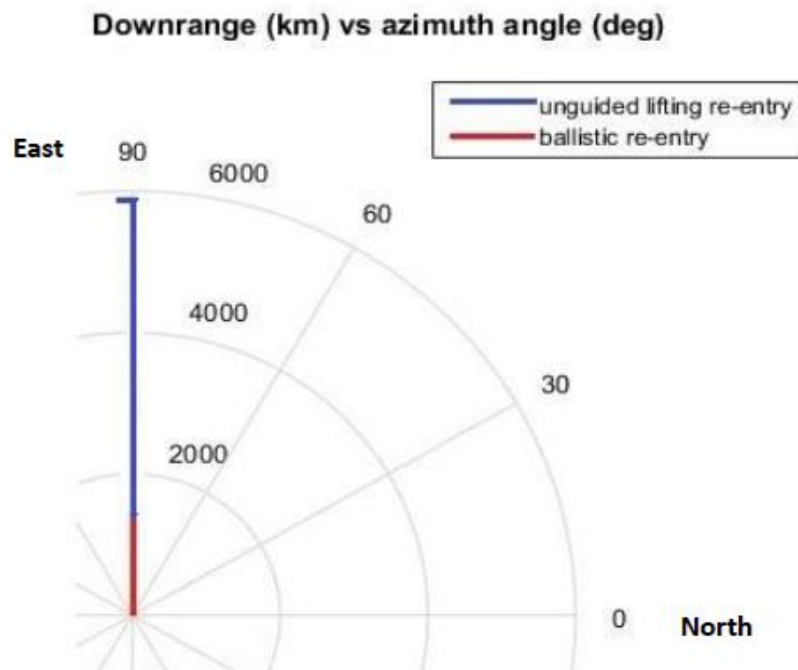


Fig. 5: Latitude vs longitude

E. Azimuth angle vs time



(a)



(b)

Fig.6: Azimuth angle variation

As vehicle is not rolled, no side force is acting on the vehicle and azimuth angle does not vary as seen in fig.6.

F. Downrange vs time

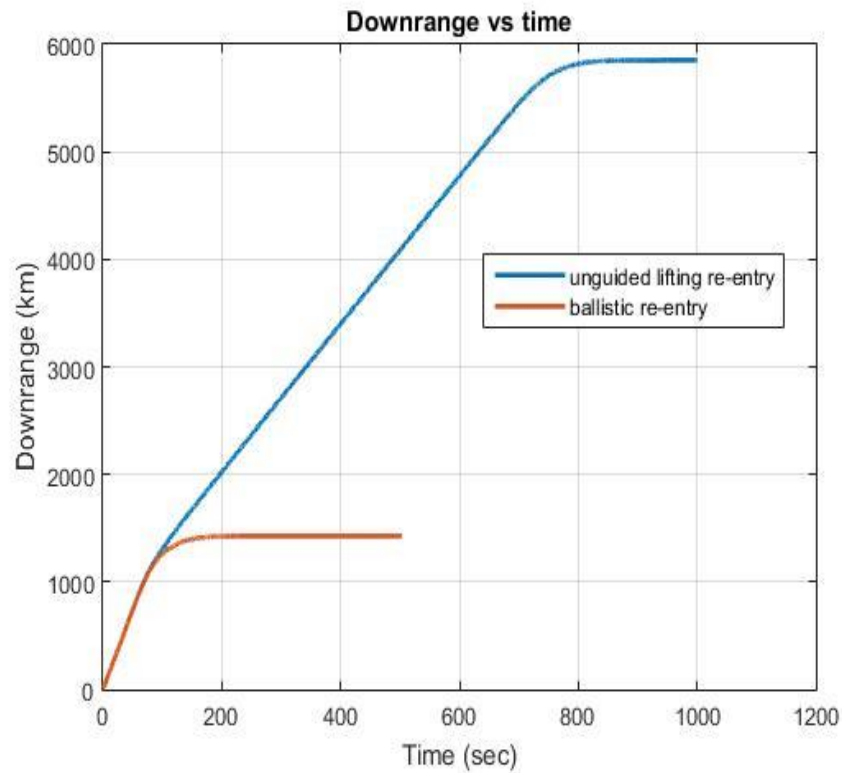


Fig.7: Downrange variation

From above plot it is seen that downrange covered in case of unguided lifting reentry is greater than that of ballistic trajectory.

G. Aerodynamic coefficients vs time

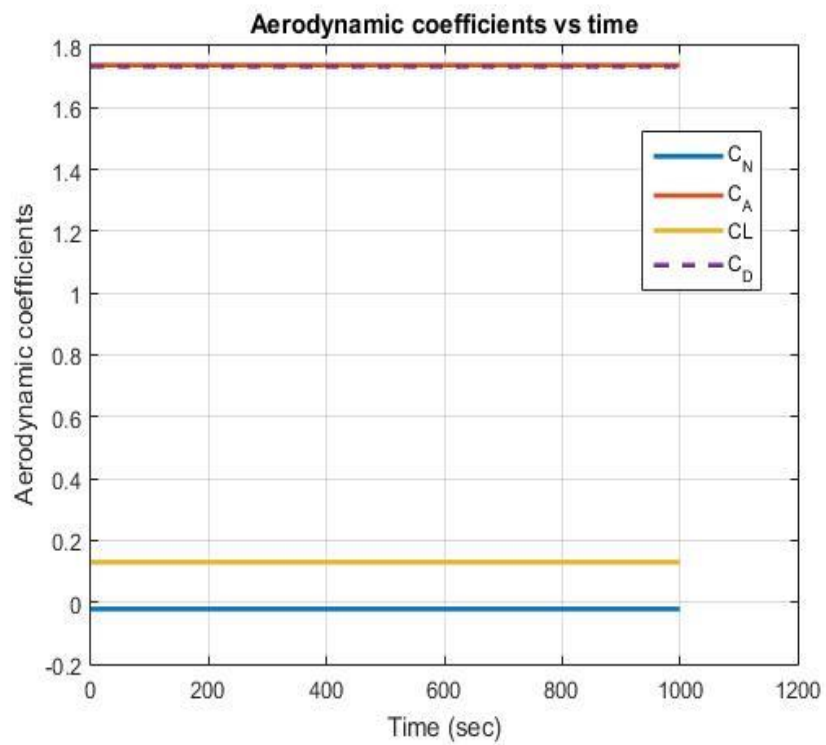


Fig.8: Aerodynamic coefficients vs time

As angle of attack is constant as -5° throughout flight, aerodynamic coefficients are constants as seen in above fig. 8.

H. Generated L/D vs time

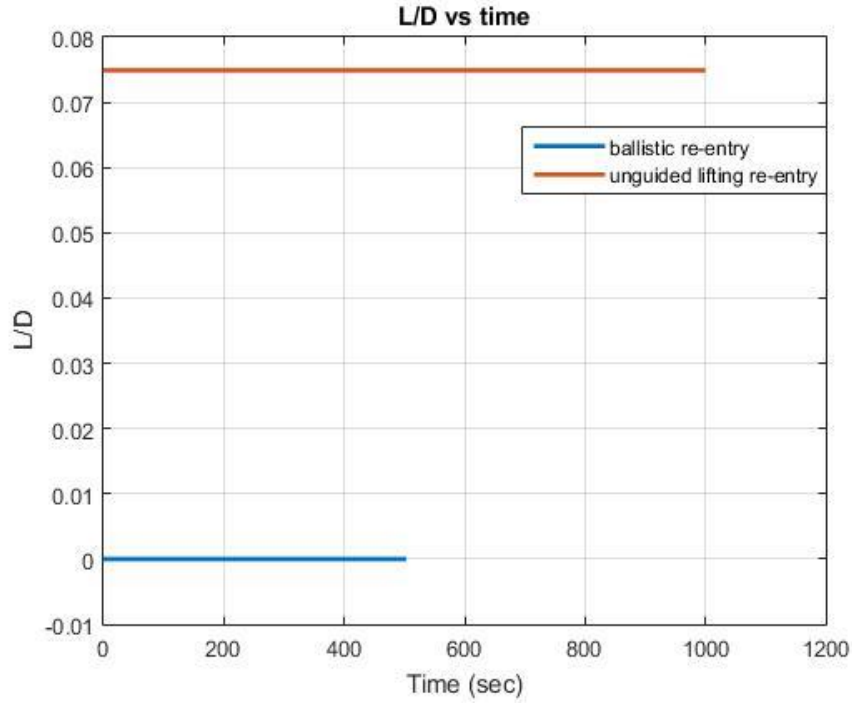


Fig.9: L/D vs time

L/D corresponding to angle of attack -5° is 0.0749 as seen in fig. 9.

I. Aerodynamic acceleration vs time

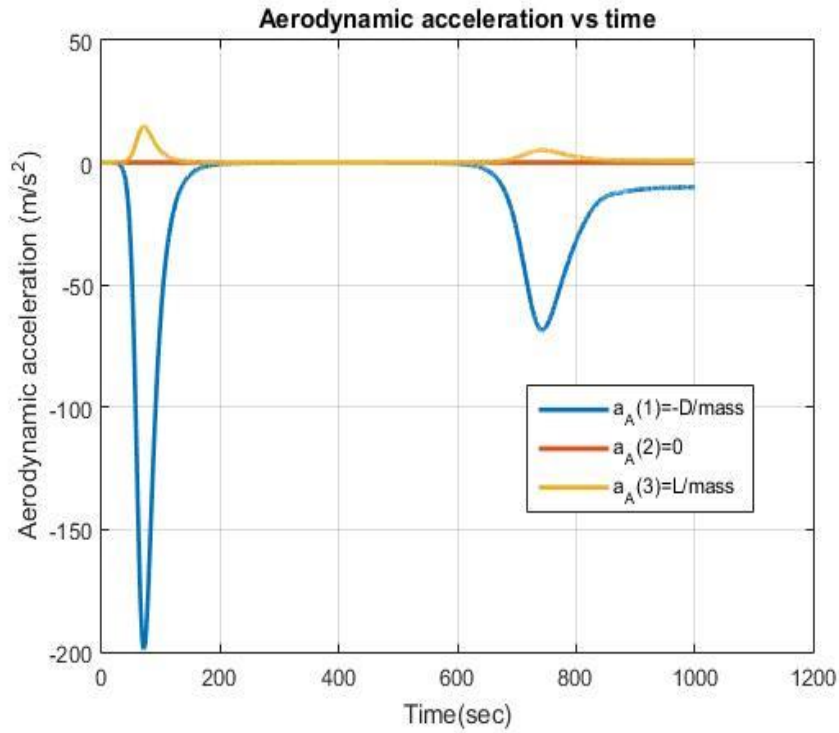


Fig. 10: Aerodynamic acceleration (wind axes frame) vs time

From fig.10, aerodynamic acceleration generated is very high and unacceptable so vehicle is to be guided to control aerodynamic acceleration.

J. Gravitational acceleration vs time

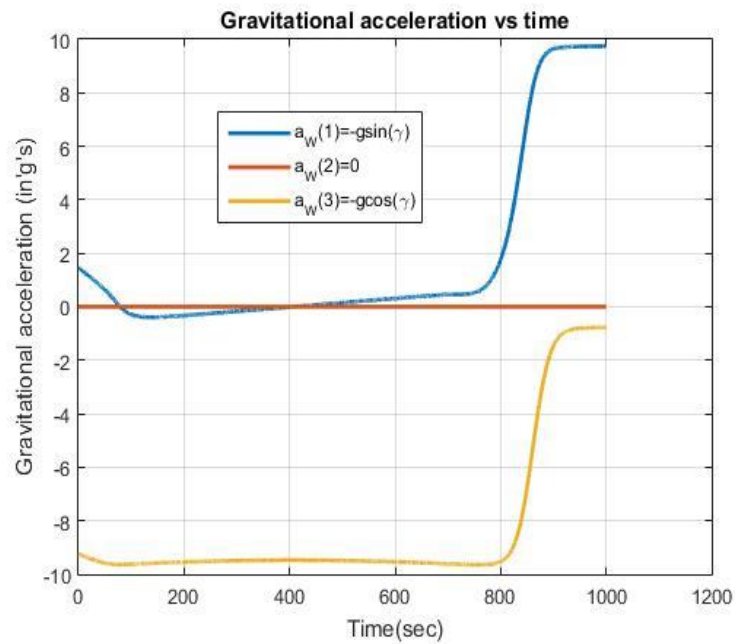


Fig.11: Gravitational acceleration (wind axes frame) vs time

K. Heat generated vs time

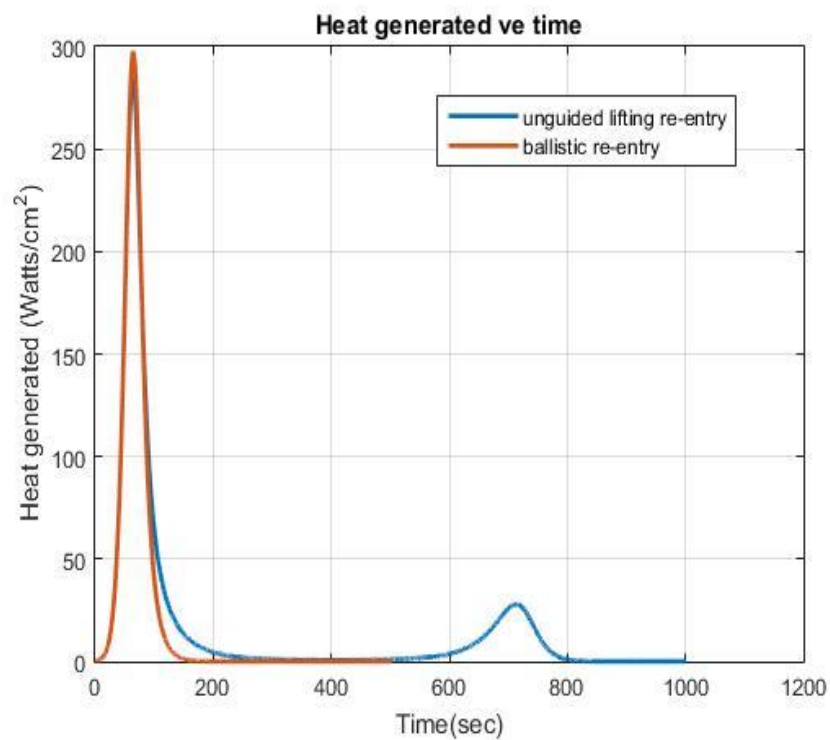


Fig.12: Heat generated vs time

4.3. Guided low-lifting reentry

Unmanned Apollo command module is guided to follow heat constraint and maximum 'g' load constraint. It follows classical approach of implementing guidance laws: constant drag deceleration

law followed by constant descent rate law. Based on guidance laws nominal trajectory is generated with zero deg roll angle which is the reference trajectory to be followed. Closed-loop guidance is implemented to follow nominal trajectory by rolling lift vector roll angle β about velocity vector and reference roll command is obtained.

4.3.1. Guidance laws

Unmanned Apollo command module is guided by two guidance laws: constant drag deceleration law followed by constant descent rate law as explained below.

Governing eqns. (Eqns. (7-9) and (19-22)) as explained in section 3.2 in simplified form by neglecting rotational effect of the earth are given below:

$$\begin{aligned}\dot{R} + \dot{h} &= v \sin \gamma \dots \dots \text{from eqn. (9) in section 2.2 of chapter} \\ \dot{h} &= v \sin \gamma \dots \dots \dots (5)\end{aligned}$$

$$\begin{aligned}\dot{v} &= \frac{1}{m} (-mg \sin \gamma - D) \dots \dots \dots \text{from eqn. (19) in section 2.2} \\ \dot{v} &= \frac{1}{m} \left(-g \sin \gamma - \frac{1}{2} \rho v^2 C_D S \right) \dots \dots D = \frac{1}{2} \rho v^2 C_D S \dots \dots \dots (6)\end{aligned}$$

$$S = \frac{\pi}{4} d^2$$

$$\dot{v} = \frac{-\rho v^2}{2 B_c} - g \sin \gamma$$

$$B_c = \frac{m}{C_D S} \text{ at zero angle of attack}$$

$$\dot{\gamma} = \frac{v}{r} \cos \gamma + \frac{1}{mv} (-mg \cos \gamma + L \cos \beta) \dots \dots \text{from eqn. (21) in section 2.2}$$

$$\begin{aligned}&= \frac{v}{r} \cos \gamma - \frac{g \cos \gamma}{v} + \frac{L \cos \beta}{mv} \\ &= \frac{v}{r} \cos \gamma - \frac{g \cos \gamma}{v} + \frac{1}{2} \rho v^2 C_L S \frac{\cos \beta}{mv} \dots \dots \text{Lift force: } L = \frac{1}{2} \rho v^2 C_L S \\ &= \frac{v}{r} \cos \gamma - \frac{g \cos \gamma}{v} + \frac{1}{2} \rho v C_D \frac{L}{D} S \frac{\cos \beta}{m} \\ &= \frac{v}{r} \cos \gamma - \frac{g \cos \gamma}{v} + \frac{1}{2} \frac{\rho v}{B_c} \left(\frac{L}{D} \right) \cos \beta\end{aligned}$$

For $\beta = 0^\circ$ (no roll)

$$\begin{aligned}\dot{\gamma} &= \frac{v}{r} \cos \gamma - \frac{g \cos \gamma}{v} + \frac{1}{2} \frac{\rho v}{B_c} \left(\frac{L}{D} \right) \\ &= \frac{g \cos \gamma}{v} \left(\frac{v^2}{gr} - 1 \right) + \frac{1}{2} \frac{\rho v}{B_c} \left(\frac{L}{D} \right) \\ &= \frac{g \cos \gamma}{v} \left(\frac{v^2 r}{gr^2} - 1 \right) + \frac{1}{2} \frac{\rho v}{B_c} \left(\frac{L}{D} \right) \\ &= \frac{g \cos \gamma}{v} \left(\frac{v^2 r}{\mu} - 1 \right) + \frac{1}{2} \frac{\rho v}{B_c} \left(\frac{L}{D} \right) \\ \dot{\gamma} &= \frac{g \cos(\gamma)}{v} \left(\frac{v^2}{v_c^2} - 1 \right) + \frac{\rho v}{2 B_c} \left(\frac{L}{D} \right)_{10r2} \dots \dots \dots (7)\end{aligned}$$

$$v_c = \sqrt{\frac{\mu}{r}}$$

Other governing eqns. are the same as mentioned in eqns. (19-22) section 2.2 but rotational effect of earth is neglected.

$$\dot{\tau} = \frac{v \cos \gamma \sin \chi}{r \cos \delta} \dots \dots \dots (8)$$

$$\dot{\delta} = \frac{v \cos \gamma \cos \chi}{r} \dots \dots \dots (9)$$

$$\dot{\chi} = \frac{1}{m v \cos \gamma} \left(\frac{1}{2} \rho v^2 C_L S \sin \beta \right) \dots \dots \dots (10)$$

$$\dot{\alpha} = v \cos \gamma \dots \dots \dots (11)$$

Above governing eqns. (5-11) are used to compute L/D required to follow particular law as explained below.

A. Constant drag deceleration law:

This guidance law maintains constant drag deceleration d_0 . This law is applied for higher altitude to dissipate most of the kinetic energy in upper atmosphere. L/D for constant drag deceleration law in vertical plane is calculated as below.

$$\text{Drag deceleration: } d = \frac{1}{2 B_c} \rho v^2$$

Differentiating drag deceleration equation w.r.t .time,

$$\dot{d} = \frac{\rho}{B_c} v \dot{v} + \frac{v^2}{2 B_c} \dot{\rho} \dots \dots \dots (12)$$

For constant drag deceleration law $\dot{d} = 0$.

$$\frac{\dot{\rho}}{\rho} = \frac{-2 \dot{v}}{v} \dots \dots \dots (13) \text{ From eqn. (12)}$$

$$\rho = \rho_0 e^{-(h/h_s)} \dots \dots \dots \text{ Exponential density model}$$

$$\dot{\rho} = \rho_0 e^{-(h/h_s)} \left(\frac{-\dot{h}}{h_s} \right)$$

$$\frac{\dot{\rho}}{\rho} = \left(\frac{-\dot{h}}{h_s} \right) = \frac{-2 \dot{v}}{v} \dots \dots \dots \text{ From eqn. (13)}$$

Putting values from eqns. (5-6)

$$\frac{v \sin(\gamma)}{h_s} = \frac{2}{v} \left(\frac{-\rho v^2}{2 B_c} - g \sin(\gamma) \right)$$

$$\frac{v^2 \sin(\gamma)}{h_s} = \frac{-\rho v^2}{B_c} - 2 g \sin(\gamma)$$

$$\left(2g + \frac{v^2}{h_s}\right) \sin(\gamma) = \frac{-\rho v^2}{B_c} = -2d_0 \dots (14)$$

Differentiating w.r.t. time,

$$2gh_s \dot{\gamma} \cos(\gamma) + v^2 \dot{\gamma} \cos(\gamma) + 2v \sin(\gamma) \dot{v} = 0$$

$$v \dot{\gamma} = \frac{-2v^2 \dot{v} \sin(\gamma)}{(2gh_s + v^2) \cos(\gamma)} \dots (15)$$

From eqn. (7) and definition of drag deceleration,

$$\frac{L}{D} = \frac{1}{d_0} \left(v \dot{\gamma} - g \cos(\gamma) \left(\frac{v^2}{v_c^2} - 1 \right) \right)$$

Putting value of $v \dot{\gamma}$ from eqn. (15)

$$\frac{L}{D} = \frac{1}{d_0} \left(\frac{-2v^2 \dot{v} \sin(\gamma)}{(2gh_s + v^2) \cos(\gamma)} - g \cos(\gamma) \left(\frac{v^2}{v_c^2} - 1 \right) \right)$$

Putting value of eqn. (6),

$$\frac{L}{D} = \frac{1}{d_0} \left(\frac{2v^2 \sin(\gamma) (d_0 + g \sin(\gamma))}{(2gh_s + v^2) \cos(\gamma)} - g \cos(\gamma) \left(\frac{v^2}{v_c^2} - 1 \right) \right)$$

$$\left(\frac{L}{D} \right)_1 = \left(\frac{2v^2 \tan \gamma_1 \left(1 + \frac{g \sin \gamma_1}{d_0} \right)}{(2gh_s + v^2)} - \frac{g \cos \gamma_1}{d_0} \left(\frac{v^2}{v_c^2} - 1 \right) \right) \dots (16)$$

Flight path angle γ_1 in eqn. (16) is calculated as:

$$\gamma_1 = \sin^{-1} \left(\frac{-2d_0 h_s}{(2gh_s + v^2)} \right) \dots (17)$$

$$v_c = \sqrt{\mu/r}$$

B. Constant descent rate law:

The descent rate at the end of constant drag deceleration law is maintained constant in constant descent rate law and applied in lower atmosphere to control aerodynamic acceleration generated.

$$\gamma = \sin^{-1} \left(\frac{h_0}{v} \right) \dots \text{From eqn. (5)}$$

Rearranging equation (7)

$$\dot{\gamma} - \frac{g \cos(\gamma)}{v} \left(\frac{v^2}{v_c^2} - 1 \right) = \frac{\rho v}{2B_c} \left(\frac{L}{D} \right)$$

$$\frac{L}{D} = \frac{2B_c}{\rho v} \left(\dot{\gamma} - \frac{g \cos(\gamma)}{v} \left(\frac{v^2}{v_c^2} - 1 \right) \right)$$

$$\frac{L}{D} = \frac{2B_c}{\rho v^2} (\dot{\gamma}v) - \frac{2B_c g}{\rho v^2 \cos(\gamma)} \left(\frac{v^2 \cos^2(\gamma)}{v_c^2} - \cos^2(\gamma) \right)$$

$$\frac{L}{D} = \frac{2B_c}{\rho v^2} (\dot{\gamma}v) - \frac{2B_c g}{\rho v^2 \cos(\gamma)} \left(\frac{v^2 \cos^2(\gamma)}{v_c^2} - 1 + \sin^2(\gamma) \right)$$

$$\frac{L}{D} = \frac{2B_c}{\rho v^2} \left(\dot{\gamma}v - \frac{g \sin^2(\gamma)}{\cos(\gamma)} \right) + \frac{2B_c g}{\rho v^2 \cos(\gamma)} \left(1 - \frac{v^2 \cos^2(\gamma)}{v_c^2} \right) \dots \dots \dots (18)$$

Differentiating $v \sin(\gamma) = \dot{h}_0$

$$v \cos(\gamma) \dot{\gamma} = -\dot{v} \sin(\gamma)$$

$$v \dot{\gamma} = -\dot{v} \tan(\gamma)$$

$$v \dot{\gamma} = -\dot{v} \tan \gamma \dots 7.43$$

$$\dot{\gamma}v - \frac{g \sin^2(\gamma)}{\cos(\gamma)} = -\tan(\gamma) \dot{v} - g \tan(\gamma) \sin(\gamma)$$

$$= -\tan(\gamma) (\dot{v} + g \sin(\gamma))$$

$$\dot{\gamma}v - \frac{g \sin^2(\gamma)}{\cos(\gamma)} = \tan(\gamma) \frac{\rho v^2}{2B_c} \dots \dots \dots \text{From eqn. (6)}$$

Eqn. (14) becomes,

$$\left(\frac{L}{D} \right)_2 = \tan \gamma_2 + \frac{g}{d \cos \gamma_2} \left(1 - \frac{v^2 \cos^2 \gamma_2}{v_c^2} \right) \dots \dots \dots (19)$$

Flight path angle γ_2 calculated as:

$$\gamma_2 = \sin^{-1} \left(\frac{\dot{h}_0}{v} \right) \dots \dots \dots (20)$$

4.3.2. Generation of nominal trajectory

4.3.2.1. Governing equations and initial state vector

Ballistic reentry is followed with zero angle of attack. Governing eqns. (eqns. (7-9) and (19-22)) explained in section 2.2 are numerically integrated using RK4 method on Matlab with time step of 0.01 sec. The roll angle is considered to be zero during generation of nominal trajectory. In these equations aerodynamic coefficients are computed using eqn. (1-4) as explained above with angle of attack $\alpha = 0^\circ$. In all computation exponential density model is utilised which is given in section 3.3. For calculation of the gravitational constant (g) gravitational model is used and the earth's radius is calculated using earth's radius model as explained in section 3.3.

4.3.2.2. Implementation of constant drag deceleration law

The drag deceleration required to follow constant drag deceleration law (given by first term of eqn. (14)) and drag deceleration actually generated (given by second term of eqn. (14)) are monitored at every time step during ballistic trajectory. The altitude at which both the terms match, constant drag deceleration law is implemented as shown in figure below:

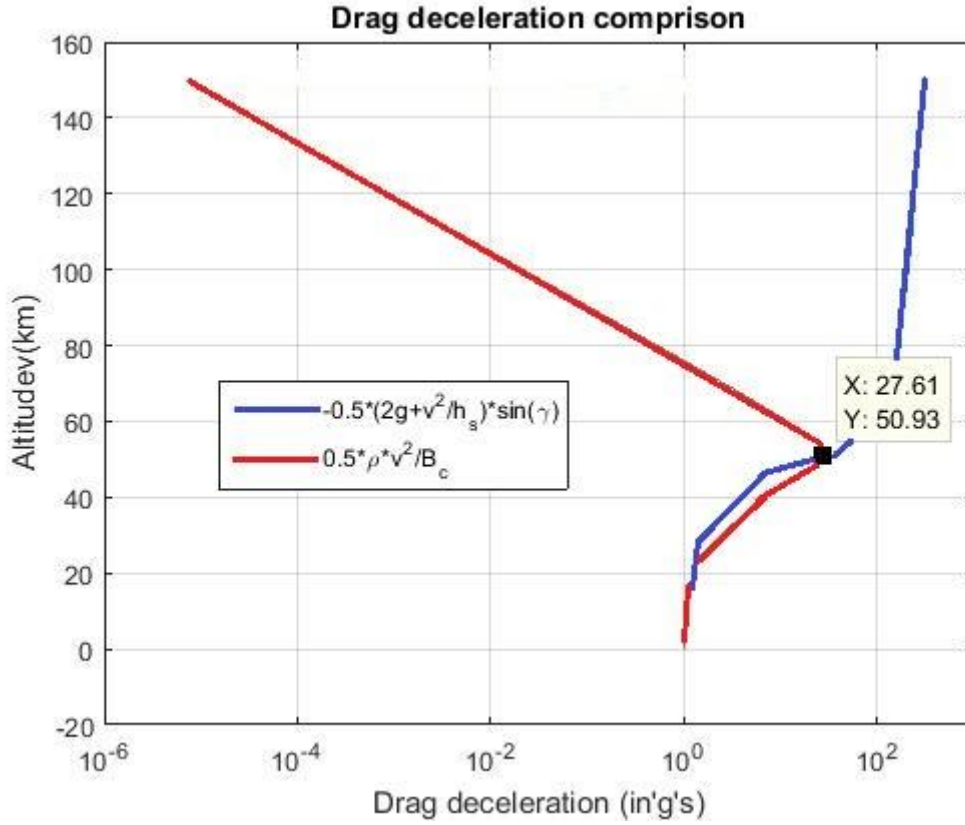


Fig. 13: Drag deceleration comparison to switch to constant drag deceleration law

From above figure, at altitude of 50.93 km, drag deceleration required to implement constant drag deceleration matches with drag deceleration generated. So constant drag deceleration law is implemented at altitude 50.93km. The state vector corresponding to the end of ballistic trajectory is given as:

$$X = [9.31^\circ \quad 0.18^\circ \quad (50.93 + 6378.14) \text{ km} \quad 10533.855 \text{ m/s} \quad 88.92^\circ \quad -1.72^\circ]^T$$

During this phase the angle of attack is calculated at every time step based on desired L/D using aerodynamic eqns. 23-26 from section 2.3. The desired L/D to follow this law is calculated using eqn. 16-17. Simulation results for this phase are given below:

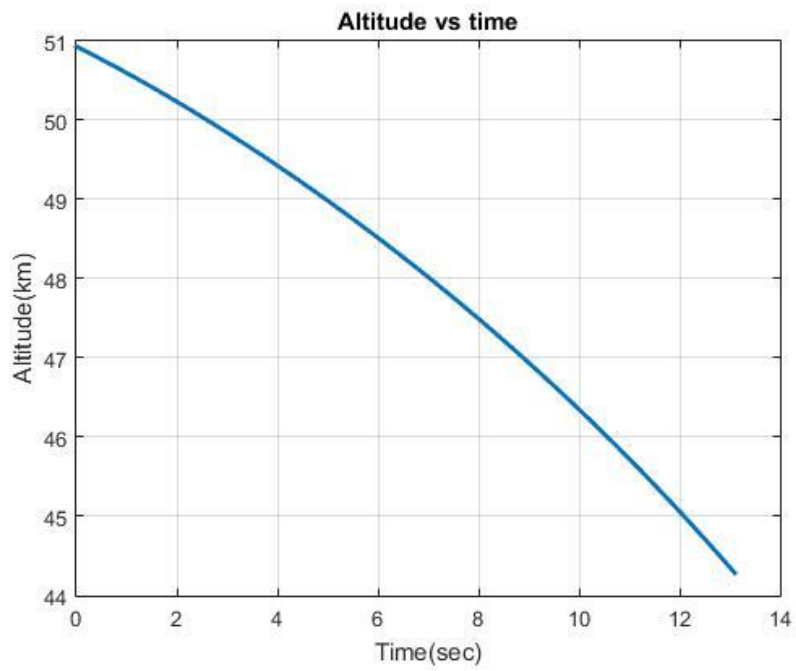


Fig. 14: Altitude vs time

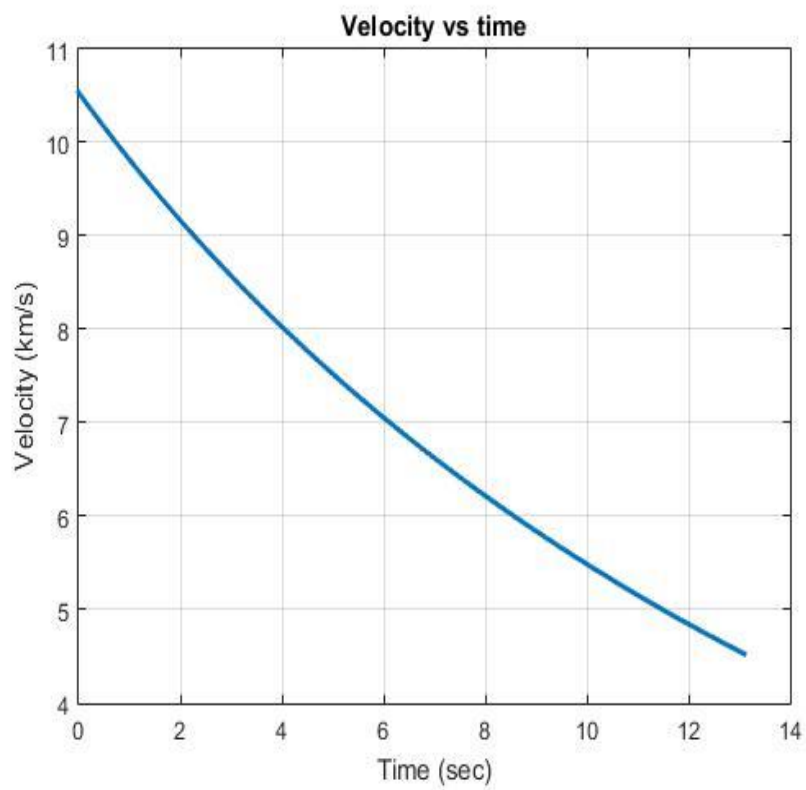


Fig. 15: Velocity vs time

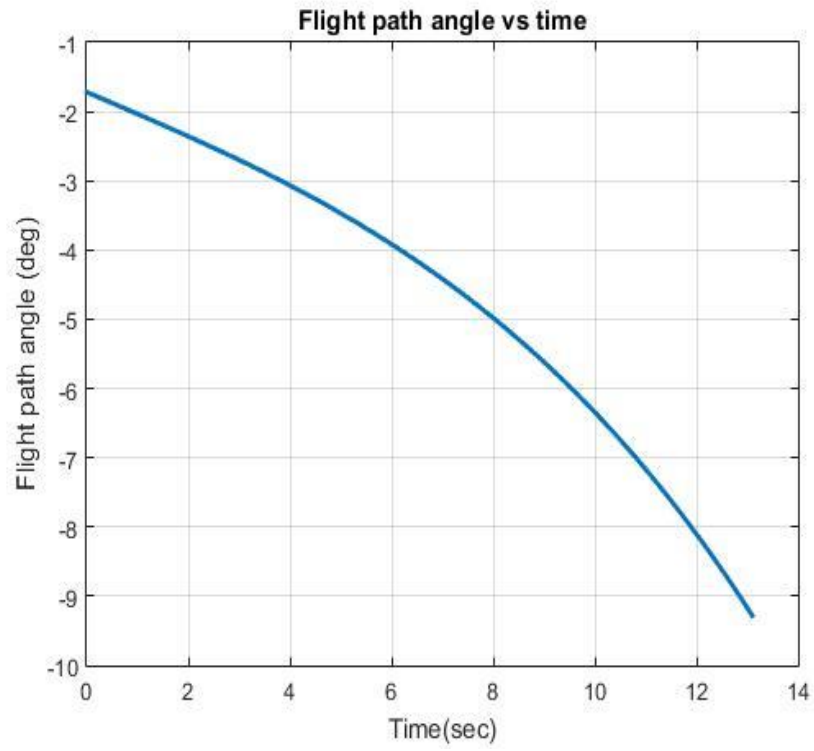


Fig. 16: Flight path angle vs time

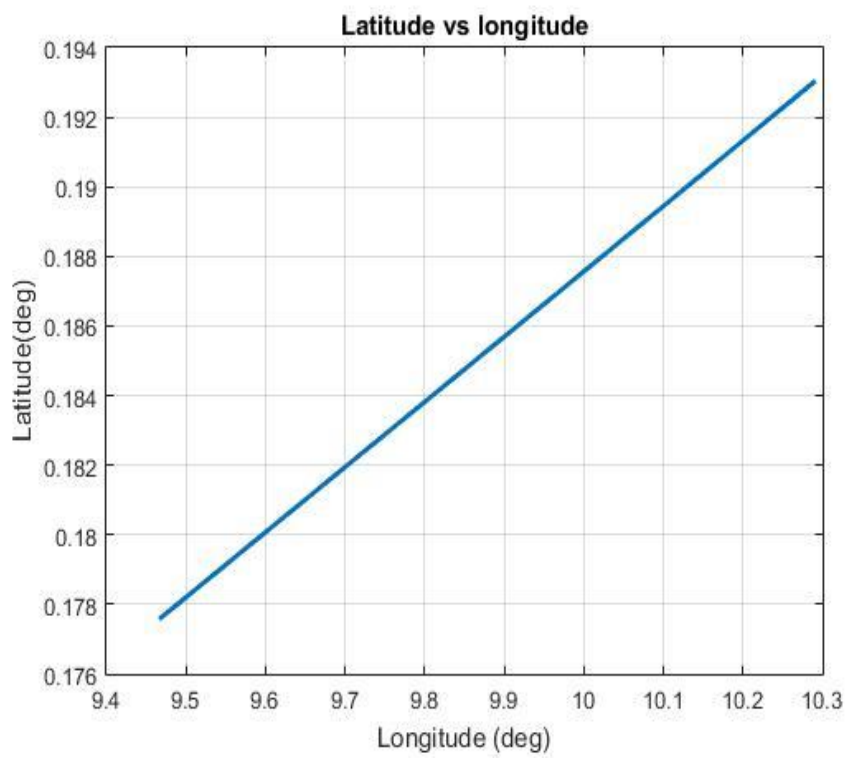


Fig. 17: Latitude vs longitude

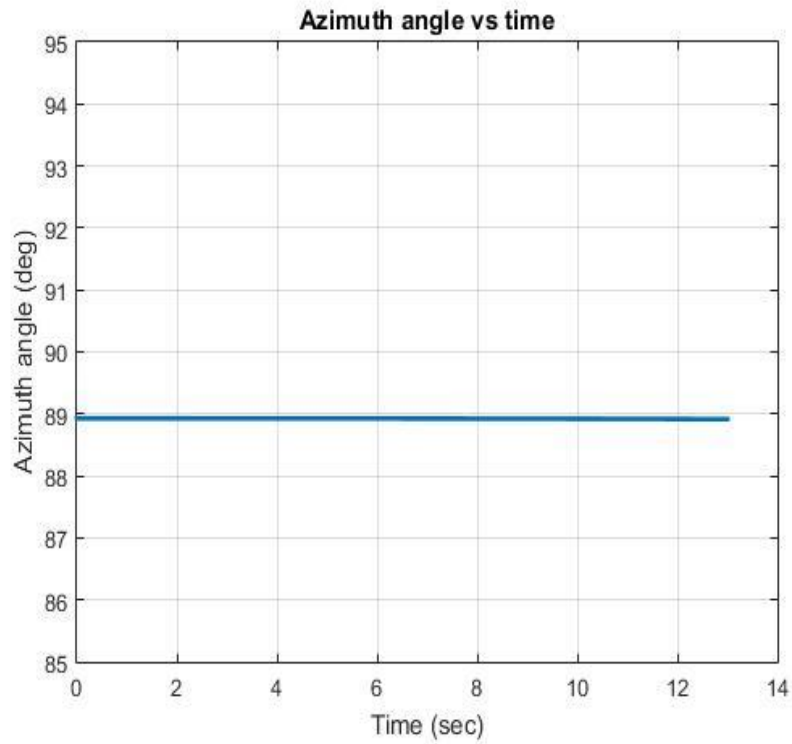


Fig. 18: Azimuth angle vs time

From above figure it is seen that azimuth angle is constant because of no roll condition during generation of nominal trajectory.

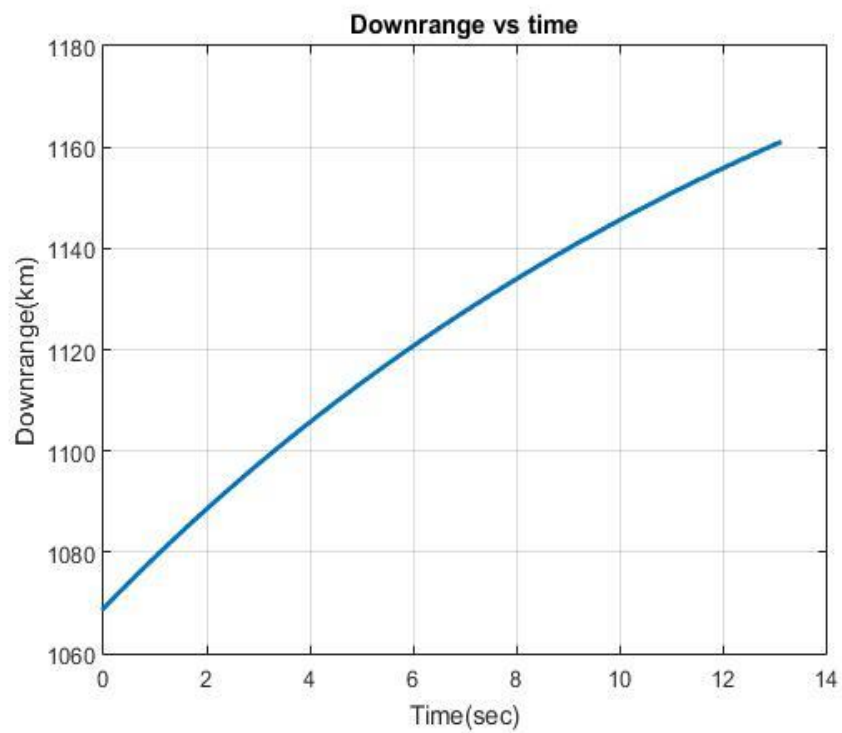


Fig. 19: Downrange vs time

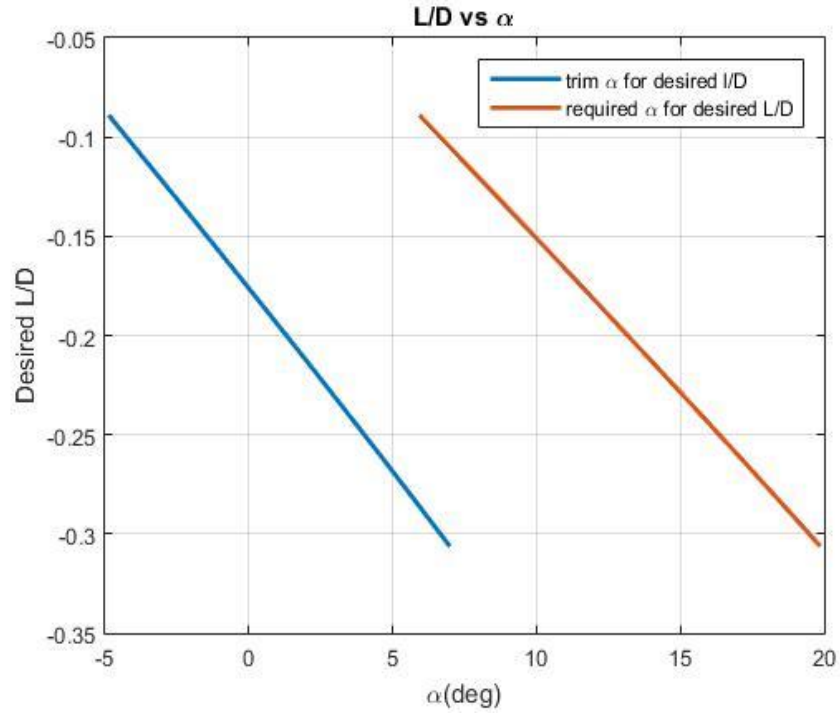


Fig. 20: Comparison of required angle of attack and trim angle of attack for desired L/D

The required angle of attack profile and trim angle of attack profile are different as seen in above plot. It shows that there is pitching moment.

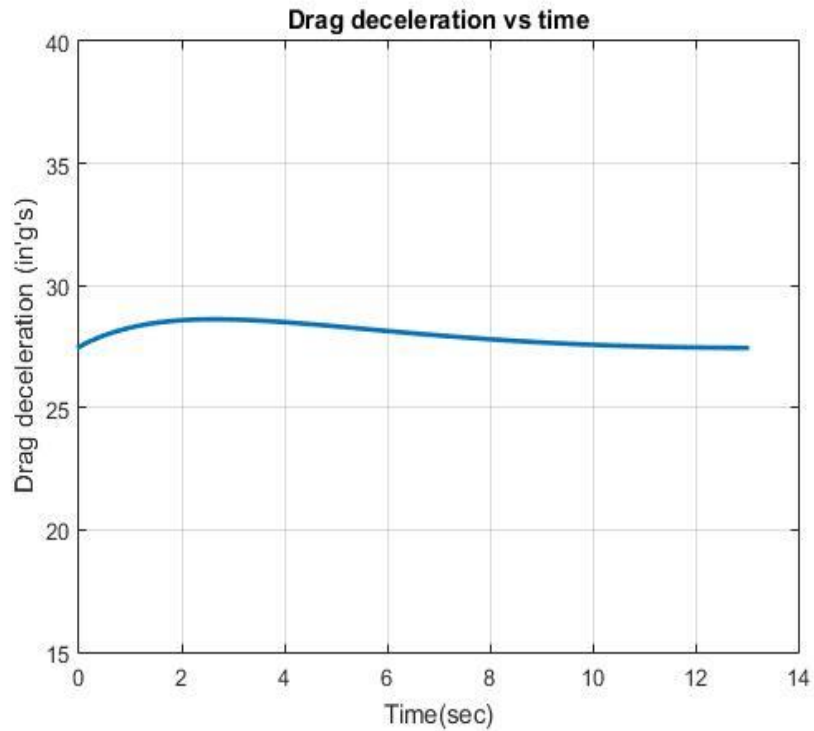


Fig. 21: Drag deceleration vs time for constant drag deceleration phase

Drag deceleration is constant and equal to 27.61g as shown in above figure, it constant drag deceleration law is followed.

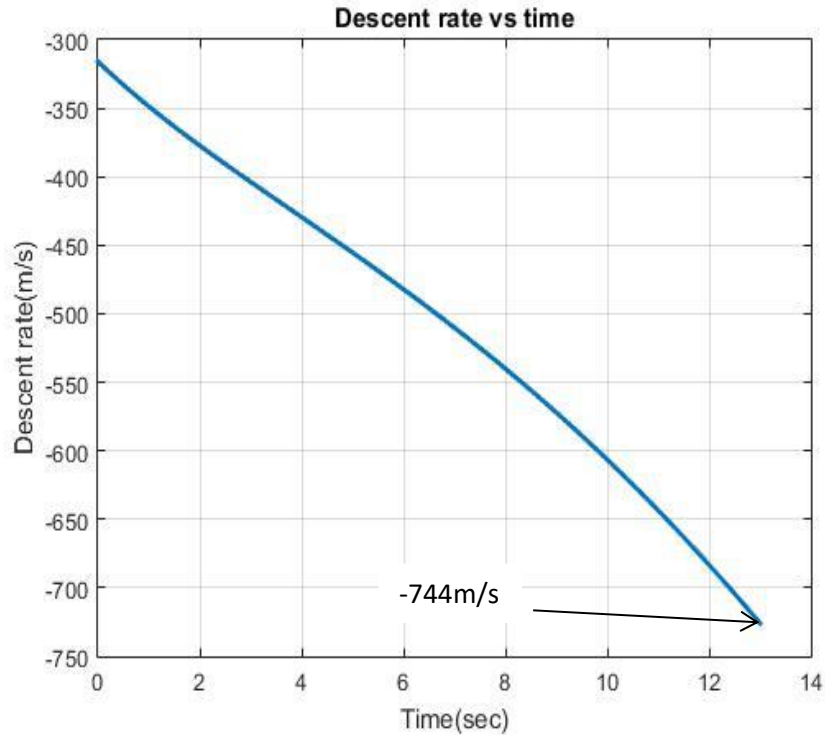


Fig. 22: Descent rate vs time

4.3.2.3. Implementation constant descent rate law

The L/D desired to follow this law is calculated using eqn. 19-20. The desired L/D ratio following constant drag deceleration law and constant descent rate law is restricted to ± 0.3 so as to have low lifting reentry of unmanned Apollo command module. The altitude at which desired L/D to follow constant drag deceleration law exceeds lower or upper range, the constant descent rate law is implemented.

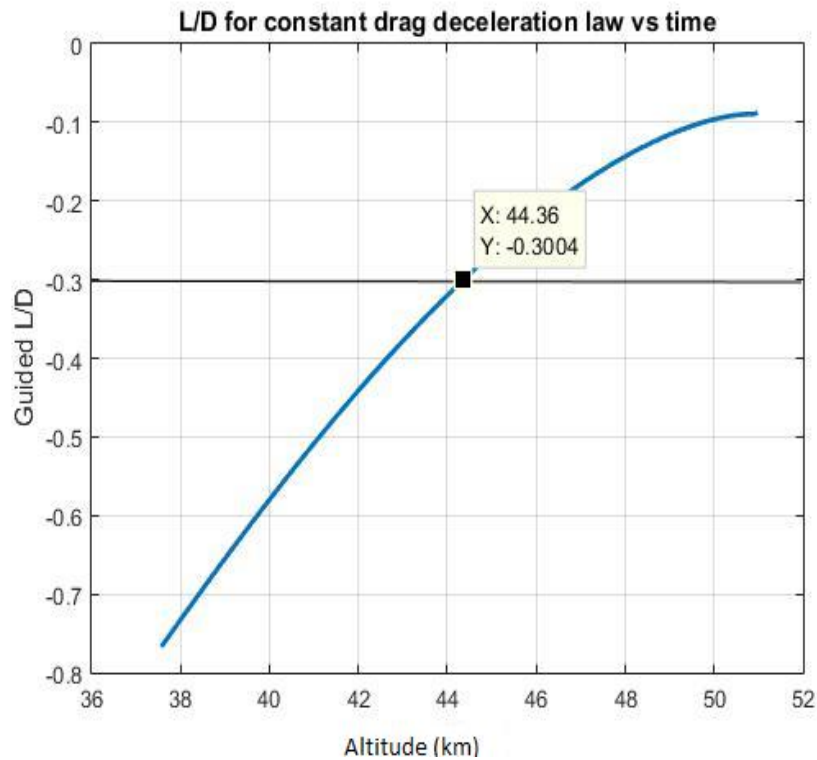


Fig. 23: Guided L/D for constant drag deceleration law vs altitude

Above figure shows that at altitude of 44.36 km, desired L/D for constant drag deceleration law exceeds the limit. So at the altitude of 44.36 km constant descent rate law is implemented. During this phase the angle of attack is calculated at every time step based on desired L/D using aerodynamic eqns.23-26 from section 2.3. The desired L/D to follow this law is calculated using eqn. 19-20. The constant descent rate value is the descent rate at the end of constant drag deceleration phase shown in fig.22. The simulation plots for this phase are given as:

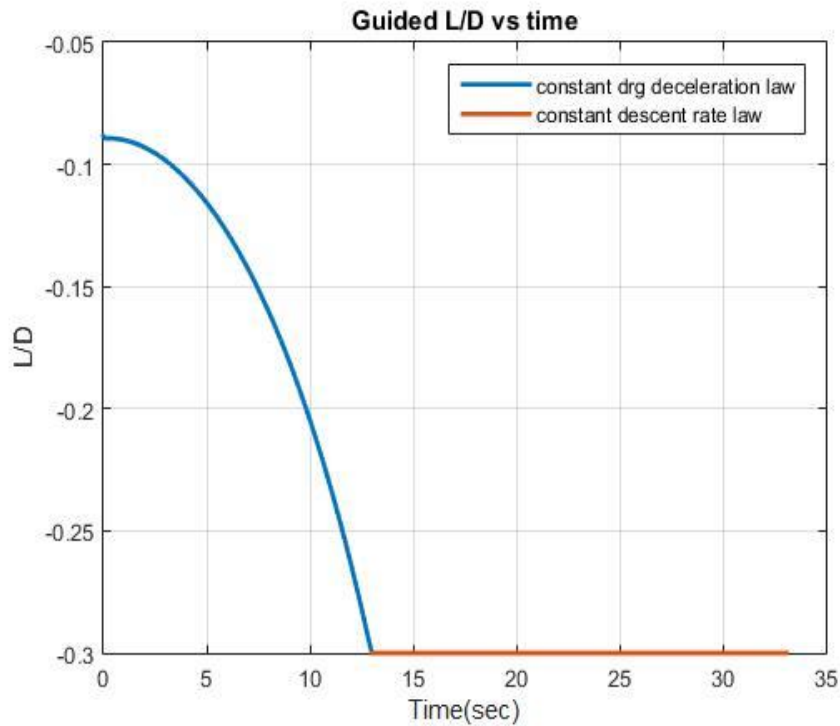


Fig. 24: Guided L/D for constant drag deceleration law and constant descent rate law

The desired L/D ratio required to follow constant descent rate phase is exceeding -0.3 limit. So it is restricted to lower limit of -0.3 as shown in above figure.

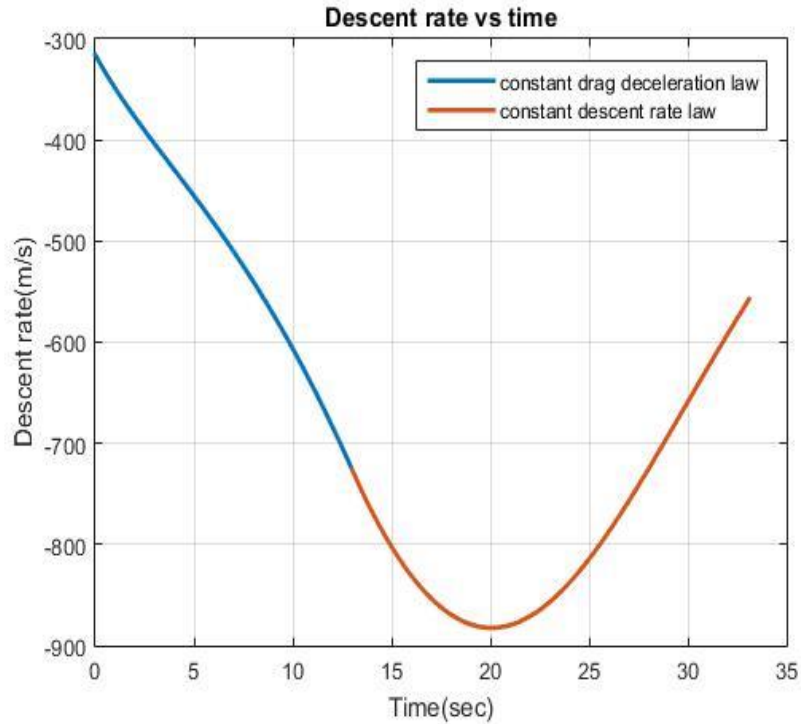


Fig. 25: Descent rate vs time for constant descent rate law

The descent rate is not maintained constant because the desired L/D to follow constant descent rate law is different than restricted L/D. This shows it is difficult to meet condition of desired L/D values and restricted L/D values.

During each time step magnitude of aerodynamic acceleration and descent rate is calculated for nominal trajectory as:

$$a_{\text{nom}} = \frac{|A|}{m}$$

$$= \frac{1}{m} \sqrt{L^2 + D^2}$$

$$a_{\text{nom}} = \frac{1}{2} \rho v^2 / B_c \sqrt{1 + \left(\frac{L}{D}\right)^2} \dots \dots \dots (21)$$

$\left(\frac{L}{D}\right)_v$: Desired L/D calculated using eqns. 16-17 and eqns. 19-20 for particular phase

$$\dot{h}_{\text{nom}} = v \sin \gamma \dots \dots \dots (22)$$

4.4.Closed-loop guidance

Closed-loop guidance is implemented to follow the nominal trajectory generated based on guidance laws as explained above.

4.4.1. Closed-loop guidance law

Closed-loop guidance is based on magnitude of total aerodynamic acceleration (acc) and descent rate ($\dot{h}$) as they are easily available from measurement.

$(L/D)_i$, $i = 1, 2$ is nominal L/D ratio for particular altitude i i.e. reference L/D and term.

$(acc - a_{nom})$ is the acceleration error term and term $(\dot{h} - \dot{h}_{nom})$ is the descent rate error term. Closed loop guidance law is based on these guidance errors.

For constant drag deceleration law phase:

L/D ratio in vertical plane is [12],

$$(L/D)_v = (L/D)_1 + C_1(acc - a_{nom}) + C_2(\dot{h} - \dot{h}_{nom}) \dots \dots \dots (23)$$

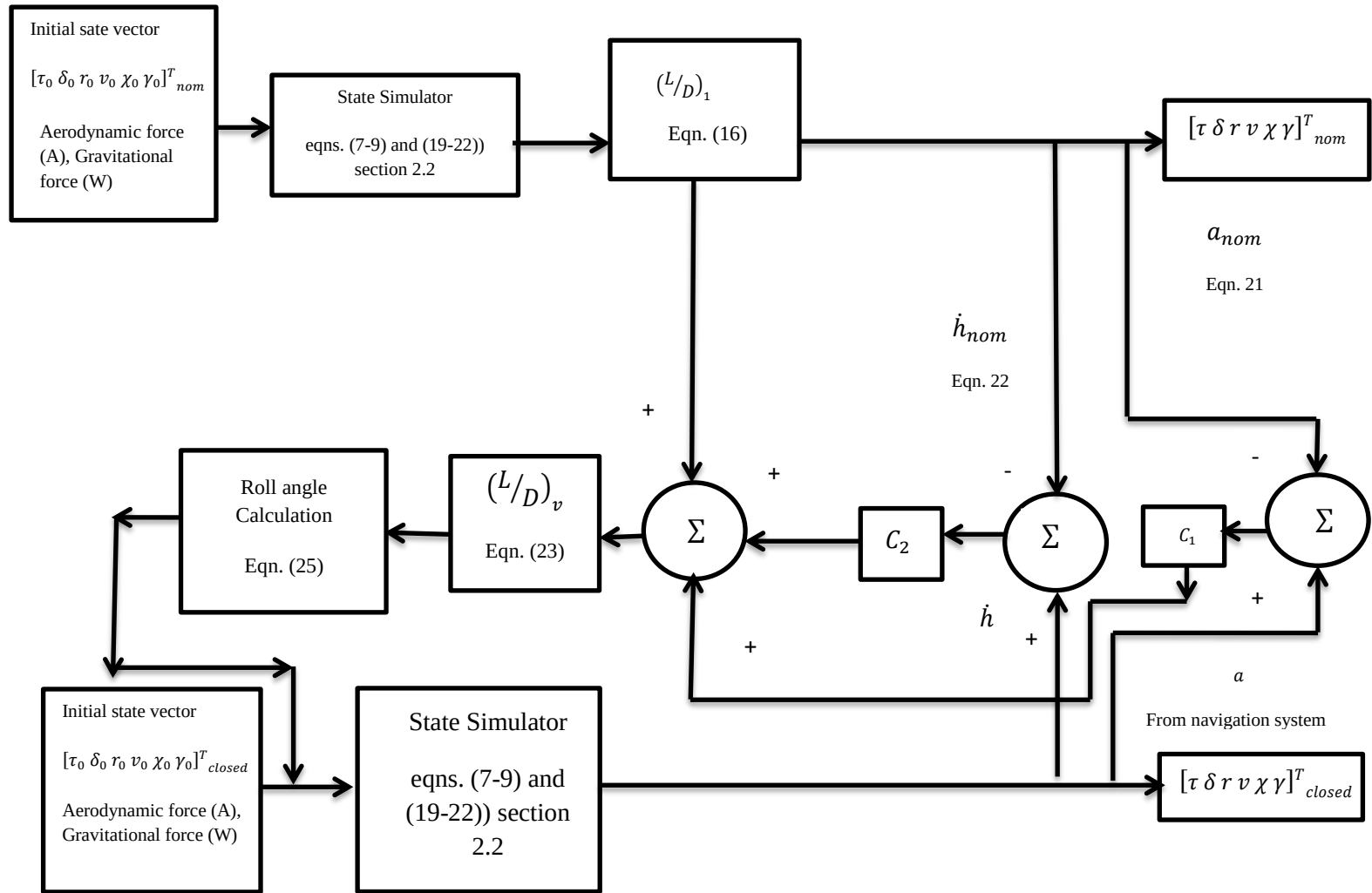


Fig.26: Closed-loop guidance law: constant drag deceleration law

For constant descent rate law phase:

L/D ratio in vertical plane is [12],

$$(L/D)_v = (L/D)_2 + C_2(\dot{h} - \dot{h}_{nom}) \dots \dots \dots (24)$$

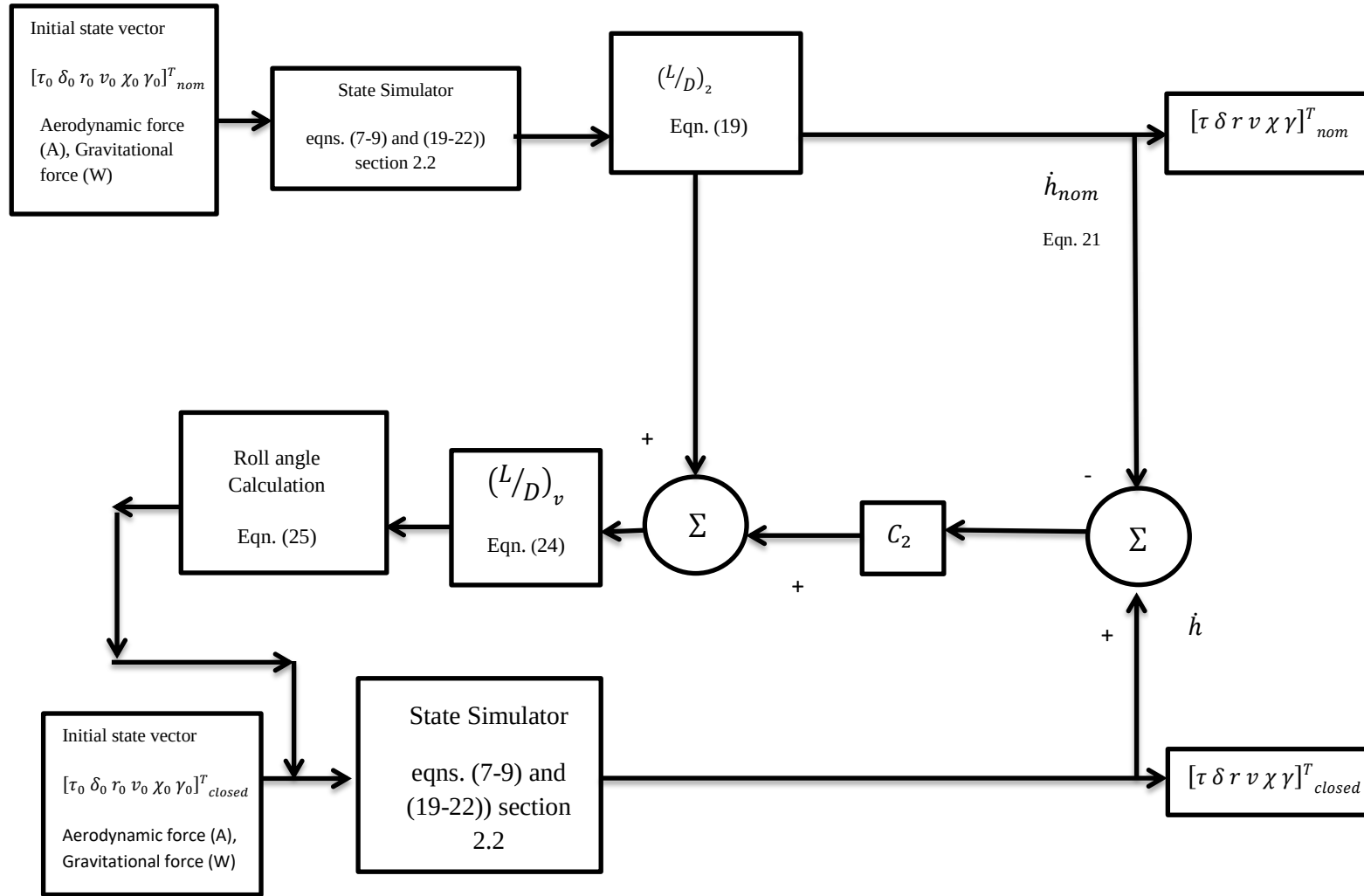


Fig.27: Closed-loop guidance law: constant descent rate law

$(L/D)_j : j = 1,2$ is obtained from nominal trajectory. a_{nom} and $\dot{h}_{nom}$ are obtained from nominal trajectory for corresponding altitude and given by eqn. (21-22). C_1 and C_2 are sensitivity gains corresponding to total aerodynamic acceleration error and descent rate error respectively.

4.4.2. Determination of C_1 and C_2 [21]

From eqns.(223-24), C_1 and C_2 are sensitivity gains to reduce acceleration error and descent rate error respectively. We can calculate these gains as,

$$C_1 = \frac{d(L/D)_j}{d(acc_{nom})} = \frac{(L/D)_{j,i+1} - (L/D)_{j,i}}{acc_{nom,i+1} - acc_{nom,i}} \dots \dots j = 1$$

$$C_2 = \frac{d(L/D)_j}{d(\dot{h}_{nom})} = \frac{(L/D)_{j,i+1} - (L/D)_{j,i}}{\dot{h}_{nom,i+1} - \dot{h}_{nom,i}} \dots j = 1,2$$

4.4.3. Roll angle modulation

In order to follow nominal trajectory, direction of the lift vector is changed by rolling the vehicle. The roll angle is calculated at every time step as follows:

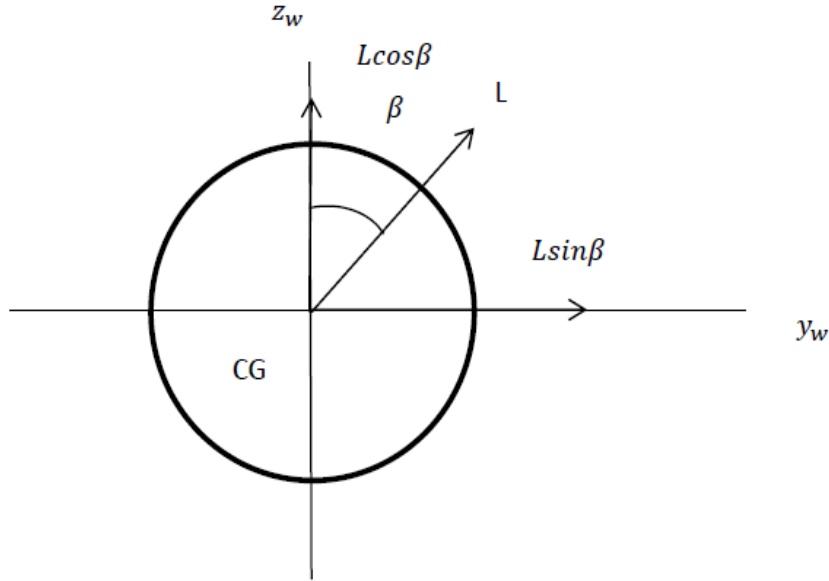


Fig.28: Lift force vector components

Vertical component of L/D is given as:

$$(L/D)_v = (L/D)_{generated} \cos \beta \dots \dots \dots \text{From fig.15}$$

The roll angle is given as,

$$\beta_c = \cos^{-1} \left[\frac{(L/D)_v}{(L/D)_{generated}} \right] \dots \dots \dots (25)$$

For constant drag deceleration law phase $(L/D)_v$ is given by eqn.24 and for constant descent rate law phase $(L/D)_v$ is given by eqn. 26 at every time step.

$(L/D)_{generated}$ is generated due to angle of attack and it is calculated using eqn. (23-26) of section 2.3 as:

$$(L/D)_{generated} = \frac{C_L}{C_D} \dots \dots \dots (26)$$

4.5. Rigid body dynamics

The Apollo command module is considered as rigid body and rotational dynamics of reentry is modelled by pitch angle (ϕ), yaw angle (ψ) and roll angle (ξ) and corresponding pitch rate ($\dot{\phi}$), yaw rate ($\dot{\psi}$) and roll rate ($\dot{\xi}$).

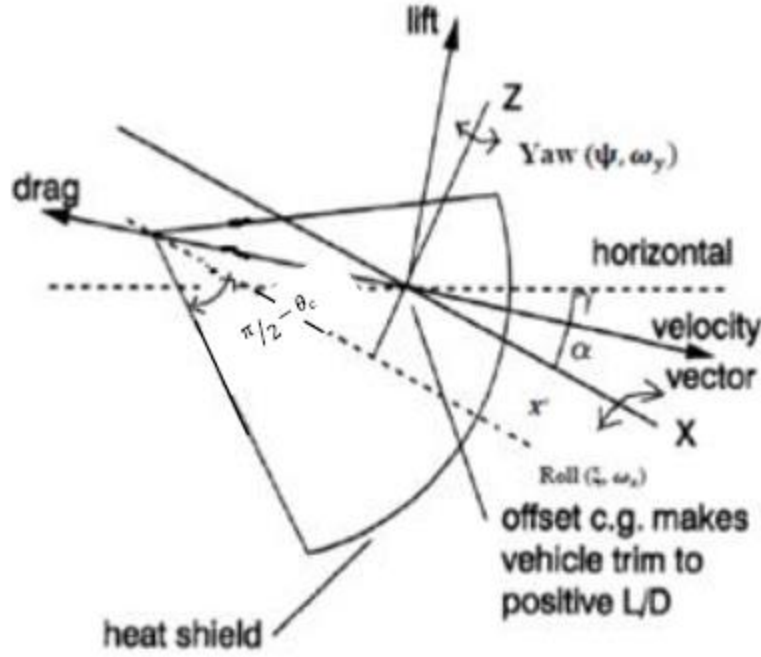


Fig.29: Rotational Dynamics of Apollo

Governing equations are as follows [13]:

For symmetry of vehicle about XZ plane [9],

$$\text{Roll rate } (\dot{\xi}): I_x \dot{\xi} - (I_y - I_z) \omega_y \omega_z - I_{xz} (\omega_x \omega_y + \dot{\omega}_z) = L \dots \dots (27)$$

$$\text{Pitch rate } (\dot{\phi}): I_y \dot{\phi} - (I_x - I_z) \omega_x \omega_z - I_{xz} (\omega_x^2 - \omega_z^2) = M \dots \dots (28)$$

$$\text{Yaw rate } (\dot{\psi}): I_z \dot{\psi} - (I_x - I_y) \omega_y \omega_z + I_{xz} (\omega_z \omega_y - \dot{\omega}_x) = N \dots \dots (29)$$

L, M, N are roll, pitch and yaw moment.

I_x, I_y, I_z are moment of inertia about X, Y, Z axes respectively.

I_{xz} is product of inertia about X and Z axes.

The numerical integration of above equations gives actual rigid body dynamics of Apollo command module. Based on actual dynamics and reference roll command obtained from closed-loop guidance, control system is designed.

CHAPTER 5

X-33 Vehicle Reentry Problems

In previous chapters, unmanned Apollo command module is considered as reentry vehicle and reference trajectory is generated based on guidance laws. Closed-loop guidance is implemented to closely follow the reference trajectory. This chapter deals with X-33, winged rocket plane like reentry vehicle. In this chapter; the reentry problem is solved as optimization problem.

5.1. Equations of Motion for Reentry:

X-33 reentry vehicle has winged rocket plane structure as shown in figure 1. It was world's first single-stage to orbit (SSTO) spacecraft-the Lockheed –Martin Venture star. The purpose was to have safer and inexpensive launch as compared to Space Shuttle program. The program was cancelled in 2001 because of technical instabilities in fuel tanks.

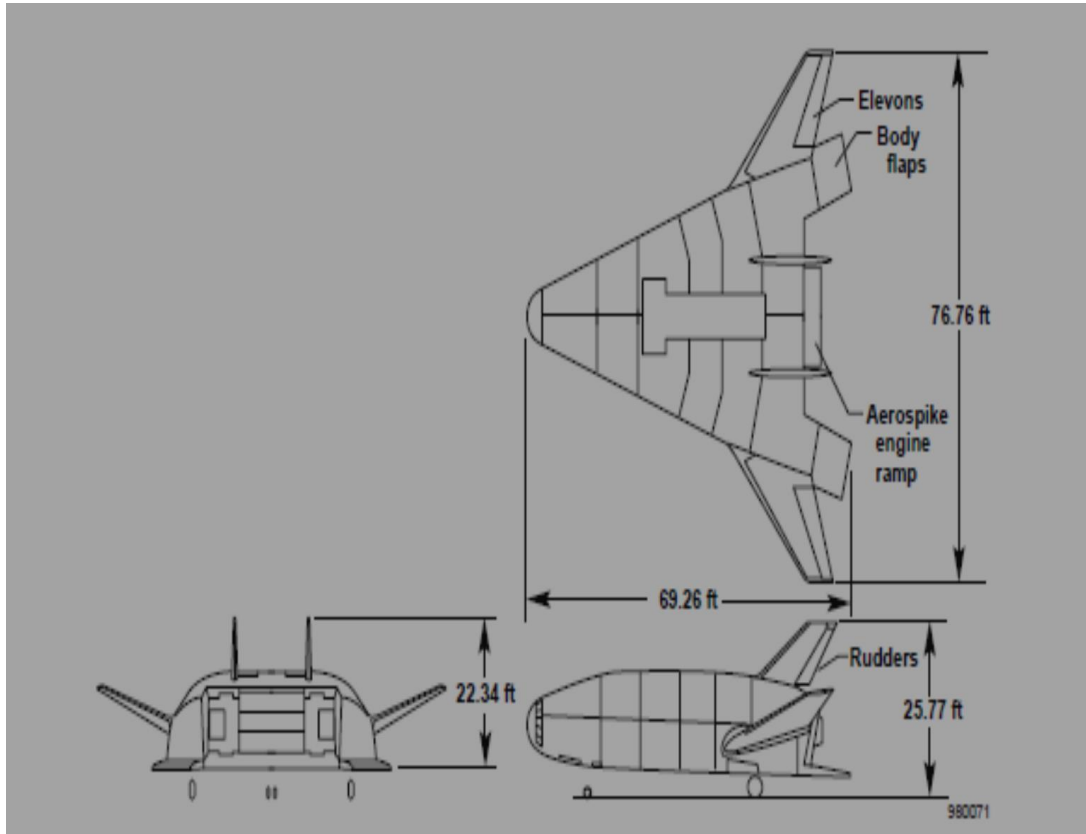


Fig.1: Three-view of X-33 Vehicle [1]

In order to describe complete state of the reentry vehicle, 6 state parameters are required. The state vector is given as [11]:

$$X = [r \quad \mu \quad \lambda \quad v \quad \Upsilon \quad \xi]^T$$

r : Radial distance from the earth's center ($r = R_e + h$)

μ : Geocentric longitude

λ : Geocentric latitude

v : Vehicle Speed

Υ : Flight path angle

ξ : Heading angle measured from east

Reentry vehicle dynamics is governed by following nonlinear differential equations [11].

Radial distance:

$$\dot{r} = v \sin \Upsilon \dots \dots \dots (1)$$

Geocentric longitude:

$$\dot{\mu} = \frac{v \cos \Upsilon \cos \xi}{r \cos \lambda} \dots \dots \dots (2)$$

Geocentric latitude:

$$\dot{\lambda} = \frac{v \cos \Upsilon \sin \xi}{r} \dots \dots \dots (3)$$

Vehicle Speed relative to rotating earth:

$$\dot{v} = -\frac{D}{m} - g \sin \Upsilon + \Omega^2 r \cos \lambda [\sin \Upsilon \cos \lambda - \cos \Upsilon \sin \lambda \sin \xi] \dots \dots \dots (4)$$

where

$$g = \frac{GM}{r^2}, \Omega: \text{Earth's angular velocity}$$

Flight path angle:

$$\dot{\Upsilon} = \frac{L \cos \sigma}{mv} + \left(\frac{v}{r} - \frac{g}{v} \right) \cos \Upsilon + 2\Omega \cos \lambda \cos \xi + \frac{\Omega^2}{r} \cos \lambda [\cos \Upsilon \cos \lambda + \sin \Upsilon \sin \lambda \sin \xi] \dots \dots \dots (5)$$

σ : Bank angle

Heading angle measured from east:

$$\dot{\xi} = \frac{L \sin \sigma}{mv \cos \Upsilon} - \frac{v}{r} \cos \Upsilon \cos \xi \tan \lambda + 2\Omega [\tan \Upsilon \cos \lambda \sin \xi - \sin \lambda] - \frac{\Omega^2 r}{v \cos \Upsilon} \sin \lambda \cos \lambda \cos \xi \dots \dots \dots (6)$$

Lift and drag forces are calculated at every time step as follows:

$$L = \frac{\rho(h) v^2 C_L(\alpha, M) S_{ref}}{2}$$

$$D = \frac{\rho(h) v^2 C_D(\alpha, M) S_{ref}}{2}$$

Atmospheric density ($\rho(h)$) is function of altitude so at each altitude (h) density is interpolated with the help of standard atmosphere data. Coefficient of lift (C_L) and drag (C_D) are functions of angle of attack (α) and Mach no. (M). Aerodynamic coefficients calculated as [4]:

$$C_L = 0.1577 - 0.0005225 \alpha^2 + 0.03506 \alpha - 0.04857M$$

$$C_D = 0.2204 + 0.0001432 \alpha^2 + 0.00558 \alpha - 0.01048M$$

Mach no. is defined as a ratio of velocity of body in a medium to velocity of sound in that medium. Velocity of vehicle is calculated at every altitude and corresponding velocity of sound(C) is calculated as:

$$C = \sqrt{\gamma RT(h)}$$

where

γ -adibatic constant of medium (for air $\gamma = 1.4$)

R-specific gas constant ($R_{\text{air}} = 287 \text{ J/kgK}$)

T-temperature of medium (K)

Temperature of medium is dependent on altitude and it is interpolated at every altitude with the help of standard atmospheric data.

5.2. Parameters for Simulation:

Data required for simulation is taken from [11]. The reentry vehicle considered is X-33 model as shown below.

No.	Data	Value
1.	mass	35828.03 kg
2.	S_{ref}	149.58 m ²
3.	R_e	6371000 m

Table 1: X-33 parameters

• Initial Parameters :

No.	Parameter	Values
1.	Altitude	51000 m
2.	Geocentric longitude	-85°
3.	Geocentric latitude	30°
4.	Velocity	2600 m/s
5.	Flight path angle	-1.3°
6.	Heading angle	0°

Table 2: Initial parameters

5.3. Simulation Results

The governing eqns. (1-6) are numerically integrated using Runge-Kutta (RK4) method on MATLAB R2014b with step size of 0.05sec. During computation, earth's rotational effect is not considered as it is not affecting much. Following are plots for variation of 6 reentry parameters with respect to time for different angle of attack and for different bank angle which are control parameters of reentry dynamics.

5.3.1. Variation of angle of attack (α)

With bank angle $\sigma = 0^\circ$ angle of attack is varied as $10^\circ, 20^\circ, 30^\circ, 40^\circ$ and its effect on trajectory is simulated on MATLAB R2014b for a specific period (800sec). Plots for variation of all states are given below:

A. Variation of altitude with time

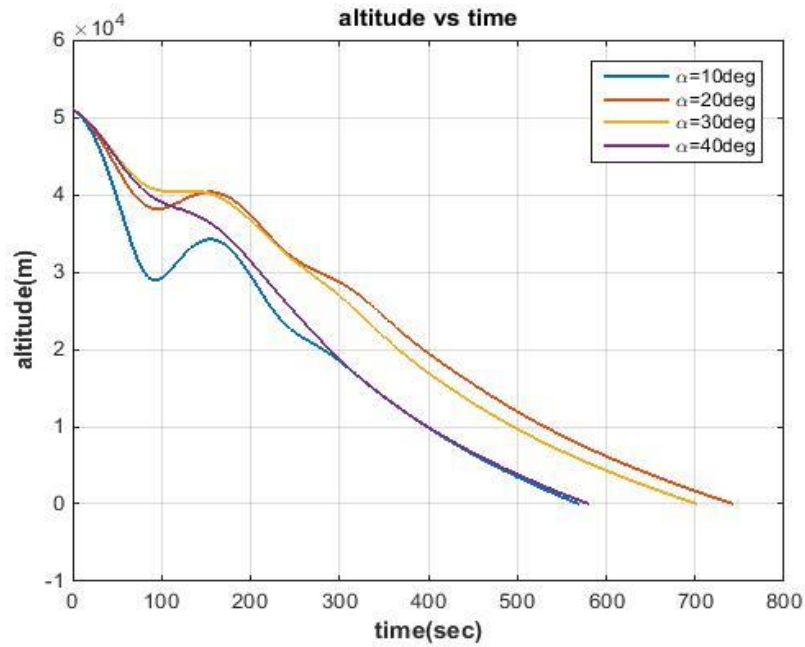


Fig.2: Altitude vs time

During simulation it is found that altitude decreases and then increases for some period because of dominance of lift force.

B. Variation of velocity with time

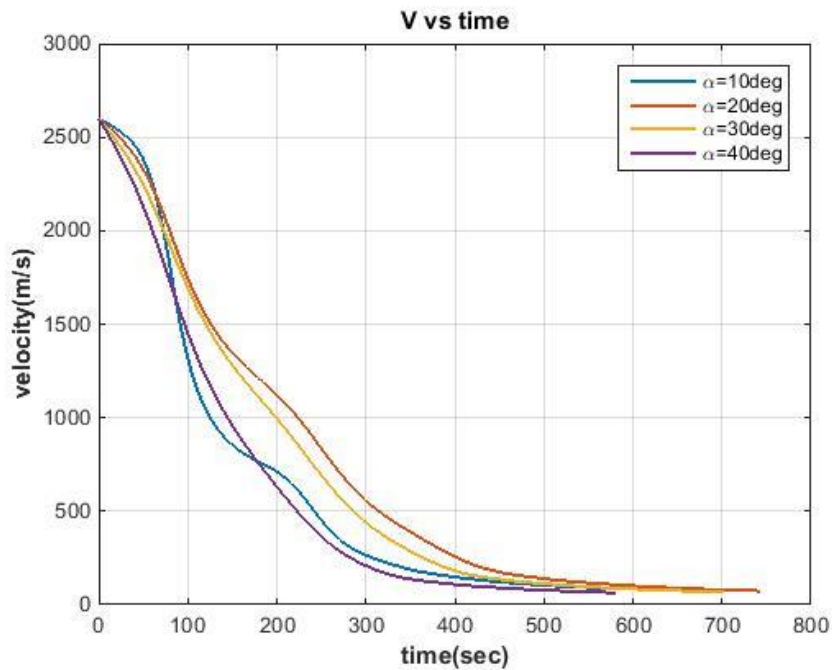


Fig.3: Velocity vs time plot

Velocity of vehicle gradually decreases with time depending upon angle of attack value and at the end there is no significant effect of α on range and flight duration.

C. Variation of flight path angle with time

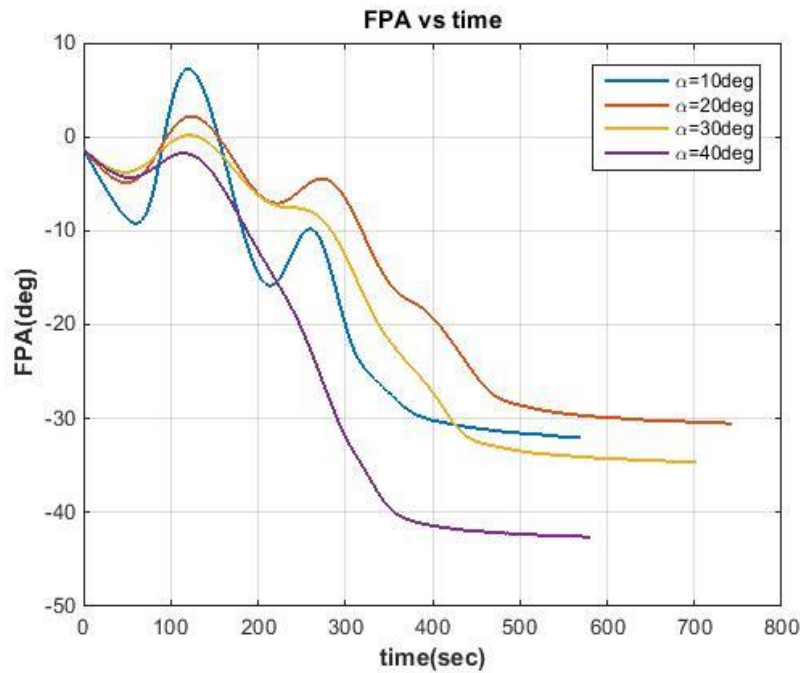


Fig.4: Flight path angle vs time

Flight path angle varies in the range of 10° to -50° and it stabilizes after some time t depending on the angle of attack value. During stabilization, gliding condition is achieved to have constant flight path angle. Gliding flight is flight with power off of vehicle where vehicle is acted upon by drag force, lift force and weight force only. Equilibrium glide is achieved for X-33 vehicle at flight path angle and it is given as:

$$\tan(\gamma) = \frac{1}{(L/D)}$$

In the following table values in column 4 and 6 are nearly equal so it shows that gliding condition is satisfied.

No.	α ($^\circ$)	L/D	$1/(L/D)$	γ ($^\circ$) (ccw)	$\tan(\gamma)$
1.	10	1.54	0.64	32	0.6249
2.	20	1.62526	0.6153	30	0.5898
3.	30	1.4182	0.7051	34.6	0.6899
4	40	1.0657	0.9381	42.5996	0.919

Table 3: Gliding condition for various α

D. Variation of heading angle with time

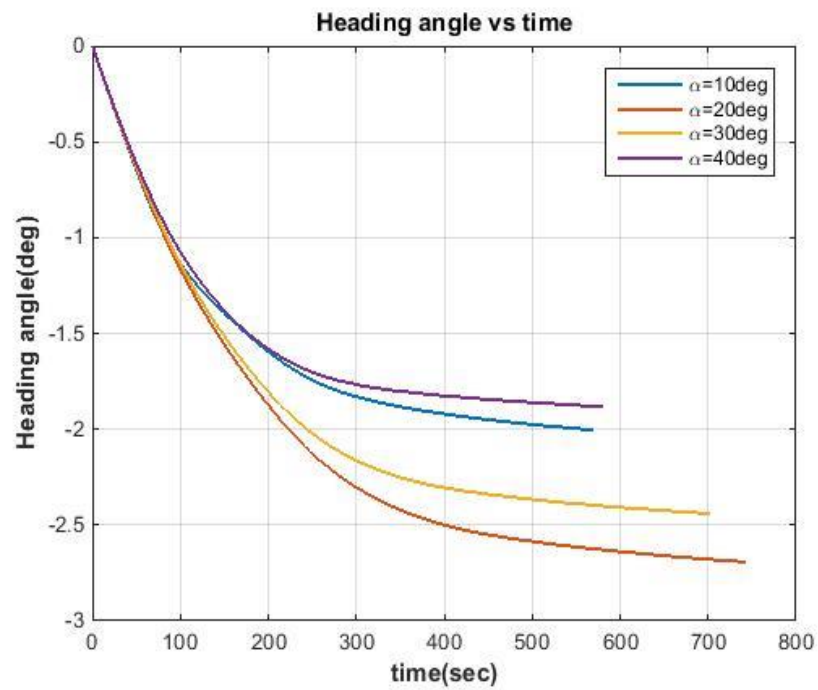


Fig.5: Heading angle vs time

Bank angle is zero deg for simulation and heading angle varies from 0° to -3° depending on angle of attack.

E. Flight trajectory

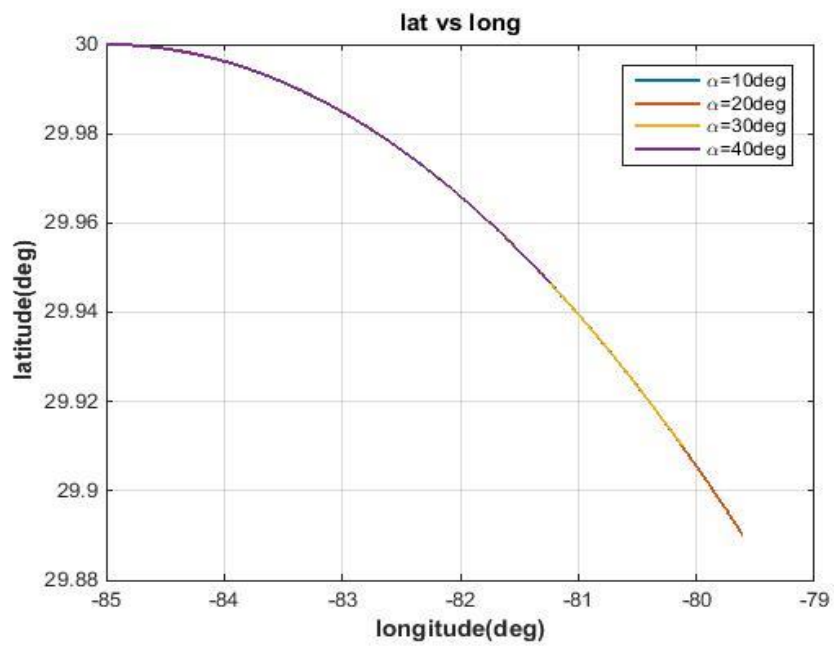


Fig.6: Latitude vs longitude plot

Latitude-longitude plot shows that latitude doesn't change much which is not a case of longitude. This is because, the vehicle initial velocity is due east with azimuth angle as 90° and there is no lateral force acting on it. So the angle of attack changes longitude and it doesn't produce lateral force. So this can be considered to have specific landing with adjustment of longitude according to requirement.

5.4. Optimization Problem

5.4.1. Optimization of single angle of attack for whole flight to land the vehicle at particular location

Flight trajectory varies according angle of attack as ensured from simulation results .From plot of geocentric latitude vs. geocentric longitude, with change in angle of attack there is no much change in latitude because the vehicle did not roll and azimuth angle is zero deg.

5.4.1.1. Problem Definition

For specific landing at particular longitude angle of attack can be optimized to have minimized error objective function. This problem considers only one angle of attack for whole flight say α_{req} . This angle of attack is optimized angle of attack using optimization techniques to have desired longitude at end. Optimization technique applied to get α_{req} is steepest descent.

5.4.1.2. Objective Function

Objective function is error for longitude attained at end of flight given as,

$$J = (\mu - \mu_{desired})^2 \times 10^5$$

10^5 factor is multiplied because order of longitude error is too small.

Desired longitude considered is -80° (-1.39626 rad) at the end of flight. From MATLAB R2014b solver, range of angle of attack to achieve desired longitude is obtained as given below:

No.	Angle of attack (α)	Longitude (μ)
1.	28°	-1.3952 rad
2.	30°	-1.3983 rad

Table 4: Details of optimization

5.4.1.3. Optimization Technique

Steepest Descent Method:

It is an iterative method of finding minimum of function when it is impossible to find it analytically. It is gradient method in which step size is chosen to achieve maximum amount of decrease in objective function at each step.

Quadratic Steepest Descent Method:

The objective function J is quadratic function of angle of attack α as,

$$J = a_0 + a_1\alpha + a_2\alpha^2 \dots (7)$$

For optimum angle of attack α_{opt} ,

$$\frac{\partial J}{\partial \alpha} = 0$$

$$a_1 + 2a_2\alpha_{opt} = 0 \dots \dots \text{from eqn. (7)}$$

$$\alpha_{opt} = \frac{-a_1}{2a_2}$$

Optimal (minima) objective function is given as,

$$J(\alpha_{opt}) = a_0 + a_1\alpha_{opt} + a_2\alpha_{opt}^2 = 0$$

Taylor's series expansion of objective function is,

$$J(\alpha) = J(\alpha_{opt}) + \frac{\partial J}{\partial \alpha} \delta \alpha \dots \dots \dots (8)$$

At $\alpha = \alpha_{opt} + \delta \alpha$

$$J(\alpha) = 0 + \frac{\partial J}{\partial \alpha} \delta \alpha$$

$$= g \delta \alpha$$

$$\delta \alpha = \frac{J(\alpha)}{\left(\frac{\partial J}{\partial \alpha}\right)} = \frac{J(\alpha)}{g}$$

Optimal angle of attack is given by,

$$\alpha_{opt} = \alpha - \delta \alpha = \alpha - \frac{J(\alpha)}{g} \dots \dots \dots (9)$$

Steepest Descent approach with simple linear interpolation:

The desired longitude is $l = -80^\circ$ i.e. 1.39626 rad. It lies in between the longitudes $l_1 = -1.3952$ rad and $l_2 = -1.3983$ rad corresponding to angle of attacks $\alpha_1 = 28^\circ$ and $\alpha_2 = 30^\circ$ respectively.

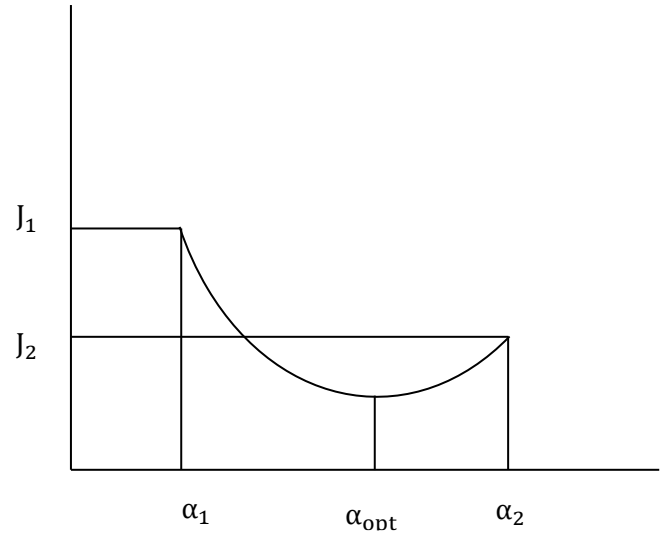


Fig. 7: Quadratic steepest descent method

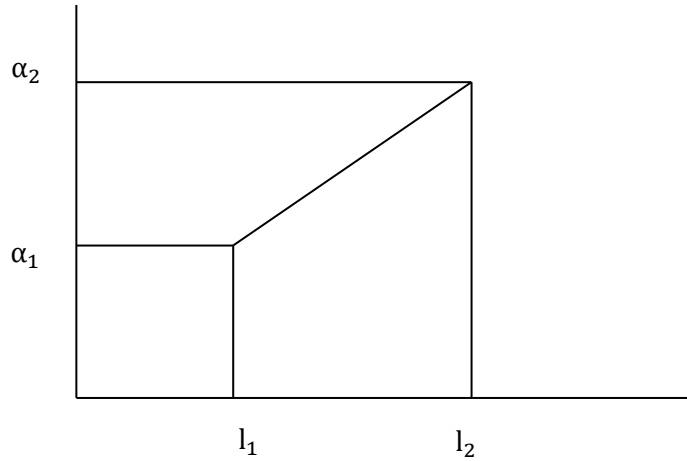


Fig. 8: Steepest descent method with simple linear interpolation

The angle of attack corresponding to desired longitude is given as,

$$\alpha = ml + c \dots \dots \dots (10)$$

From above figure,

$$\alpha_1 = ml_1 + c \dots \dots (11)$$

$$\alpha_2 = ml_2 + c \dots \dots (12)$$

Slope of the line m is given as,

$$m = \frac{\alpha_2 - \alpha_1}{l_2 - l_1} \dots \dots \dots (13)$$

From eqn. 11 and 13,

$$\begin{aligned} c &= \alpha_1 - \frac{\alpha_2 - \alpha_1}{l_2 - l_1} l_1 \\ &= \frac{\alpha_1 l_2 - \alpha_2 l_1}{l_2 - l_1} \end{aligned}$$

So angle of attack for desired longitude is given as,

$$\alpha = \left(\frac{\alpha_2 - \alpha_1}{l_2 - l_1} \right) l + \frac{\alpha_1 l_2 - \alpha_2 l_1}{l_2 - l_1} \dots \dots \dots (14)$$

Algorithm for steepest descent method:

- A. Initial guess α_0 ($\alpha_0 = 28^\circ$)
- B. Calculate minimum step size and gradient (g)
- C. Angle of attack at every step is calculated as,

$$\alpha_{i+1} = \alpha_i + (\text{step size})_{\min} \times g$$

D. Iterate till $J \leq 10^{-3} \text{rad}^2$

Minimum step size calculation: Minimum step size is calculated by curve fitting between objective function J and step size given by following equation,

$$J = a_0 + a_1(\text{step size}) + a_2(\text{step size})^2$$

Step size is taken as 0.5/g, 1/g and 2/g and corresponding objective functions are calculated. By solving following system of equations coefficients a_0, a_1, a_2 are calculated.

$$J_1 = [a_0 + a_1(\text{step size}) + a_2(\text{step size})^2]_1$$

$$J_2 = [a_0 + a_1(\text{step size}) + a_2(\text{step size})^2]_2$$

$$J_3 = [a_0 + a_1(\text{step size}) + a_2(\text{step size})^2]_3$$

$$A = \begin{bmatrix} 1 & (\text{step size})_1 & (\text{step size})_1^2 \\ 1 & (\text{step size})_2 & (\text{step size})_2^2 \\ 1 & (\text{step size})_3 & (\text{step size})_3^2 \end{bmatrix}$$

$$B = \begin{bmatrix} J_1 \\ J_2 \\ J_3 \end{bmatrix}, x = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix}$$

By solving equation $Ax=B$, a_0, a_1, a_2 can be calculated. Minimum step size is obtained by differentiating J with respect to step size and then equating to 0.

$$(\text{step size})_{\min} = -\frac{a_1}{2a_2}$$

Gradient (g) calculation: Gradient is calculated at each time step using objective function as,

$$g = \frac{dJ}{d\alpha} = \frac{J_2 - J_1}{\alpha_2 - \alpha_1}$$

5.4.1.4. Results

As explained above steepest descent method is applied using MATLAB R2014b and the optimized angle of attack obtained is 28.74° .

No.	α ($^\circ$)	μ (rad)	J (rad^2)
1.	28	-1.3952	0.1216
2.	28.74	-1.3963	4.63×10^{-7}
3.	30	-1.3983	0.4059

Table 5: Optimized α details

The desired longitude lies in between the longitude obtained for angle of attacks 28° and 30° from reentry dynamics at the end of flight. So the corresponding should in between 28° and 30° . Above table shows that optimised angle of attack computed using steepest descent method lies in between angle of attacks 28° and 30° .

5.4.2. Optimization of angle of attack using *fmincon* to maximize range

MATLAB R2014b has in-built optimizer *fmincon*. It finds minimum of constrained nonlinear multivariable function based on Sequential Quadratic Programming (SQP) algorithm. *fmincon* solves the problem in form of:

Minimum $f(x)$ subject to,

$$\begin{cases} Ax \leq B \\ A_{eq}x = B_{eq} \end{cases} \dots\dots\dots \text{Linear constraints}$$

$$\begin{cases} c(x) \leq 0 \\ c_{eq}(x) = 0 \end{cases} \dots\dots\dots \text{Non linear constraints}$$

$$LB \leq x \leq UB \dots\dots\dots \text{Bounding of variable}$$

Common Syntax:

$$x = \text{fmincon}(\text{fun}, x_0, A, B, A_{eq}, B_{eq}, LB, UB, \text{noncon})$$

where

x_0 = Initial guess for minimum (initial condition of α)

noncon= M-file function call for nonlinear constraints

5.4.2.1. Problem Definition

Angle of attack is optimized to have maximum range attained at end of flight. Bank angle is kept constant as zero deg. Reentry equations considered are same as in previous section. Simulation data used is also same as above sections.

5.4.2.2. Objective Function

Objective function is to maximize range that is longitude attained at end of flight. It is given as,

$$J = -|\mu|$$

5.4.2.3. Nonlinear Constraints

There are no linear constraints in this problem. But reentry parameters are constrained by nonlinear inequality constraints, as reentry parameters velocity, altitude, latitude, flight path angle and heading angle are functions of angle of attack. Nonlinear inequality constraints for problem are on state parameters $[r - r_e \quad \lambda \quad v \quad \gamma \quad \xi]^T$ with respect to desired values of the same $[h_c \quad \lambda_c \quad v_c \quad \gamma_c \quad \xi_c]$. At the end of flight non-linear constraints to be satisfied are:

$$\text{Altitude error: } [(r - R_e) - h_c] \times 10^{-2} \leq 0.05 \text{ m}$$

$$\text{Latitude error: } \lambda - \lambda_c \leq 0.001 \text{ rad}$$

$$\text{Velocity error: } (v - v_c) \times 10^{-1} \leq 0.05 \text{ m/s}$$

$$\text{Flight path angle error: } \gamma - \gamma_c \leq 0.05 \text{ rad}$$

$$\text{Heading angle error: } \xi - \xi_c \leq 0.05 \text{ rad}$$

where

All values below are states obtained by numerical integration of governing equations (1-6) with zero deg bank angle and 30° angle of attack at the end of the flight.

h_c = Constrained altitude (9.21m)

λ_c = Constrained latitude (0.5220 rad)

v_c = Constrained velocity (67.3671 m/s)

γ_c = Constrained flight path angle (-0.6040 rad)

ξ_c = Constrained heading angle (-0.0426 rad)

5.4.2.4. Initialization & Boundary Conditions

Boundary condition for angle of attack and flight period is:

$$0^\circ \leq \alpha \leq 40^\circ$$

Initial value for angle of attack is considered as 30° .

5.4.2.5. Optimization Technique

Problem is nonlinear constrained minimization problem, so MATLAB R2014b inbuilt solver *fmincon* is used to have optimized alpha value α_{opt} to maximize range as J_{opt} . Tolerances for function, constraints, x value are modified from default values by using *options* command in MATLAB R2014b as,

- A. Tolx= 10^{-12}
- B. Tolcon=0.05
- C. Tolfun= 10^{-7}

5.4.2.6. Results

Optimizer found out local minima with satisfaction of constraints by showing message. It gave following results:

- A. $\alpha_{opt} = 30.0184^\circ$
- B. $J_{opt} = -1.3983 \text{ rad} = 80.12^\circ$

Simulation Plots for α_{opt} are given as:

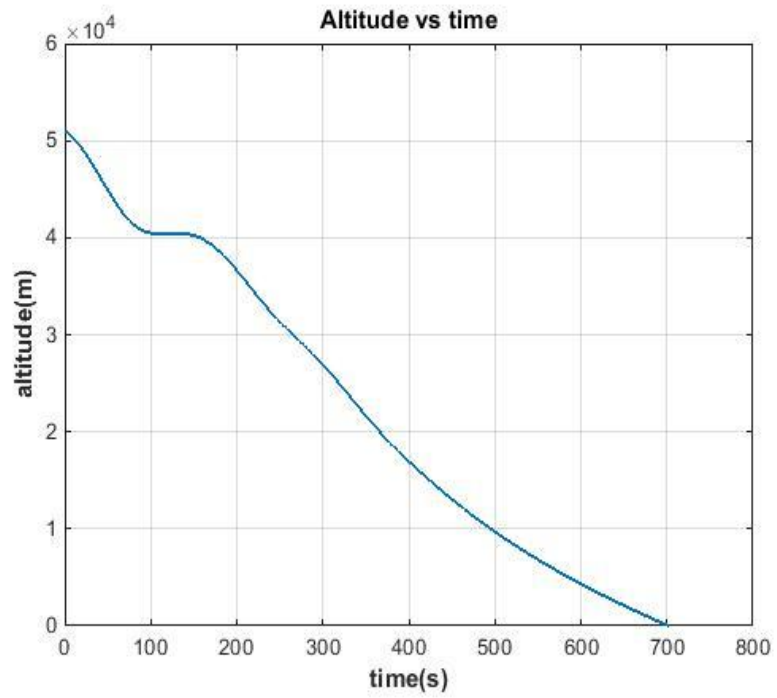


Fig.9: Altitude vs time

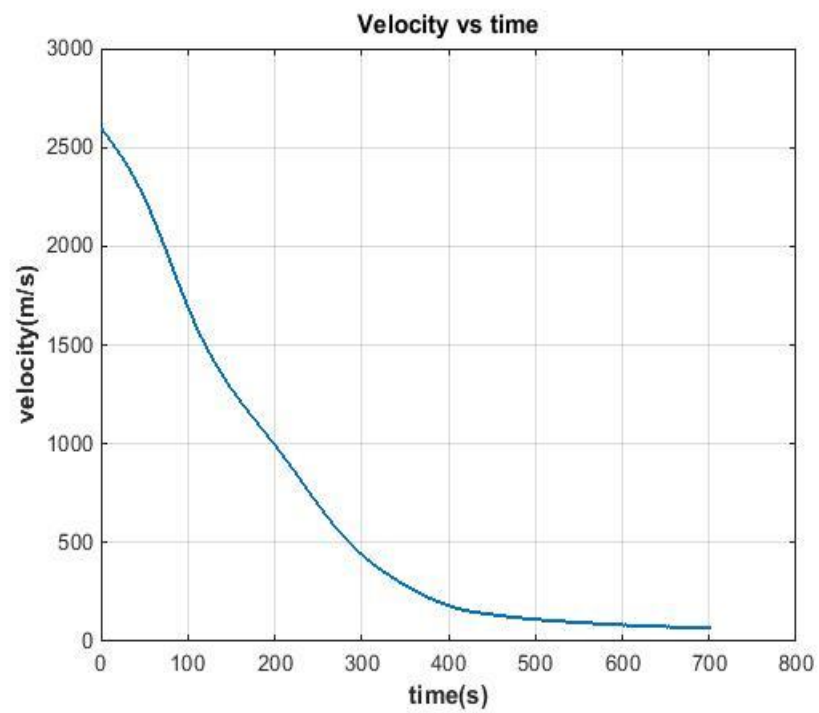


Fig.10: Velocity vs time

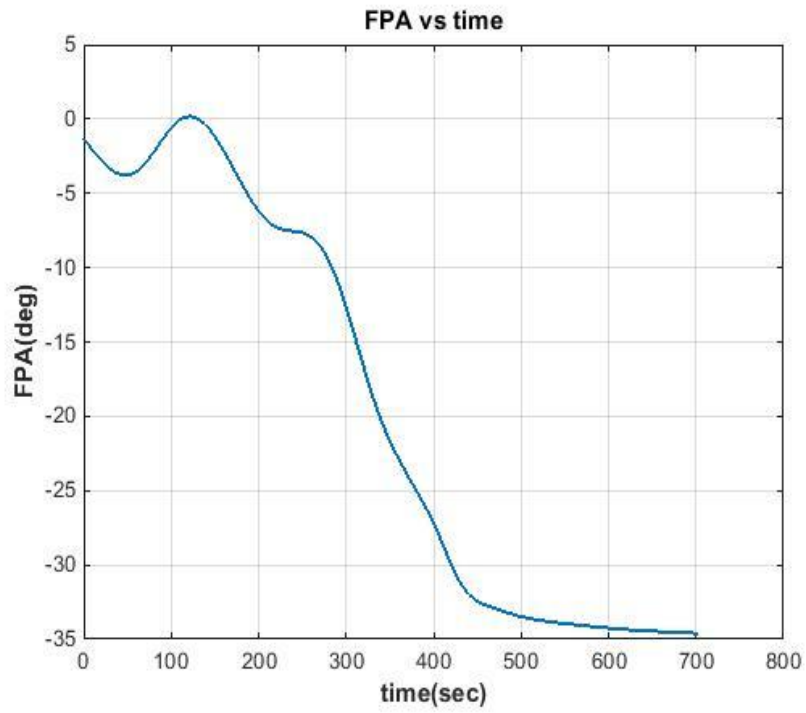


Fig.11: Flight path angle vs time

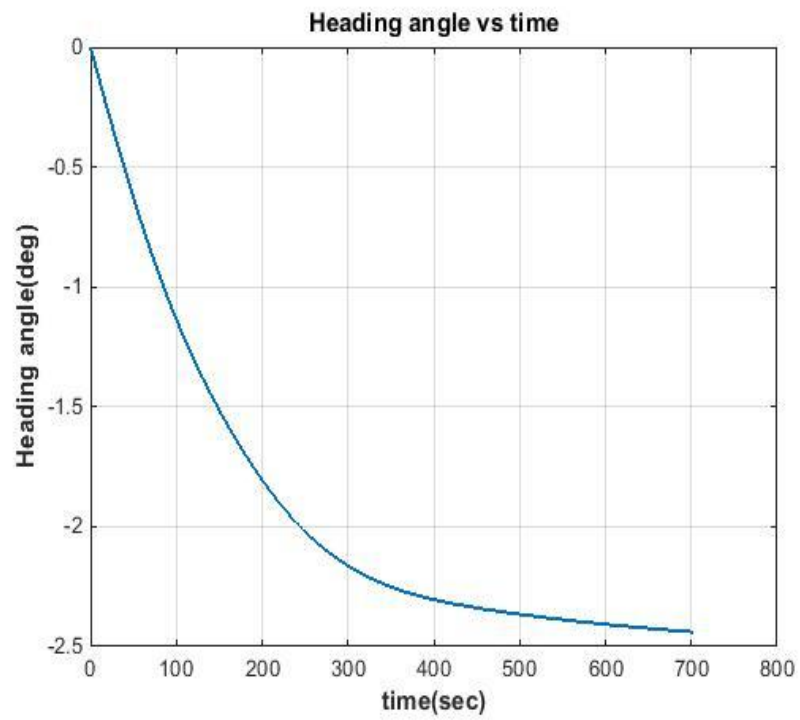


Fig.12: Heading angle vs time

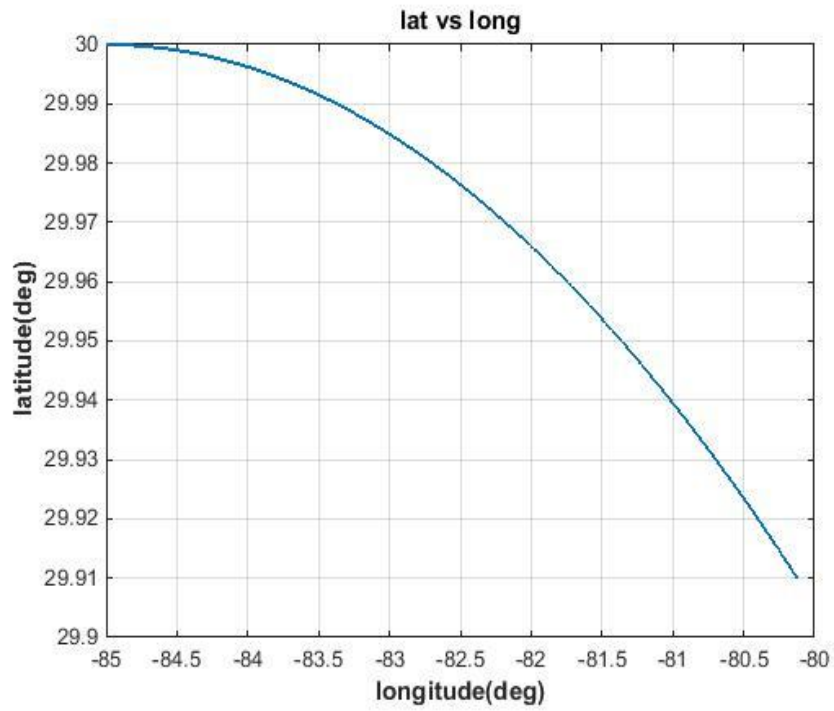


Fig.13: Flight Trajectory

From above plot it is seen that the longitude obtained at the end of is equal to maximized objective function i.e. 80.12°

CHAPTER 6

Conclusion and Future Work

Formulation of reentry dynamics relative to the local geocentric tangent frame of the rotating earth is developed. Comparative study of entry corridors for manned and unmanned cases is done to determine manned mission is critical to design. Reentry of Apollo command module is illustrated. The need of guidance is shown by comparing ballistic reentry with a lifting, ricocheting reentry at a constant angle of attack.

Classic guidance laws of constant drag deceleration followed by constant descent rate have inherent limitations because they ignore the dependence of drag deceleration and descent rate on initial flight path angle, and also because the lift is limited. The inherent limitations of the classic guidance laws are not apparent from the literature. These complexities led to the application of two-point boundary value - dynamic optimization technique to design the reentry trajectory that meets the aerodynamic heating and drag deceleration load limits.

The optimisation approach to reentry problem is developed to maximise the downrange and to land at particular location. The optimization techniques are computationally more challenging, though, because they require sophisticated optimization software.

The future avenue will be implementing various constrained optimisation techniques to reentry problem and identify the most feasible technique. The control system will be designed to follow the optimal trajectory generated based on actual dynamics.

References:

1. Brent R. Cobleigh , "*Development of X-33 Aerodynamic Uncertainty Model*", NASA/TP-1998-206544
2. Claude A.Graves and Jon C Harpold, "*Shuttle Entry Guidance*", Mission Planning and Analysis Division, NASA, JSC-14694,February 1979
3. Claude A.Graves and Jon C Harpold, "*REENTRY TARGETING PHILOSOPHYAND FLIGHT RESULTS FROM APOLLO 10 AND 11*", Mission Planning and Analysis Division,8th Aerospace Science Meeting, NASA,1970
4. D. E. Boylan B. J. Griffith, "Simulation of the Apollo Command Module Aerodynamics at High Altitudes", Journal of Spacecraft, Vol.6 No.3,pp 334-336, 1969
5. Fang Wang, Qun Zong, and Bailing Tian "*Adaptive Backstepping Finite Time Attitude Control of Reentry RLV with Input Constraint*", Hindawi Publishing Corporation Mathematical Problems in Engineering, Volume 2014, Article ID 801747, 19 pages, January 2014
6. F J Regan, " Reentry Vehicle Dynamics", AIAA 1984
7. Howard Curtis, "Orbital Mechanics for Engineering Students", Elsevier Aerospace Engineering Series, 2015
8. J R Wertz, " Spacecraft Attitude Determination and Control", Reidel,1980
9. James N. Moss, Christopher E. Glass and Francis A Greene, "*Blunt Body Aerodynamics for hypersonic low density flows*", 25th International Symposium on Rarefied Gas Dynamics , 2006
10. John D. Anderson, " Introduction to Flight" ,McGraw-Hill Book Company, Third Edition,1989
11. Kevin P. Bollino, David D. Doman, "*Optimal Nonlinear Feedback Guidance for Reentry Vehicles*", AIAA Guidance, Navigation and Control Conference and Exhibit, August 06
12. Maxwell Noton, "Spacecraft Navigation and Guidance", Springer, 2012
13. Michael V. Cook ,Flight Dynamics Principle, Elsevier Aerospace Engineering Series, Second Edition,2007
14. Oliver Montenbruck, Eberhard Gill, Satellite Orbits: Models , Methods and Applications, Springer, Second Edition, 2001
15. Muhammad Imran Afzal, Design and analysis of vehicle and guidance concept for interplanetary return mission, M.Tech Dissertation, Universität Stuttgart, 2010
16. Patrick Gallias, "Atmospheric Reentry Vehicle Mechanics", Springer, First edition,2007
17. Peng Wang, Luhua Liu, Jie Wu, Wei Zheng , "*Reentry Vehicle Modeling and Sliding Mode Controller Design*", 5th International Conference on Recent Advances in Space Technologies ,IEEE, 2011
18. Sudin Kadam, "Reentry Optimal Guidance, Control and Footprint Error Analysis for Range Maximization and Targeted Splashing", Master of Technology Thesis, IIT Bombay, July 2014
19. William S. Widnall, "Some Dynamics and Control challenges that occurred during the Apollo project" ,M.I.T., Aeronautics and Astronautics Course, Lecture 32, Fall 2009
20. Peter Zipfel, Modelling and Simulation of Aerospace Vehicle Dynamics , AIAA Education Series, 2000
21. Pierre T. Kadamba, Anouck R. Gernard, Fundamentals of Aerospace Navigation and Guidance, Cambridge Aerospace Series, First Edition, 2014

Appendix A

Standard Atmosphere Model

Altitude(m)	Temperature(K)	Density(kg/m ³)	Pressure (Pa)
0	300.600006	1.169	100900
1000	294.600006	1.064	89980
2000	288.399994	0.9672	80060
3000	283.600006	0.873	71070
4000	278.399994	0.7881	62970
5000	272.899994	0.7106	55650
6000	267.200012	0.6397	49060
7000	261.399994	0.5749	43140
8000	254.899994	0.5169	37820
9000	247.899994	0.4643	33030
10000	240.399994	0.4166	28740
11000	232.399994	0.3732	24890
12000	224.199997	0.3334	21450
13000	215.899994	0.2967	18390
14000	207.699997	0.2627	15660
15000	200.800003	0.2302	13270
16000	195.600006	0.1991	11180
17000	194.199997	0.1686	9395
18000	197.300003	0.1395	7901
19000	201.699997	0.1152	6667
20000	206.600006	0.09526	5647
21000	209.100006	0.07994	4798
22000	212.199997	0.06709	4086
23000	215.5	0.0564	3488
24000	218.800003	0.04753	2985
25000	221.300003	0.0403	2560
26000	223.800003	0.03423	2199
27000	226	0.02915	1892
28000	228	0.02491	1631
29000	230.300003	0.02129	1407
30000	232.399994	0.01824	1216
31000	234.399994	0.01564	1052
32000	236.300003	0.01344	911.900024
33000	238.399994	0.01156	791.099976
34000	240.699997	0.009946	687.299988
35000	243.399994	0.008559	597.900024
36000	246.100006	0.007375	521
37000	248.899994	0.006364	454.700012
38000	251.699997	0.005502	397.5
39000	254.199997	0.004769	348

40000	256.600006	0.004141	305
41000	258.899994	0.003603	267.700012
42000	260.600006	0.003144	235.199997
43000	262	0.00275	206.800003
44000	262.899994	0.002411	181.899994
45000	263.799988	0.002115	160.199997
46000	264.399994	0.001859	141
47000	264.700012	0.001635	124.199997
48000	264.600006	0.001441	109.400002
49000	264.100006	0.001272	96.379997
50000	263.5	0.001122	84.870003
51000	262.5	0.000992	74.709999
52000	261	0.000878	65.720001
53000	258.899994	0.000778	57.77
54000	256.399994	0.000689	50.73
55000	253.600006	0.000611	44.48
56000	251.100006	0.00054	38.959999
57000	248.5	0.000477	34.07
58000	245.300003	0.000423	29.75
59000	241.699997	0.000374	25.93
60000	238	0.00033	22.559999
61000	234.100006	0.000291	19.58
62000	230.300003	0.000257	16.950001
63000	226.899994	0.000225	14.65
64000	223.5	0.000197	12.63
65000	220.399994	0.000172	10.86
66000	217.300003	0.00015	9.327
67000	213.899994	0.00013	7.99
68000	210.199997	0.000113	6.827
69000	207.300003	0.000098	5.819
70000	204.800003	0.000084	4.95
71000	203.5	0.000072	4.204
72000	201.699997	0.000062	3.567
73000	199.5	0.000053	3.021
74000	197	0.000045	2.556
75000	194.399994	0.000039	2.157
76000	193.399994	0.000033	1.817
77000	191.399994	0.000028	1.529
78000	190.600006	0.000024	1.284
79000	188.399994	0.00002	1.078
80000	187.100006	0.000017	0.9029
81000	185.600006	0.000015	0.8508
82000	184.100006	0.000012	0.7184
83000	182.800003	0.000011	0.6061

84000	181.5	0.000009	0.5108
85000	180.300003	0.000008	0.4301
86000	179.199997	0.000006	0.3617
87000	178.199997	0.000006	0.3038
88000	177.300003	0.000005	0.2549
89000	176.5	0.000004	0.2135
90000	175.699997	0.000003	0.1786
91000	175	0.000003	0.1491
92000	174.5	0.000002	0.1243
93000	174	0.000002	0.1035
94000	173.600006	0.000002	0.08597
95000	173.300003	0.000001	0.07135
96000	173.100006	0.000001	0.05916
97000	173	0.000001	0.04903
98000	172.899994	0.000001	0.04065
99000	173.399994	0.000001	0.03373
100000	174.800003	0	0.02805